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Escuela de Ciencias Matemáticas y Computacionales

TÍTULO: ANÁLISIS COMPARATIVO DE BASES FUNCIONALES PARA SERIES ECONÓMICAS VOLÁTILES

Trabajo de integración curricular presentado como requisito para la obtención del título de Matemático

Autor/a:

Quiña Molina Michael Esteban

Tutor/a:

Ph.D. Isidro Amaro

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Resumen

El resumen es una presentación abreviada. Se debe usar una extensión máxima de 250 palabras. Se recomienda que este resumen sea analítico, es decir que sea completo, con información cuantitativa y cualitativa.

Palabras Clave:
Al final del resumen se deben usar palabras claves tomadas del texto, las cuales permiten la recuperación de la información, (Mínimo 3 y máximo 7 palabras, preferiblemente use lenguaje técnico-científico). No se deberá usar siglas, ni formulas con caracteres especiales.

Abstract

Es el mismo resumen, pero traducido al idioma inglés.

Keywords:
Al final del resumen se deben usar palabras claves tomadas del texto, las cuales permiten la recuperación de la información, (Mínimo 3 y máximo 7 palabras, preferiblemente use lenguaje técnico-científico), en idioma inglés. No se deberá usar siglas, ni formulas con caracteres especiales.

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Chapter 1

Introduction

1.1 Background

Functional Data Analysis (FDA) treats each observed trajectory—rather than each individual measurement—as the fundamental unit of statistical analysis. A discretely sampled time series is first converted into a continuous function by expressing it as a linear combination of basis elements, and all subsequent analysis (smoothing, principal component analysis, regression) is carried out on these functional objects [1]. This basis-expansion step is not a neutral preprocessing choice: the basis system determines which features of the underlying process are preserved, which are treated as noise and suppressed, and which patterns become visible to downstream methods such as Functional Principal Component Analysis (FPCA).

Most applications of FDA to macroeconomic time series adopt smooth, globally supported bases—typically B-splines or truncated Fourier series—implicitly assuming that the underlying trajectories belong to a Sobolev space of functions with bounded, uniformly distributed roughness [1]. This approach is applied by [2] to annual GDP growth rates of Latin American economies, representing each country's growth trajectory with a B-spline basis before conducting FPCA to study cross-country modes of variation. This choice is reasonable for economies whose growth process evolves smoothly, but Latin American GDP growth series are also punctuated by abrupt, localized shocks—the 1982 debt crisis, the 1999 regional currency crises, the 2008 global financial crisis, and the 2020 COVID-19 contraction—whose magnitude and timing differ sharply across countries.

An alternative to globally smooth bases is provided by wavelets, which form an unconditional basis for Besov spaces $B^s_{p,q}$—function classes that, unlike Sobolev spaces, tolerate spatially inhomogeneous regularity, i.e. functions that are smooth over most of their domain but exhibit sharp, localized transitions elsewhere [3]. Because wavelets localize simultaneously in time and frequency, they can in principle represent a smooth trend and a sharp crisis episode within the same trajectory without forcing a single global smoothness parameter to compromise between the two. This theoretical property motivates a systematic empirical comparison between globally smooth bases (B-splines, Fourier) and a multiresolution basis (wavelets) for representing volatile economic series.

1.2 Problem statement

The choice of functional basis in FDA applications to economic time series is rarely treated as an object of empirical scrutiny in its own right: a single basis system—most often B-splines—is selected on convenience or precedent, fitted, and used as the input to FPCA or other downstream analyses, without assessing whether its underlying smoothness assumptions are appropriate for the data at hand. This is problematic for series such as GDP growth in volatile emerging economies, where country-level heterogeneity in volatility (from Guatemala's comparatively stable trajectory to Venezuela's crisis-dominated one) and time-varying, crisis-driven variance call into question the adequacy of a single global smoothness criterion applied uniformly across countries.

Two consequences follow when this choice is left unexamined. First, a basis that oversmooths may achieve a favorable reconstruction error while silently discarding the crisis episodes that are often the most economically meaningful features of the series, so that low error is mistaken for a faithful representation. Second, because FPCA is computed directly on the fitted functional objects, any distortion introduced at the basis-expansion stage propagates into the estimated modes of variation, so that the resulting principal components may reflect artifacts of the smoothing choice rather than genuine cross-country structure. Existing work extending the study of [2] has not systematically compared alternative basis families under a common, fair evaluation protocol that jointly considers reconstruction fidelity, coefficient economy (sparsity), and the interpretability of the re-

sulting FPCA. This thesis addresses that gap by comparing B-spline, Fourier, and wavelet basis expansions on an extended panel of Latin American GDP growth series (1960–2024).

1.3 Objectives

1.3.1 General Objective

To comparatively evaluate B-spline, Fourier, and wavelet basis expansions for representing the GDP growth trajectories of volatile Latin American economies, in terms of reconstruction fidelity, coefficient economy, and their effect on the interpretability of the resulting functional principal component analysis.

1.3.2 Specific Objectives

1. To extend the annual GDP growth dataset of [2] for 20 Latin American countries through 2024, incorporating the COVID-19 contraction and subsequent recovery period.

2. To implement three families of functional basis expansions for the resulting panel—B-spline, Fourier, and wavelet (Haar, DB2, Sym4)—under a common, reproducible computational pipeline.

3. To define and compute comparison metrics—reconstruction error (RMSE), sparsity, and FPCA variance explained—that allow the three basis families to be evaluated at matched levels of reconstruction quality.

4. To assess, for each basis family, the extent to which crisis episodes and country-specific volatility structure are preserved or suppressed in the functional reconstruction.

5. To characterize how the choice of basis affects the parsimony and economic interpretability of the resulting FPCA modes of variation, relative to the original results of [2].

Chapter 2

Theoretical Framework

2.1 Overview

This chapter develops, with full mathematical rigor, the theoretical machinery underlying the functional principal component analysis (FPCA) pipeline implemented in Chapter 4: smoothing of discretely observed trajectories, estimation of the mean function and covariance operator, spectral decomposition of the covariance operator, and asymptotic validation of the resulting estimators. The exposition follows [4], whose treatment of functional data analysis through the lens of Hilbert space and linear operator theory provides, at present, the most complete rigorous account of this pipeline. Definitions, theorem statements, and proofs throughout this chapter are our own exposition of that material, adapted to the notation and scope of this thesis, with [4] cited at the relevant points.

The chapter is organized to mirror the logical dependency of the FPCA pipeline rather than the chronological order in which a reader might first encounter the topics. Section 2.2 fixes notation and recalls, without proof, the elementary Hilbert space and bounded-operator concepts (inner products, orthonormal bases, the projection theorem, reproducing kernel Hilbert spaces, Sobolev spaces, adjoint operators) that everything else is built on. Section 2.3 develops the spectral theory of compact operators, culminating in Mercer's Theorem, which identifies the covariance kernel of a stochastic process with the spectral decomposition of its covariance operator. Section 2.4 develops perturbation bounds for the eigenvalues and eigenfunctions of a compact self-adjoint operator, which are the technical tool later used to transfer consistency of an estimated covariance operator into consis-

tency of its estimated eigenelements. Section 2.5 treats smoothing and regularization, in particular smoothing splines, which underlie the adaptive B-spline basis used in Chapter 4. Section 2.6 develops the probabilistic foundations of FDA: the mean element and covariance operator of a Hilbert-space-valued random element, and the Karhunen–Loève Theorem, which is the population version of FPCA. Section 2.7 studies the large sample behavior of estimators of the mean function and covariance operator, and Section 2.8 closes the chapter with the asymptotic theory of the FPCA estimators themselves — consistency and asymptotic normality of the estimated eigenvalues and eigenfunctions — which is the theoretical justification for the FPCA results reported in Chapter 5.

A scope note is in order. The source [4] develops this entire pipeline for general basis systems and, concretely, for smoothing splines (equivalently, B-spline bases with a roughness penalty), which matches the adaptive B-spline smoothing used as the baseline in Section 4.2.1. It does not, however, treat the Fourier or wavelet basis systems used as the comparison bases in this thesis. The theoretical treatment of those two basis systems — in particular, the approximation-theoretic and minimax properties that motivate the use of wavelets for spatially inhomogeneous (Besov-space) signals such as the ones studied in Chapter 5 — will be added from a second source in a subsequent revision of this chapter.

Throughout, $\mathbb{H}$, $\mathbb{H}_1$, $\mathbb{H}_2$ denote real Hilbert spaces with inner products $\langle \cdot, \cdot \rangle$ (or $\langle \cdot, \cdot \rangle_1$, $\langle \cdot, \cdot \rangle_2$) and induced norms $\| \cdot \|$. To avoid a notational clash with the number of retained basis coefficients K used in Chapter 4, we denote the covariance operator of a random element by Γ and its kernel by $\gamma(s, t)$, rather than the letter K used for this purpose in [4].

2.2 Mathematical Preliminaries

This section intents to provide a fast review of the elementary facts about metric, normed, Hilbert, and reproducing kernel Hilbert spaces, and about bounded linear operators, that are used throughout the rest of the chapter. All results in this section are standard; proofs may be found in [4], as well as in [5], [6], and [7].

2.2.1 Metric, Normed, and Banach Spaces

Definition 2.1. *A* metric *on a set* $\mathbb{M}$ *is a function* $d : \mathbb{M} \times \mathbb{M} \to \mathbb{R}$ *satisfying, for all* $x_1, x_2, x_3 \in \mathbb{M}$*:*
(i) $d(x_1, x_2) \geq 0$*;*
(ii) $d(x_1, x_2) = 0$ *iff* $x_1 = x_2$*;*
(iii) $d(x_1, x_2) = d(x_2, x_1)$*;*
(iv) $d(x_1, x_3) \leq d(x_1, x_2) + d(x_2, x_3)$ *(triangle inequality).*
The pair $(\mathbb{M}, d)$ *is called a* metric space.

From now on, we will denote a metric space simply by $\mathbb{M}$, with the metric d understood. A sequence $\{x_n\} \subset \mathbb{M}$ converges to $x \in \mathbb{M}$ if $d(x_n, x) \to 0$ as $n \to \infty$.

Definition 2.2. *A sequence* $\{x_n\} \subset \mathbb{M}$ *is* Cauchy *if for each* $\varepsilon > 0$ *there exists* $N \in \mathbb{N}$ *such that for all* $m, n \geq N$, $d(x_m, x_n) < \varepsilon$*. The metric space* $(\mathbb{M}, d)$ *is* complete *if every Cauchy sequence converges, and* separable *if it has a countable dense subset.*

Definition 2.3. *Let* $\mathbb{V}$ *be a vector space over* $\mathbb{R}$*. A* norm *on* $\mathbb{V}$ *is a function* $\|\cdot\| : \mathbb{V} \to \mathbb{R}$ *such that forall* $v, v_1, v_2 \in \mathbb{V}$, $a \in \mathbb{R}$*:*
(i) $\|v\| \geq 0$ *with equality iff* $v = 0$*;*
(ii) $\|av\| = |a|\|v\|$*; and*
(iii) $\|v_1 + v_2\| \leq \|v_1\| + \|v_2\|$.
Every norm induces a metric $d(x, y) := \|x - y\|$*. A* Banach space *is a normed vector space that is complete under this metric.*

Definition 2.4. *Let* $(E, \mathcal{B}, \mu)$ *be a measure space and* $p \in [1, \infty)$*. The space* $\mathbb{L}^p(E, \mathcal{B}, \mu)$ *consists of (equivalence classes, under equality* μ*-a.e., of) measurable functions* f *on* E *with* $\int_E |f|^p \, d\mu < \infty$*, equipped with the norm* $\|f\|_p = \left(\int_E |f|^p \, d\mu \right)^{1/p}$*. When* $E = [0, 1]$, $\mathcal{B}$ *is the Borel* σ*-field, and* μ *is Lebesgue measure, we write* $\mathbb{L}^p[0, 1]$.

Theorem 2.1 (Riesz–Fischer Theorem)**.** *Each* $\mathbb{L}^p(E, \mathcal{B}, \mu)$, $p \geq 1$, *is a Banach space.*

2.2.2 Hilbert Spaces and Orthonormal Bases

Definition 2.5. *An* inner product *on a vector space* $\mathbb{V}$ *is a function* $\langle \cdot, \cdot \rangle : \mathbb{V} \times \mathbb{V} \to \mathbb{R}$ *that is symmetric, bilinear, and satisfies* $\langle v, v \rangle \geq 0$ *with equality iff* $v = 0$*. It induces the norm* $\|v\| = \langle v, v \rangle^{1/2}$*, for which the Cauchy–Schwarz inequality* $|\langle v_1, v_2 \rangle| \leq \|v_1\|\|v_2\|$ *holds. A* Hilbert space *is a complete inner-product space.*

The space $\mathbb{L}^2(E, \mathcal{B}, \mu)$ of Definition 2.4 is a Hilbert space with $\langle f_1, f_2 \rangle = \int_E f_1 f_2 \, d\mu$; this is the setting most relevant for FDA, with $E = [0, 1]$ after rescaling of the observation window $\mathcal{T}$.

Definition 2.6. *Elements x_1, x_2 of a Hilbert space are* orthogonal *if $\langle x_1, x_2 \rangle = 0$. A countable collection $\{e_j\}$ is an* orthonormal sequence *if $\|e_j\| = 1$ for all j and the e_j are pairwise orthogonal. An orthonormal sequence $\{e_j\}$ in a Hilbert space $\mathbb{H}$ is an* orthonormal basis*, or complete orthonormal system (CONS), if $\overline{\text{span}}\{e_j\} = \mathbb{H}$, where $\overline{\text{span}}$ denotes closed linear span; equivalently, if $\langle x, e_j \rangle = 0$ for all j forces $x = 0$.*

Theorem 2.2 (Fourier expansion / Parseval's relation)**.** *If $\{e_j\}$ is a CONS for a Hilbert space $\mathbb{H}$, every $x \in \mathbb{H}$ satisfies $x = \sum_{j=1}^{\infty} \langle x, e_j \rangle e_j$ and $\|x\|^2 = \sum_{j=1}^{\infty} \langle x, e_j \rangle^2$. A Hilbert space is separable if and only if it admits a countable orthonormal basis; every infinite-dimensional separable Hilbert space is congruent (isometrically isomorphic) to ℓ^2, the space of square-summable real sequences.*

2.2.3 The Projection Theorem

Theorem 2.3 (Projection Theorem)**.** *Let $\mathbb{M}$ be a closed linear subspace of a Hilbert space $\mathbb{H}$. For every $x \in \mathbb{H}$ there is a unique element $\hat{x} \in \mathbb{M}$, called the* projection *of x onto $\mathbb{M}$, minimizing $\|x - y\|$ over $y \in \mathbb{M}$; it is characterized by $\langle x - \hat{x}, y \rangle = 0$ for all $y \in \mathbb{M}$.*

Consequently, writing $\mathbb{M}^{\perp} := \{x \in \mathbb{H} : \langle x, y \rangle = 0 \text{ for all } y \in \mathbb{M}\}$ for the orthogonal complement of $\mathbb{M}$, every $x \in \mathbb{H}$ decomposes uniquely as $x = x_1 + x_2$ with $x_1 \in \mathbb{M}$, $x_2 \in \mathbb{M}^{\perp}$, written $\mathbb{H} = \mathbb{M} \oplus \mathbb{M}^{\perp}$.

2.2.4 Reproducing Kernel Hilbert Spaces

Definition 2.7. *Let $\mathbb{H}$ be a Hilbert space of real-valued functions on a set E. A bivariate function K on $E \times E$ is a* reproducing kernel *for $\mathbb{H}$ if:*
(i) $K(\cdot, t) \in \mathbb{H}$ for every $t \in E$, and
(ii)$f(t) = \langle f, K(\cdot, t) \rangle$ for every $f \in \mathbb{H}$, $t \in E$ (the reproducing property).
A Hilbert space possessing a reproducing kernel is a reproducing kernel Hilbert space *(RKHS), denoted $\mathbb{H}(K)$.*

The evaluation functional $\mathcal{T}_t : f \mapsto f(t)$ has representer $K(\cdot, t)$, and $\mathbb{H}$ is an RKHS if and only if every evaluation functional $\mathcal{T}_t$, $t \in E$, is bounded.

A reproducing kernel is necessarily unique, symmetric, and nonnegative definite, i.e. $\sum_{i=1}^{n}\sum_{j=1}^{n} a_i a_j K(t_i, t_j) \geq 0$ for any n, $\{a_i\} \subset \mathbb{R}$, $\{t_i\} \subset E$. Conversely, the Moore–Aronszajn Theorem guarantees that every symmetric, nonnegative definite K on $E \times E$ is the reproducing kernel of a unique RKHS $\mathbb{H}(K)$. If K_1, K_2 are reproducing kernels for RKHSs $\mathbb{H}(K_1), \mathbb{H}(K_2)$ of functions on E with $\mathbb{H}(K_1) \cap \mathbb{H}(K_2) = \{0\}$, then $K = K_1 + K_2$ is the reproducing kernel of $\mathbb{H}(K) = \mathbb{H}(K_1) \oplus \mathbb{H}(K_2)$ (the two summand spaces are then orthogonal subspaces of $\mathbb{H}(K)$). RKHS norm convergence implies pointwise (and, under a uniform bound on K, uniform) convergence, which is what distinguishes RKHSs from general $\mathbb{L}^2$-type function spaces and makes them convenient for smoothing problems (Section 2.5) and for representing covariance kernels (Section 2.6).

2.2.5 Sobolev Spaces

Definition 2.8. *For an integer $q \geq 1$, the* Sobolev space $\mathbb{W}^q[0,1]$ *consists of functions f on $[0,1]$ that are $(q-1)$-times differentiable with $f^{(q-1)}$ absolutely continuous and $f^{(q)} \in \mathbb{L}^2[0,1]$. Equipped with the inner product*

$$\langle f, g\rangle'_{\mathbb{W}^q[0,1]} := \langle f, g\rangle_2 + \langle f^{(q)}, g^{(q)}\rangle_2, \tag{2.1}$$

where $\langle\cdot,\cdot\rangle_2$ is the $\mathbb{L}^2[0,1]$ inner product, $\mathbb{W}^q[0,1]$ is a reproducing kernel Hilbert space.

The Sobolev space $\mathbb{W}^q[0,1]$ formalizes the notion of a function with bounded, uniformly distributed roughness of order q: it is precisely the class of functions whose smoothing-spline estimator is studied in Section 2.5, and the assumption that GDP growth trajectories $X_i \in \mathbb{W}^q[0,1]$ underlies the smoothing-spline justification for the B-spline baseline of Section 4.2.1.

2.2.6 Bounded Linear Operators and Adjoints

Definition 2.9. *Let $\mathbb{X}_1, \mathbb{X}_2$ be normed vector spaces. A linear transformation $\mathcal{T} : \mathbb{X}_1 \to \mathbb{X}_2$ is* bounded *if there is a finite $C > 0$ with $\|\mathcal{T}x\|_2 \leq C\|x\|_1$ for all $x \in \mathbb{X}_1$.*

Boundedness is equivalent to (uniform) continuity of $\mathcal{T}$. The set $\mathcal{B}(\mathbb{X}_1, \mathbb{X}_2)$ of bounded linear transformations from $\mathbb{X}_1$ to $\mathbb{X}_2$ is itself a normed space under the *operator norm* $\|\mathcal{T}\| := \sup_{\|x\|_1=1} \|\mathcal{T}x\|_2$, and is a Banach space whenever $\mathbb{X}_2$ is. We write $\mathcal{B}(\mathbb{X})$ for $\mathcal{B}(\mathbb{X}, \mathbb{X})$.

Theorem 2.4 (Adjoint operator)**.** *Let $\mathbb{H}_1, \mathbb{H}_2$ be Hilbert spaces. For every $\mathcal{T} \in \mathcal{B}(\mathbb{H}_1, \mathbb{H}_2)$ there is a unique $\mathcal{T}^* \in \mathcal{B}(\mathbb{H}_2, \mathbb{H}_1)$, the* adjoint *of $\mathcal{T}$, satisfying*

$$\langle \mathcal{T}x_1, x_2\rangle_2 = \langle x_1, \mathcal{T}^* x_2\rangle_1, \qquad x_1 \in \mathbb{H}_1,\ x_2 \in \mathbb{H}_2. \tag{2.2}$$

If $\mathbb{H}_1 = \mathbb{H}_2 = \mathbb{H}$, $\mathcal{T}$ is self-adjoint *if $\mathcal{T}^* = \mathcal{T}$. The adjoint satisfies $(\mathcal{T}^*)^* = \mathcal{T}$, $\|\mathcal{T}^*\| = \|\mathcal{T}\|$, and $\|\mathcal{T}^*\mathcal{T}\| = \|\mathcal{T}\|^2$.*

Definition 2.10. *A self-adjoint operator $\mathcal{T} \in \mathcal{B}(\mathbb{H})$ is* nonnegative definite *if $\langle \mathcal{T}x, x\rangle \geq 0$ for all $x \in \mathbb{H}$.*

Note that for any $\mathcal{T} \in \mathcal{B}(\mathbb{H}_1, \mathbb{H}_2)$, the operator $\mathcal{T}^*\mathcal{T}$ is nonnegative definite on $\mathbb{H}_1$, since it is self-adjoint and $\langle \mathcal{T}^*\mathcal{T}x, x\rangle = \|\mathcal{T}x\|^2 \geq 0$.

This fact is the operator-theoretic reason that covariance operators (Section 2.6), which have precisely this $\mathcal{T}^*\mathcal{T}$-type structure, are always nonnegative definite.

2.3 Spectral Theory of Compact Operators

This section develops the eigenvalue/eigenvector decomposition of a compact self-adjoint operator, its non-self-adjoint (singular value) counterpart, and Mercer's Theorem, which identifies the spectral decomposition of an integral operator with a series expansion of its kernel. This machinery is what will let us pass, in Section 2.6, from a covariance kernel $\gamma(s,t)$ to the eigenvalues/eigenfunctions that define FPCA. Throughout, all Hilbert spaces are assumed separable, which entails no loss of generality for compact operators (Theorem 2.7 below). Definitions, theorems, and proofs follow [4], Chapter 4.

2.3.1 Compact Operators

Definition 2.11. *A linear transformation $\mathcal{T}$ from a normed space $\mathbb{X}_1$ into a normed space $\mathbb{X}_2$ is* compact *if for every bounded sequence $\{x_n\} \subset \mathbb{X}_1$, $\{\mathcal{T}x_n\}$ contains a subsequence that converges in $\mathbb{X}_2$.*

A compact linear transformation is necessarily bounded: if $\mathcal{T}$ were unbounded there would be a bounded sequence $\{x_n\}$ with $\|\mathcal{T}x_n\|_2 \geq n$, so $\{\mathcal{T}x_n\}$ could have no convergent subsequence. Compactness is a genuine restriction, however.

Theorem 2.5. *The identity operator on an infinite-dimensional normed space is not compact.*

Proof. Consider first a Hilbert space $\mathbb{H}$ with CONS $\{e_j\}$. Since $\|e_i - e_j\| = \sqrt{2}$ for $i \neq j$, the bounded sequence $\{e_j\}$ has no Cauchy (hence no convergent) subsequence, so $\{Ie_j\}$ has none either.

For a general infinite-dimensional normed space $\mathbb{X}$, it suffices to construct a bounded sequence $\{e_j\}$ with $\inf_{i\neq j} \|e_i - e_j\| > 0$; the same argument then applies. Let e_1 have unit norm and $\mathbb{Y}_1 = \operatorname{span}\{e_1\}$. Pick any $x \notin \mathbb{Y}_1$ and set $d = \inf\{\|x - y\| : y \in \mathbb{Y}_1\} > 0$ (finite since $\mathbb{Y}_1$ is closed, being finite-dimensional). For any $\alpha \in (1, \infty)$ choose $z \in \mathbb{Y}_1$ with $\|x - z\| < \alpha d$ and set $e_2 = (x - z)/\|x - z\|$. For any $y \in \mathbb{Y}_1$,

$$\|e_2 - y\| = \frac{\|x - (z + \|x - z\|y)\|}{\|x - z\|} > \frac{d}{\alpha d} = \alpha^{-1},$$

since $z + \|x - z\|y \in \mathbb{Y}_1$; in particular $\|e_1 - e_2\| > \alpha^{-1}$. Repeating this construction with $\mathbb{Y}_2 = \operatorname{span}\{e_1, e_2\}$, and inductively thereafter, produces $\{e_j\}$ with $\|e_i - e_j\| > \alpha^{-1}$ for all $i \neq j$. □

Theorem 2.6. *Let $\mathcal{T}$ be a compact operator between normed spaces $\mathbb{X}_1, \mathbb{X}_2$. Then:*
(1) $\overline{\operatorname{Im}(\mathcal{T})}$ is separable;
(2) any finite-rank bounded operator is compact;
(3) the composition of two operators is compact if either factor is compact;
(4) if $\mathbb{X}_2$ is a Banach space, the set of compact operators in $\mathcal{B}(\mathbb{X}_1, \mathbb{X}_2)$ is closed under the operator norm.

Proof. (1) Let $B(0; r) = \{x \in \mathbb{X}_1 : \|x\|_1 \leq r\}$ and $\{y_n\} \subset \mathcal{T}(B(0; r))$. Choose $x_n \in B(0; r)$ with $\|y_n - \mathcal{T}x_n\|_2 < n^{-1}$. Compactness of $\mathcal{T}$ gives a convergent subsequence of $\{\mathcal{T}x_n\}$, hence of $\{y_n\}$, so $\mathcal{T}(B(0; r))$ is sequentially compact, hence compact and separable. Since $\operatorname{Im}(\mathcal{T}) \subset \bigcup_{r=1}^{\infty} \mathcal{T}(B(0; r))$, $\operatorname{Im}(\mathcal{T})$, and thus its closure, is separable.

(2) If $\operatorname{Im}(\mathcal{T})$ is finite-dimensional and $\{x_n\}$ is bounded in $\mathbb{X}_1$, the Bolzano–Weierstrass Theorem gives a convergent subsequence of $\{\mathcal{T}x_n\}$.

(3) Let $\mathcal{T}_1 \in \mathcal{B}(\mathbb{X}_1, \mathbb{X}_2)$, $\mathcal{T}_2 \in \mathcal{B}(\mathbb{X}_2, \mathbb{X}_3)$ and $\{x_n\}$ bounded in $\mathbb{X}_1$. If $\mathcal{T}_1$ is compact, $\{\mathcal{T}_1 x_n\}$ has a convergent subsequence, whose image under the continuous map $\mathcal{T}_2$ also converges. If instead $\mathcal{T}_2$ is compact, then since $\{\mathcal{T}_1 x_n\}$ is bounded, $\{\mathcal{T}_2\mathcal{T}_1 x_n\}$ has a convergent subsequence.

(4) Let $\{\mathcal{T}_n\}$ be compact with $\|\mathcal{T}_n - \mathcal{T}\| \to 0$. Let $\{x_k\}$ be bounded in $\mathbb{X}_1$. By a diagonal argument, extract nested subsequences $\{x_{nk}\}_k \subset \{x_{(n-1)k}\}_k$ (with $\{x_{0k}\} := \{x_k\}$) such that $\mathcal{T}_n x_{nk} \to y_n$ as $k \to \infty$, for each n. Choose k_n increasing in n with $\|\mathcal{T}_n x_{nk} - y_n\|_2 < n^{-1}$

for $k \geq k_n$. For $n' > n$,

$$y_n - y_{n'} = (y_n - \mathcal{T}_n x_{n'k_{n'}}) + (\mathcal{T}_{n'} x_{n'k_{n'}} - y_{n'}) + (\mathcal{T}_n - \mathcal{T})x_{n'k_{n'}} + (\mathcal{T} - \mathcal{T}_{n'})x_{n'k_{n'}},$$

so $\|y_n - y_{n'}\|_2 \leq n^{-1} + n'^{-1} + \|\mathcal{T}_n - \mathcal{T}\| + \|\mathcal{T}_{n'} - \mathcal{T}\|$, which shows $\{y_n\}$ is Cauchy, hence convergent to some y (as $\mathbb{X}_2$ is complete). Finally

$$\|\mathcal{T} x_{nk_n} - y\|_2 \leq \|\mathcal{T} - \mathcal{T}_n\| \|x_{nk_n}\|_1 + \|\mathcal{T}_n x_{nk_n} - y_n\|_2 + \|y_n - y\|_2 \to 0,$$

so $\{\mathcal{T} x_{nk_n}\}$ converges and $\mathcal{T}$ is compact. □

Theorem 2.7. *Let $\mathbb{H}_1, \mathbb{H}_2$ be Hilbert spaces and $\mathcal{T} \in \mathcal{B}(\mathbb{H}_1, \mathbb{H}_2)$. Then:*
(1) $\mathcal{T}$ is compact if and only if there exist a sequence offinite-rank operators $\{\mathcal{T}_n\}$ with $\|\mathcal{T}_n - \mathcal{T}\| \to 0$ as $n \to \infty$; and
(2)$\mathcal{T}$ is compact if and only if $\mathcal{T}^*$ is compact.

Proof. (1) ($\Longleftarrow$) Assume there exists a sequence of finite-rank operators $\{\mathcal{T}_n\}$ with $\|\mathcal{T}_n - \mathcal{T}\| \to 0$ as $n \to \infty$. The result follows from parts (2) and (4) of Theorem 2.6. ($\Longrightarrow$) Assume $\mathcal{T}$ is compact. Part (1) of Theorem 2.6 gives a CONS $\{e_j\}$ for $\overline{\mathrm{Im}(\mathcal{T})}$; define $\mathcal{T}_n x := \sum_{j=1}^{n} \langle \mathcal{T}x, e_j \rangle_2 e_j = \mathcal{P}_n \mathcal{T} x$ with $\mathcal{P}_n$ the projection onto $\mathrm{span}\{e_1, \ldots, e_n\}$. By contradiction, suppose that $\|\mathcal{T}_n - \mathcal{T}\| \not\to 0$, then there is $\epsilon > 0$ and unit-norm $\{x_{n'}\}$ with $\|(\mathcal{T} - \mathcal{T}_{n'})x_{n'}\|_2 > \epsilon$. By compactness $\mathcal{T} x_{n''} \to y$ along a subsequence, and

$$(\mathcal{T} - \mathcal{T}_{n''})x_{n''} = (I - \mathcal{P}_{n''})y + (I - \mathcal{P}_{n''})(\mathcal{T} x_{n''} - y) \to 0,$$

a contradiction; hence $\|\mathcal{T}_n - \mathcal{T}\| \to 0$.

(2) $\mathcal{T}_n^*$ is finite-rank (Theorem 2.4) with $\|\mathcal{T}_n^* - \mathcal{T}^*\| = \|\mathcal{T}_n - \mathcal{T}\|$, so if $\mathcal{T}$ is compact, so is $\mathcal{T}^*$ by the first part of the theorem; apply this twice, using $(\mathcal{T}^*)^* = \mathcal{T}$, for the converse. □

As a consequence of Theorem 2.7 and part (1) of Theorem 2.6, for a compact $\mathcal{T} \in \mathcal{B}(\mathbb{H}_1, \mathbb{H}_2)$, $\overline{\mathrm{Im}(\mathcal{T}^*)}$ is separable, and since $\mathbb{H}_1 = \mathrm{Ker}(\mathcal{T}) \oplus \overline{\mathrm{Im}(\mathcal{T}^*)}$, there is no loss of generality in assuming both $\mathbb{H}_1, \mathbb{H}_2$ are separable, an assumption we maintain throughout.

2.3.2 Spectral Decomposition of Compact Self-Adjoint Operators

Definition 2.12. *Let $\mathcal{T} \in \mathcal{B}(\mathbb{H})$, $\lambda \in \mathbb{R}$ and nonzero $e \in \mathbb{H}$ with $\mathcal{T}e = \lambda e$. Then λ and e are, respectively, an* eigenvalue *and corresponding* eigenvector *(or* eigenfunction*) of $\mathcal{T}$.* $\mathrm{Ker}(\mathcal{T} - \lambda I)$ *is the* eigenspace *of λ.*

Theorem 2.8. *Let $\mathcal{T} \in \mathcal{B}(\mathbb{H})$ and $e_j \in \mathrm{Ker}(\mathcal{T} - \lambda_j I)$, $j = 1, 2, \ldots$, with the λ_j distinct and nonzero. Then:*
(1) the e_j are linearly independent;
(2) if $\mathcal{T}$ is self-adjoint, the e_j are mutually orthogonal.

Proof. (1) We have to show that if there are scalars $c_1, \ldots, c_n$ such that $\sum_{j=1}^{n} c_j e_j = 0$ and $\sum_{j=1}^{n} \lambda c_j e_j = 0$, then $c_j = 0$ for all j. We proceed by induction on n. For $n = 2$: if $c_1 e_1 + c_2 e_2 = 0$ then also $\lambda_1 c_1 e_1 + \lambda_2 c_2 e_2 = 0$, so $\lambda_1(c_1 e_1 + c_2 e_2) - (\lambda_1 c_1 e_1 + \lambda_2 c_2 e_2) = (\lambda_1 - \lambda_2) c_2 e_2 = 0$, forcing $c_2 = 0$ (as $\lambda_1 \neq \lambda_2$) and then $c_1 = 0$. Assume the claim for $n = k$ and suppose $\sum_{j=1}^{k+1} c_j e_j = 0$. Then $\lambda_{k+1} \sum_{j=1}^{k+1} c_j e_j - \sum_{j=1}^{k+1} \lambda_j c_j e_j = \sum_{j=1}^{k} (\lambda_{k+1} - \lambda_j) c_j e_j = 0$, so $c_j = 0$ for $j \leq k$ by the induction hypothesis, and then $c_{k+1} = 0$.

(2) $\langle e_i, e_j \rangle = \langle e_i, \lambda_j^{-1} \mathcal{T} e_j \rangle = \lambda_j^{-1} \langle \mathcal{T} e_i, e_j \rangle = \lambda_j^{-1} \lambda_i \langle e_i, e_j \rangle$, which forces $\langle e_i, e_j \rangle = 0$ since $\lambda_i \neq \lambda_j$. □

Theorem 2.9. *Let $\mathcal{T} \in \mathcal{B}(\mathbb{H})$ be compact. Then:*
(1) $\mathrm{Ker}(\mathcal{T} - \lambda I)$ is finite-dimensional for every $\lambda \neq 0$;
(2) for every $\epsilon > 0$, only finitely many distinct eigenvalues of $\mathcal{T}$ satisfy $|\lambda| \geq \epsilon$;
(3) the set of nonzero eigenvalues of $\mathcal{T}$ is countable.

Proof. (1) By contradiction, suppose that $\mathrm{Ker}(\mathcal{T} - \lambda I)$ is infinite-dimensional for some $\lambda \neq 0$, then the construction of Theorem 2.5 yields unit-norm $\{e_j\} \subset \mathrm{Ker}(\mathcal{T} - \lambda I)$ with $\inf_{i \neq j} \|e_i - e_j\| > 0$, so $\inf_{i \neq j} \|\mathcal{T} e_i - \mathcal{T} e_j\| = |\lambda| \inf_{i \neq j} \|e_i - e_j\| > 0$, contradicting compactness.

(2) By contradiction, suppose there are infinitely many distinct eigenvalues $\lambda_1, \lambda_2, \ldots$ with $|\lambda_j| \geq \epsilon$, for any $\epsilon > 0$. Denote $\mathbb{M}_n = \mathrm{span}\{e_1, \ldots, e_n\}$ for $e_j \in \mathrm{Ker}(\mathcal{T} - \lambda_j I)$, $\mathbb{M}_0 := \{0\}$. By part (1) of Theorem 2.8 the e_j are linearly independent, so $\mathbb{M}_n \cap \mathbb{M}_{n-1}^{\perp}$ is one-dimensional; Gram–Schmidt produces an orthonormal sequence $\{x_n\}$ with $x_n \in \mathrm{Ker}(\mathcal{T} - \lambda_n I) \cap \mathbb{M}_{n-1}^{\perp}$, and for $n > m$, $\|\mathcal{T} x_n - \mathcal{T} x_m\|^2 = \|\mathcal{T} x_n\|^2 + \|\mathcal{T} x_m\|^2 \geq 2\lambda^2$, which contradicts compactness.

(3) Immediate from (2). □

We now state and prove the central result of this section: every compact self-adjoint operator admits an eigenvalue/eigenvector decomposition that is the direct infinite-dimensional analog of the spectral decomposition of a symmetric positive semidefinite matrix.

Theorem 2.10 (Spectral Theorem for compact self-adjoint operators)**.** *Let $\mathcal{T}$ be a compact, self-adjoint operator on $\mathbb{H}$. The nonzero eigenvalues of $\mathcal{T}$ form a finite set or a sequence tending to 0, each with finite-dimensional eigenspace, and eigenvectors for distinct*

eigenvalues are orthogonal. Order the eigenvalues $|\lambda_1| \geq |\lambda_2| \geq \cdots$ *with corresponding orthonormal eigenvectors* $e_1, e_2, \ldots$ *(obtained via Gram–Schmidt within repeated eigenvalues). Then* $\{e_j\}$ *is a CONS for* $\overline{\mathrm{Im}(\mathcal{T})}$ *and*

$$\mathcal{T} = \sum_{j\geq 1} \lambda_j \, e_j \otimes e_j, \qquad i.e. \qquad \mathcal{T}x = \sum_{j\geq 1} \lambda_j \langle x, e_j\rangle e_j, \quad x \in \mathbb{H}. \tag{2.3}$$

Proof. Finiteness/decay of the eigenvalue sequence, finite multiplicity, and orthogonality of eigenvectors are Theorems 2.8 and 2.9. It remains to prove (2.3). Since $\mathcal{T}$ is self-adjoint, Theorem **??**, part (3) with $\mathcal{T}^* = \mathcal{T}$ gives $\mathbb{H} = \mathrm{Ker}(\mathcal{T}) \oplus \mathrm{Im}(\mathcal{T})$, and (2.3) clearly holds for $x \in \mathrm{span}\{e_j : j \geq 1\}$ and for $x \in \mathrm{Ker}(\mathcal{T})$. So it suffices to show

$$\mathrm{Im}(\mathcal{T}) = \overline{\mathrm{span}}\{e_j : j \geq 1\}. \tag{2.4}$$

For any finite n and $c_1, \ldots, c_n \in \mathbb{R}$, $\sum_{j=1}^n c_j e_j = \mathcal{T}\Big(\sum_{j=1}^n \lambda_j^{-1} c_j e_j\Big) \in \mathrm{Im}(\mathcal{T})$, so $\overline{\mathrm{span}}\{e_j : j \geq 1\} \subset \overline{\mathrm{Im}(\mathcal{T})} = \mathrm{Im}(\mathcal{T})$ (the closure equality using Theorem 2.6(1) and completeness of finite-rank projections, as in Theorem 2.7). Write, by Theorem 2.3, $\mathrm{Im}(\mathcal{T}) = \overline{\mathrm{span}}\{e_j : j \geq 1\} \oplus \mathbb{N}$ for $\mathbb{N}$ the orthogonal complement of $\overline{\mathrm{span}}\{e_j\}$ within $\mathrm{Im}(\mathcal{T})$. For $x \in \mathbb{N}$, $y \in \overline{\mathrm{span}}\{e_j\}$, self-adjointness gives $\langle \mathcal{T}x, y\rangle = \langle x, \mathcal{T}y\rangle = 0$, so $\mathcal{T}$ maps $\mathbb{N}$ into $\mathbb{N}$. Let $\mathcal{T}_{\mathbb{N}}$ denote this restriction. Any nonzero eigenvalue of $\mathcal{T}_{\mathbb{N}}$ is an eigenvalue of $\mathcal{T}$; but every nonzero eigenvalue of $\mathcal{T}$ already appears among $\{\lambda_j\}$ with eigenvector in $\mathrm{span}\{e_j\}$, so $\mathcal{T}_{\mathbb{N}}$ can have no nonzero eigenvalue. Since $\mathcal{T}_{\mathbb{N}}$ is a compact self-adjoint operator on $\mathbb{N}$, if it were not the zero operator then $\pm\|\mathcal{T}_{\mathbb{N}}\|$ would be one of its eigenvalues (Theorem 2.11 below), a contradiction; hence $\mathcal{T}_{\mathbb{N}} = 0$ and $\mathbb{N} \subset \mathrm{Im}(\mathcal{T}) \cap \mathrm{Ker}(\mathcal{T}) = \{0\}$, proving (2.4). □

Every eigenvalue of a nonnegative definite operator is nonnegative, since $\langle \mathcal{T}e, e\rangle = \lambda\|e\|^2$; conversely, by Theorem 2.10, a compact self-adjoint $\mathcal{T}$ is nonnegative definite if and only if all its eigenvalues are nonnegative. This is the property that will identify covariance operators (Section 2.6) as having entirely nonnegative spectra.

Theorem 2.11. *If* $\mathcal{T}$ *is compact and nonnegative definite with eigenpairs* $\{(\lambda_j, e_j)\}$ *ordered* $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$*, then for every* $k \geq 1$*,*

$$\lambda_k = \max_{e \in \mathrm{span}\{e_1,\ldots,e_{k-1}\}^\perp} \frac{\langle \mathcal{T}e, e\rangle}{\|e\|^2}, \tag{2.5}$$

with $\mathrm{span}\{e_1, \ldots, e_{k-1}\}^\perp = \mathbb{H}$ *when* $k = 1$*; more generally, for a compact self-adjoint* $\mathcal{T}$

with eigenpairs ordered $\lambda_1^2 \geq \lambda_2^2 \geq \cdots$,

$$|\lambda_k| = \max_{e \in \operatorname{span}\{e_1,\ldots,e_{k-1}\}^\perp} \frac{\|\mathcal{T}e\|}{\|e\|}, \qquad \textit{so that} \qquad \|\mathcal{T}\| = \max_j |\lambda_j|. \tag{2.6}$$

Proof. For $e = \sum_{j\geq k} a_j e_j \in \operatorname{span}\{e_1,\ldots,e_{k-1}\}^\perp$, $\langle \mathcal{T}e, e\rangle/\|e\|^2 = \sum_{j\geq k} \lambda_j a_j^2 \big/ \sum_{j\geq k} a_j^2 \leq \lambda_k$ (as $\lambda_j \leq \lambda_k$ for $j \geq k$), with equality at $e = e_k$; this proves (2.5). For (2.6), apply (2.5) to the nonnegative definite compact operator $\mathcal{T}^2$ (whose eigenpairs are (λ_j^2, e_j), since $\mathcal{T}^2 = \sum_j \lambda_j^2 e_j \otimes e_j$ by (2.3) applied twice) to get $\lambda_k^2 = \max_{e \perp \operatorname{span}\{e_1,\ldots,e_{k-1}\}} \|\mathcal{T}e\|^2/\|e\|^2$, and take square roots. □

Theorem 2.12 (Courant–Fischer minimax principle)**.** *Let* $\mathcal{T}$ *be a nonnegative, compact operator on* $\mathbb{H}$ *with eigenvalues* $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$ *and eigenvectors* $\{e_j\}$*. Then, for every* $k \geq 1$*,*

$$\lambda_k = \max_{v_1,\ldots,v_k \in \mathbb{H}} \min_{v \in \operatorname{span}\{v_1,\ldots,v_k\}} \frac{\langle \mathcal{T}v, v\rangle}{\|v\|^2} = \min_{v_1,\ldots,v_{k-1} \in \mathbb{H}} \max_{v \in \operatorname{span}\{v_1,\ldots,v_{k-1}\}^\perp} \frac{\langle \mathcal{T}v, v\rangle}{\|v\|^2}, \tag{2.7}$$

with both extrema attained at $v_k = e_k$*.*

Proof. Let $\mathbb{M}_{k-1} = \operatorname{span}\{e_1,\ldots,e_{k-1}\}$. For the second equality in (2.7): given any $(k-1)$-dimensional $\mathbb{V}_{k-1} \subset \mathbb{H}$, the subspace $\mathbb{V}_{k-1}^\perp \cap \mathbb{M}_k$ (with $\mathbb{M}_k = \operatorname{span}\{e_1,\ldots,e_k\}$) is nontrivial; for $v = \sum_{j=1}^k a_j e_j$ in this intersection, $\langle \mathcal{T}v, v\rangle/\|v\|^2 = \sum_{j=1}^k \lambda_j a_j^2 \big/ \sum_{j=1}^k a_j^2 \geq \lambda_k$, so $\max_{v \in \mathbb{V}_{k-1}^\perp} \langle \mathcal{T}v, v\rangle/\|v\|^2 \geq \lambda_k$ for every $\mathbb{V}_{k-1}$, with equality at $\mathbb{V}_{k-1} = \mathbb{M}_{k-1}$ by Theorem 2.11.

For the first equality: given any k-dimensional $\mathbb{V}_k \subset \mathbb{H}$, the subspace $\mathbb{V}_k \cap \mathbb{M}_{k-1}^\perp$ is nontrivial; for $v = \sum_{j\geq k} a_j e_j$ in this intersection, $\langle \mathcal{T}v, v\rangle/\|v\|^2 \leq \lambda_k$ (as in the proof of Theorem 2.11), so $\min_{v \in \mathbb{V}_k} \langle \mathcal{T}v, v\rangle/\|v\|^2 \leq \min_{v \in \mathbb{V}_k \cap \mathbb{M}_{k-1}^\perp} \langle \mathcal{T}v, v\rangle/\|v\|^2 \leq \lambda_k$ for every $\mathbb{V}_k$, with equality at $\mathbb{V}_k = \mathbb{M}_k$. □

The Courant–Fischer principle yields an elementary but essential perturbation bound: the eigenvalues of a compact self-adjoint operator are a 1-Lipschitz function of the operator itself, in operator norm. This is the first (and simplest) instance of the perturbation theory that will be extended to eigenvectors in Section 2.4 and applied to the estimated covariance operator in Section 2.8.

Theorem 2.13. *Let* $\mathcal{T}, \widetilde{\mathcal{T}}$ *be nonnegative definite compact operators on* $\mathbb{H}$ *with eigenvalue sequences* $\{\lambda_j\}$*,* $\{\widetilde{\lambda}_j\}$*. Then*

$$\sup_{k\geq 0} |\lambda_{k+1} - \widetilde{\lambda}_{k+1}| \leq \|\mathcal{T} - \widetilde{\mathcal{T}}\|. \tag{2.8}$$

Proof. By Theorem 2.12, for any $v_1, \dots, v_k \in \mathbb{H}$,

$$\max_{v \in \operatorname{span}\{v_1,\dots,v_k\}^\perp} \frac{\langle \mathcal{T} v, v\rangle}{\|v\|^2} \le \max_{v \in \operatorname{span}\{v_1,\dots,v_k\}^\perp} \frac{\langle \widetilde{\mathcal{T}} v, v\rangle}{\|v\|^2} + \max_{v \in \mathbb{H}} \frac{\langle (\mathcal{T} - \widetilde{\mathcal{T}}) v, v\rangle}{\|v\|^2},$$

and the last term is bounded by $\|\mathcal{T} - \widetilde{\mathcal{T}}\|$. Taking the minimum over $v_1, \dots, v_k$ on both sides gives $\lambda_{k+1} \le \widetilde{\lambda}_{k+1} + \|\mathcal{T} - \widetilde{\mathcal{T}}\|$; exchanging the roles of $\mathcal{T}$ and $\widetilde{\mathcal{T}}$ gives the reverse inequality. □

2.3.3 The Singular Value Decomposition

For a compact operator $\mathcal{T} \in \mathcal{B}(\mathbb{H}_1, \mathbb{H}_2)$ between two (possibly distinct) Hilbert spaces, self-adjointness is not available, but a closely related expansion — the singular value decomposition (SVD) — still holds, obtained by applying the Spectral Theorem to the nonnegative definite compact operators $\mathcal{T}^*\mathcal{T}$ and $\mathcal{T}\mathcal{T}^*$.

Theorem 2.14 (Singular Value Decomposition)**.** *Let $\mathcal{T} \in \mathcal{B}(\mathbb{H}_1, \mathbb{H}_2)$ be compact. Let $\{(\lambda_j^2, f_{1j})\}$ be the eigenpairs of the compact, nonnegative definite operator $\mathcal{T}^*\mathcal{T}$ (Theorem 2.10), with $\lambda_1 \ge \lambda_2 \ge \cdots \ge 0$, and set $f_{2j} := \mathcal{T} f_{1j}/\lambda_j$ for $\lambda_j > 0$. Then $\{(\lambda_j^2, f_{2j})\}$ are eigenpairs of $\mathcal{T}\mathcal{T}^*$, $\{f_{2j}\}$ is orthonormal, $\mathcal{T}^* f_{2j} = \lambda_j f_{1j}$, and*

$$\mathcal{T} = \sum_{j \ge 1} \lambda_j \, (f_{1j} \otimes_1 f_{2j}), \qquad \textit{i.e.} \qquad \mathcal{T} x = \sum_{j \ge 1} \lambda_j \langle x, f_{1j}\rangle_1 f_{2j}, \quad x \in \mathbb{H}_1. \tag{2.9}$$

The triples $(\lambda_j, f_{1j}, f_{2j})$ are the singular system *of $\mathcal{T}$; the λ_j are its* singular values.

Proof. For $\lambda_j > 0$, $(\mathcal{T}\mathcal{T}^*)(\mathcal{T} f_{1j}) = \mathcal{T}(\mathcal{T}^*\mathcal{T} f_{1j}) = \lambda_j^2 \mathcal{T} f_{1j}$, so $f_{2j} = \mathcal{T} f_{1j}/\lambda_j$ is a unit-norm eigenvector of $\mathcal{T}\mathcal{T}^*$ for eigenvalue λ_j^2; orthonormality of $\{f_{2j}\}$ follows from Theorem 2.8(2) applied to $\mathcal{T}\mathcal{T}^*$. And $\mathcal{T}^* f_{2j} = \mathcal{T}^*\mathcal{T} f_{1j}/\lambda_j = \lambda_j f_{1j}$.

For (2.9): by Theorem **??** and Theorem 2.3, part 3 of Theorem 2.3(applied twice) gives $(\operatorname{Ker}(\mathcal{T}))^\perp = (\operatorname{Ker}(\mathcal{T}^*\mathcal{T}))^\perp = (\operatorname{Im}(\mathcal{T}^*\mathcal{T}))^{\perp\perp} = \overline{\operatorname{Im}(\mathcal{T}^*\mathcal{T})}$, which by (2.4) (applied to $\mathcal{T}^*\mathcal{T}$) equals $\overline{\operatorname{span}}\{f_{1j} : j \ge 1\}$. So it suffices to verify (2.9) for $x \in \overline{\operatorname{span}}\{f_{1j}\}$, where it holds by Theorem 2.2 and the definition of f_{2j}, and for $x \in \operatorname{Ker}(\mathcal{T})$, where both sides vanish. □

Theorem 2.15. *If $\mathcal{T} \in \mathcal{B}(\mathbb{H}_1, \mathbb{H}_2)$ is compact with singular values $\lambda_1 \ge \lambda_2 \ge \cdots \ge 0$, then $\|\mathcal{T}\| = \lambda_1$, and, more generally, for every k,*

$$\lambda_k = \max_{v_1,\dots,v_k \in \mathbb{H}_1} \; \min_{f \in \operatorname{span}\{v_1,\dots,v_k\}} \frac{\|\mathcal{T} f\|_2}{\|f\|_1} = \min_{v_1,\dots,v_{k-1} \in \mathbb{H}_1} \; \max_{f \in \operatorname{span}\{v_1,\dots,v_{k-1}\}^\perp} \frac{\|\mathcal{T} f\|_2}{\|f\|_1}. \tag{2.10}$$

Proof. Immediate from Theorems 2.11 and 2.12 applied to $\mathcal{T}^*\mathcal{T}$, since $\|\mathcal{T} f\|_2^2/\|f\|_1^2 = \langle \mathcal{T}^*\mathcal{T} f, f\rangle_1/\|f\|_1^2$ and $\mathcal{T}^*\mathcal{T}$ has eigenvalues λ_j^2. □

Theorem 2.16. *An operator $\mathcal{T} \in \mathcal{B}(\mathbb{H}_1, \mathbb{H}_2)$ is compact if and only if it admits the expansion* (2.9)*.*

Proof. Necessity is Theorem 2.14. For sufficiency, if (2.9) holds, let $\mathcal{T}_n = \sum_{j=1}^{n} \lambda_j (f_{1j} \otimes_1 f_{2j})$, which is finite-rank, hence compact. By Theorem 2.15 applied to $\mathcal{T} - \mathcal{T}_n$ (whose singular values are $\lambda_{n+1}, \lambda_{n+2}, \dots$), $\|\mathcal{T} - \mathcal{T}_n\| = \lambda_{n+1} \to 0$, so $\mathcal{T}$ is compact by Theorem 2.6(4). □

Theorem 2.17. *If $\mathcal{T} \in \mathcal{B}(\mathbb{H}_1, \mathbb{H}_2)$ is a compact operator with infinitely many nonzero singular values, then* $\mathrm{Im}(\mathcal{T})$ *is not closed.*

Proof. Let $\{(\lambda_j, f_{1j}, f_{2j})\}$ be the singular system, $\lambda_j \downarrow 0$. Choose a subsequence $\{j_k\}$ with $\lambda_{j_k} \le k^{-2}$ and set $y = \sum_{k \ge 1} \lambda_{j_k} f_{2j_k} \in \mathbb{H}_2$ (convergent since $\sum_k \lambda_{j_k}^2 < \infty$). Each partial sum lies in $\mathrm{Im}(\mathcal{T})$, since $f_{2j_k} = \mathcal{T}(f_{1j_k}/\lambda_{j_k})$, but $\sum_k \langle y, f_{2j_k} \rangle_2^2 / \lambda_{j_k}^2 = \sum_k 1 = \infty$ shows y fails the Picard condition $\sum_j \langle y, f_{2j} \rangle_2^2 / \lambda_j^2 < \infty$ that characterizes $\mathrm{Im}(\mathcal{T})$ (via $\mathcal{T}^\dagger y := \sum_j \lambda_j^{-1} \langle y, f_{2j} \rangle_2 f_{1j}$, the Moore–Penrose inverse, which converges precisely when this sum is finite). Hence $y \notin \mathrm{Im}(\mathcal{T})$, even though y is a limit of elements of $\mathrm{Im}(\mathcal{T})$. □

This last fact — that a compact operator with infinite rank never has closed range — is the root cause of the ill-posedness that pervades inverse problems in FDA (e.g. functional linear regression); it is not needed for the FPCA development that follows, but is recorded here because it is an immediate and instructive consequence of the SVD.

2.3.4 Hilbert–Schmidt and Trace-Class Operators

Within the class of compact operators, two subclasses defined by summability of the singular values are of particular importance for FDA: Hilbert–Schmidt operators, whose squared singular values are summable, and trace-class operators, whose singular values themselves are summable. Covariance operators (Section 2.6) will always be trace class, hence Hilbert–Schmidt, hence compact.

Theorem 2.18. *Let $\mathcal{T}_1, \mathcal{T}_2 \in \mathcal{B}(\mathbb{H}_1, \mathbb{H}_2)$ and suppose that for some CONSs $\{e_{1i}\}$, $\{e_{2j}\}$ of $\mathbb{H}_1, \mathbb{H}_2$, either $\sum_i (\|\mathcal{T}_1 e_{1i}\|_2^2 + \|\mathcal{T}_2 e_{1i}\|_2^2) < \infty$ or $\sum_j (\|\mathcal{T}_1^* e_{2j}\|_1^2 + \|\mathcal{T}_2^* e_{2j}\|_1^2) < \infty$. Then*

$$\sum_i \langle \mathcal{T}_1 e_{1i}, \mathcal{T}_2 e_{1i} \rangle_2 = \sum_j \langle \mathcal{T}_1^* e_{2j}, \mathcal{T}_2^* e_{2j} \rangle_1, \tag{2.11}$$

and this common value does not depend on the choice of CONSs.

Proof. First take $\mathcal{T}_1 = \mathcal{T}_2 = \mathcal{T}$. Expanding $\mathcal{T}e_{1i} = \sum_j \langle \mathcal{T}e_{1i}, e_{2j}\rangle_2 e_{2j}$ and using Parseval's relation (Theorem 2.2) twice,

$$\sum_i \|\mathcal{T}e_{1i}\|_2^2 = \sum_i \sum_j \langle \mathcal{T}e_{1i}, e_{2j}\rangle_2^2 = \sum_i \sum_j \langle e_{1i}, \mathcal{T}^* e_{2j}\rangle_1^2 = \sum_j \|\mathcal{T}^* e_{2j}\|_1^2,$$

which shows the common double-sum value does not depend on $\{e_{1i}\}$, and, by symmetry of the middle expression, not on $\{e_{2j}\}$ either. For general $\mathcal{T}_1, \mathcal{T}_2$, apply this identity to $\mathcal{T}_1 + \mathcal{T}_2$, $\mathcal{T}_1$, and $\mathcal{T}_2$ and subtract, using either summability hypothesis to justify the term-by-term subtraction. □

Definition 2.13. *$\mathcal{T} \in \mathcal{B}(\mathbb{H}_1, \mathbb{H}_2)$ is* Hilbert–Schmidt *(HS) if, for some (any) CONS $\{e_i\}$ of $\mathbb{H}_1$, $\|\mathcal{T}\|_{HS}^2 := \sum_i \|\mathcal{T}e_i\|_2^2 < \infty$. The HS inner product of $\mathcal{T}_1, \mathcal{T}_2 \in \mathcal{B}_{HS}(\mathbb{H}_1, \mathbb{H}_2)$ is $\langle \mathcal{T}_1, \mathcal{T}_2\rangle_{HS} := \sum_i \langle \mathcal{T}_1 e_i, \mathcal{T}_2 e_i\rangle_2$. By Theorem 2.18, choosing $\{e_i\}$ to be the singular vectors $\{f_{1j}\}$ of a compact $\mathcal{T}$ gives $\|\mathcal{T}\|_{HS}^2 = \sum_j \lambda_j^2$.*

Theorem 2.19. *A Hilbert–Schmidt operator is compact.*

Proof. Let $\mathcal{T}$ be HS, $\{e_{2i}\}$ a CONS for $\mathbb{H}_2$, and $\mathcal{T}_n x := \sum_{i=1}^n \langle \mathcal{T}x, e_{2i}\rangle_2 e_{2i}$, which is finite-rank. For $\|x\|_1 \le 1$, by Cauchy–Schwarz,

$$\|(\mathcal{T}_n - \mathcal{T})x\|_2^2 = \sum_{i=n+1}^{\infty} \langle \mathcal{T}x, e_{2i}\rangle_2^2 = \sum_{i=n+1}^{\infty} \langle x, \mathcal{T}^* e_{2i}\rangle_1^2 \le \sum_{i=n+1}^{\infty} \|\mathcal{T}^* e_{2i}\|_1^2 \to 0,$$

where the tail sum $\to 0$ since $\sum_i \|\mathcal{T}^* e_{2i}\|_1^2 = \|\mathcal{T}\|_{HS}^2 < \infty$ (Theorem 2.18). So $\|\mathcal{T}_n - \mathcal{T}\| \to 0$ and $\mathcal{T}$ is compact by Theorem 2.6(4). □

Theorem 2.20. *$\mathcal{B}_{HS}(\mathbb{H}_1, \mathbb{H}_2)$ is a separable Hilbert space under $\langle \cdot, \cdot \rangle_{HS}$, with CONS $\{e_{1i} \otimes_1 e_{2j}\}$ for any CONSs $\{e_{1i}\}, \{e_{2j}\}$ of $\mathbb{H}_1, \mathbb{H}_2$.*

Proof. Writing $a_{ij} = \langle \mathcal{T}e_{1i}, e_{2j}\rangle_2$, Theorem 2.18 gives $\|\mathcal{T}\|_{HS}^2 = \sum_i \sum_j a_{ij}^2$, identifying $\mathcal{B}_{HS}(\mathbb{H}_1, \mathbb{H}_2)$ isometrically with $\ell^2(\mathbb{N} \times \mathbb{N})$; completeness follows from that of ℓ^2 (Theorem 2.2/Riesz–Fischer). Orthonormality of $\{e_{1i} \otimes_1 e_{2j}\}$ is immediate. If $\langle \mathcal{T}, e_{1i} \otimes_1 e_{2j}\rangle_{HS} = \langle \mathcal{T}e_{1i}, e_{2j}\rangle_2 = 0$ for all i, j, then $\mathcal{T} = 0$ by (2.11), so $\{e_{1i} \otimes_1 e_{2j}\}$ is a CONS by Theorem 2.2. □

The next result shows that the rank-k truncation of the SVD is the best possible rank-k approximation to a Hilbert–Schmidt operator, in HS norm — the operator-theoretic form of the classical Eckart–Young theorem. This is the result that will justify truncating an estimated FPCA expansion to its leading k components as the optimal k-term summary of the estimated covariance structure (cf. the variance-explained criterion of Section 4.3).

Theorem 2.21 (Schmidt–Mirsky / Eckart–Young)**.** *Let $\mathcal{T} \in \mathcal{B}_{HS}(\mathbb{H}_1, \mathbb{H}_2)$ have singular system $\{(\lambda_j, f_{1j}, f_{2j})\}$. Then for any finite k and any $x_1, \dots, x_k \in \mathbb{H}_1$, $y_1, \dots, y_k \in \mathbb{H}_2$,*

$$\Big\| \mathcal{T} - \sum_{j=1}^{k} x_j \otimes_1 y_j \Big\|_{HS} \geq \Big\| \mathcal{T} - \sum_{j=1}^{k} \lambda_j f_{1j} \otimes_1 f_{2j} \Big\|_{HS} = \Big(\sum_{j>k} \lambda_j^2 \Big)^{1/2}. \tag{2.12}$$

Proof. It suffices to show $\|\mathcal{T} - \sum_{j=1}^{k} x_j \otimes_1 y_j\|_{HS}^2 \geq \|\mathcal{T}\|_{HS}^2 - \sum_{j=1}^{k} \lambda_j^2$. Without loss the (y_j, x_j) may be taken orthonormal (an orthonormal reparametrization of $\operatorname{span}\{x_j\}$, $\operatorname{span}\{y_j\}$ can only decrease the norm of the difference). Extending $\{y_j\}_{j=1}^{k}$ to a CONS $\{y_j\}$ of $\mathbb{H}_1$ and writing the HS norm via Definition 2.13,

$$\Big\| \mathcal{T} - \sum_{i=1}^{k} x_i \otimes_1 y_i \Big\|_{HS}^2 = \sum_{j} \Big\| \Big(\mathcal{T} - \sum_{i=1}^{k} x_i \otimes_1 y_i \Big) y_j \Big\|_2^2 = \sum_{j=1}^{k} \|\mathcal{T} y_j - x_j\|_2^2 + \sum_{j>k} \|\mathcal{T} y_j\|_2^2.$$

Since $\sum_{j>k} \|\mathcal{T} y_j\|_2^2 = \|\mathcal{T}\|_{HS}^2 - \sum_{j=1}^{k} \|\mathcal{T} y_j\|_2^2$, it remains to show $\sum_{j=1}^{k} \|\mathcal{T} y_j\|_2^2 \leq \sum_{j=1}^{k} \lambda_j^2$. Expanding $\mathcal{T} y_j = \sum_l \lambda_l \langle f_{1l}, y_j \rangle_1 f_{2l}$ gives $\|\mathcal{T} y_j\|_2^2 = \sum_l \lambda_l^2 \langle f_{1l}, y_j \rangle_1^2$, and, using that λ_l^2 is nonincreasing in l together with Parseval's relation applied to the orthonormal $\{y_j\}_{j=1}^{k}$ (so that $\sum_{j=1}^{k} \langle f_{1l}, y_j \rangle_1^2 \leq 1$ for every l, and $\sum_l \sum_{j=1}^{k} \langle f_{1l}, y_j \rangle_1^2 = k$),

$$\sum_{j=1}^{k} \|\mathcal{T} y_j\|_2^2 = \sum_{l} \lambda_l^2 \sum_{j=1}^{k} \langle f_{1l}, y_j \rangle_1^2 \leq \sum_{l=1}^{k} \lambda_l^2 + \lambda_k^2 \Big(k - \sum_{l=1}^{k} \sum_{j=1}^{k} \langle f_{1l}, y_j \rangle_1^2 \Big) \leq \sum_{l=1}^{k} \lambda_l^2,$$

where the first inequality bounds $\lambda_l^2 \leq \lambda_k^2$ for $l > k$ and adds and subtracts $\lambda_k^2 \sum_{l=1}^{k} (\cdot)$, and the second uses $\sum_{l=1}^{k} \sum_{j=1}^{k} \langle f_{1l}, y_j \rangle_1^2 \leq k$. Equality throughout at $x_j = \lambda_j f_{2j}$, $y_j = f_{1j}$ gives $\|\mathcal{T} - \sum_{j=1}^{k} \lambda_j f_{1j} \otimes_1 f_{2j}\|_{HS}^2 = \sum_{j>k} \lambda_j^2$. □

Definition 2.14. *$\mathcal{T} \in \mathcal{B}(\mathbb{H}_1, \mathbb{H}_2)$ is* trace class *if $\|\mathcal{T}\|_{TR} := \sum_j \langle (\mathcal{T}^*\mathcal{T})^{1/2} e_j, e_j \rangle_1 < \infty$ for some (any) CONS $\{e_j\}$ of $\mathbb{H}_1$, where $(\mathcal{T}^*\mathcal{T})^{1/2}$ is the nonnegative self-adjoint square root furnished by Theorem 2.10 applied with fractional powers of the eigenvalues. Equivalently, $\|\mathcal{T}\|_{TR} = \sum_j \lambda_j$, the sum of the singular values of $\mathcal{T}$.*

Theorem 2.22. *A trace-class operator is Hilbert–Schmidt (hence compact).*

Proof. If $\mathcal{T}$ is trace class then $(\mathcal{T}^*\mathcal{T})^{1/4}$ is HS, since $\|\mathcal{T}\|_{TR} = \|(\mathcal{T}^*\mathcal{T})^{1/4}\|_{HS}^2$ (both sides equal $\sum_j \lambda_j$, comparing Definitions 2.13 and 2.14 via the eigenvalues $\lambda_j^{1/2}$ of $(\mathcal{T}^*\mathcal{T})^{1/4}$), hence compact (Theorem 2.19), so $\mathcal{T}^*\mathcal{T} = ((\mathcal{T}^*\mathcal{T})^{1/4})^4$ is compact (Theorem 2.6(3)) with the same singular values λ_j as $\mathcal{T}$ (Theorem 2.14). Then $\|\mathcal{T}\|_{HS}^2 = \sum_j \lambda_j^2 \leq \lambda_1 \sum_j \lambda_j = \lambda_1 \|\mathcal{T}\|_{TR} < \infty$. □

For a self-adjoint $\mathcal{T}$ with eigenvalues $\{\lambda_j\}$, $\|\mathcal{T}\|_{TR} = \sum_j |\lambda_j|$, and if $\mathcal{T}$ is trace class we call $\operatorname{trace}(\mathcal{T}) := \sum_j \lambda_j = \sum_j \langle \mathcal{T} e_j, e_j \rangle$ (for any CONS $\{e_j\}$) its *trace*; when $\mathcal{T}$ is also

nonnegative definite, $\operatorname{trace}(\mathcal{T}) = \|\mathcal{T}\|_{TR}$. This is the quantity that Section 4.3 computes, for the empirical covariance operator, as total variance; the percentage of variance explained by the leading eigenvalues is precisely $\sum_{j\le k}\lambda_j/\operatorname{trace}(\mathcal{T})$.

The following bound — a trace-norm analog of Theorem 2.13 for the whole singular value sequence at once — will be the key technical tool for transferring HS-norm consistency of an estimated covariance operator into ℓ^2 consistency of its estimated eigenvalues in Section 2.8.

Theorem 2.23. *Let $\mathcal{T}, \widetilde{\mathcal{T}} \in \mathcal{B}_{HS}(\mathbb{H}_1, \mathbb{H}_2)$ with nonascending singular values $\{\lambda_j\}, \{\widetilde{\lambda}_j\}$. Then*

$$\sum_{j\ge 1}(\lambda_j - \widetilde{\lambda}_j)^2 \le \operatorname{trace}\Big\{(\mathcal{T}-\widetilde{\mathcal{T}})^*(\mathcal{T}-\widetilde{\mathcal{T}})\Big\} = \|\mathcal{T}-\widetilde{\mathcal{T}}\|_{HS}^2. \tag{2.13}$$

Proof. Let $\{(\lambda_j, f_{1j}, f_{2j})\}$, $\{(\widetilde{\lambda}_j, \widetilde{f}_{1j}, \widetilde{f}_{2j})\}$ be the singular systems of $\mathcal{T}, \widetilde{\mathcal{T}}$. Expanding $\widetilde{\mathcal{T}} f_{1j} = \sum_k \widetilde{\lambda}_k \langle f_{1j}, \widetilde{f}_{1k}\rangle_1 \widetilde{f}_{2k}$ and using Definition 2.13,

$$\|\mathcal{T}-\widetilde{\mathcal{T}}\|_{HS}^2 = \sum_j \lambda_j^2 + \sum_j \widetilde{\lambda}_j^2 - 2\sum_{j,k}\lambda_j\widetilde{\lambda}_k\langle f_{1j}, \widetilde{f}_{1k}\rangle_1\langle f_{2j}, \widetilde{f}_{2k}\rangle_2.$$

By Cauchy–Schwarz and Parseval's relation, for every fixed j, $\left|\sum_k \langle f_{1j}, \widetilde{f}_{1k}\rangle_1\langle f_{2j}, \widetilde{f}_{2k}\rangle_2\right| \le 1$, and likewise for every fixed k summing over j; so, writing $\mathcal{A} = \{\langle f_{1j}, \widetilde{f}_{1k}\rangle_1\langle f_{2j}, \widetilde{f}_{2k}\rangle_2\}_{j,k}$, the row and column sums of $\mathcal{A}$ are bounded by 1 in absolute value. In the finite-rank case (rank $\le n$ for both operators; the general case follows by truncating and taking limits), writing $\lambda = (\lambda_1, \dots, \lambda_n)^T$, $\widetilde{\lambda} = (\widetilde{\lambda}_1, \dots, \widetilde{\lambda}_n)^T$, (2.13) is equivalent to $\lambda^T \mathcal{A} \widetilde{\lambda} \le \lambda^T \widetilde{\lambda}$. Since $\lambda, \widetilde{\lambda}$ have nonincreasing nonnegative entries, each can be written as a nonnegative combination of the vectors $\{e_i\}_{i=1}^n$ (e_i having its first i entries equal to 1, the rest 0), so it suffices to verify $e_k^T \mathcal{A} e_l \le e_k^T e_l$ for all k, l, which holds because $|e_k^T \mathcal{A} e_l| = \left|\sum_{j\le k, i\le l}\mathcal{A}_{ji}\right| \le \min(k,l) = e_k^T e_l$, using the row/column sum bound. □

2.3.5 Integral Operators and Mercer's Theorem

We specialize now to the setting most relevant for FDA: $\mathbb{H} = \mathbb{L}^2(E, \mathcal{B}, \mu)$ for a finite measure μ on a compact metric space E (with $\mathcal{B} = \mathcal{B}(E)$ the Borel σ-field and, without loss, $\operatorname{supp}(\mu) = E$), and operators that arise as integral operators against a kernel. This is the setting in which the covariance operator of a stochastic process will live (Section 2.6), and Mercer's Theorem is what makes the covariance kernel and covariance operator two faces of the same object.

Definition 2.15. *For a measurable κ on $E \times E$ with $\iint_{E\times E} \kappa^2(s,t)\, d\mu(s) d\mu(t) < \infty$, the associated* integral operator $\mathcal{K} \in \mathcal{B}(\mathbb{L}^2(E, \mathcal{B}, \mu))$ *is*

$$(\mathcal{K}f)(\cdot) := \int_E \kappa(s, \cdot) f(s)\, d\mu(s), \qquad f \in \mathbb{L}^2(E, \mathcal{B}, \mu), \tag{2.14}$$

with κ its kernel*; by Cauchy–Schwarz, $\|\mathcal{K}\| \le \left(\iint_{E\times E} \kappa^2\, d\mu\, d\mu \right)^{1/2}$. We assume throughout that κ is continuous on $E \times E$ (hence uniformly continuous, by compactness of E).*

Lemma 2.1. *For every $f \in \mathbb{L}^2(E, \mathcal{B}, \mu)$, $(\mathcal{K}f)(\cdot)$ is uniformly continuous.*

Proof. Given $\epsilon > 0$, uniform continuity of κ gives $\delta > 0$ with $|\kappa(s, t_2) - \kappa(s, t_1)| < \epsilon$ whenever $|t_2 - t_1| < \delta$, for all s. Then $|(\mathcal{K}f)(t_2) - (\mathcal{K}f)(t_1)| \le \int_E |\kappa(s, t_2) - \kappa(s, t_1)||f(s)|\, d\mu(s) \le \epsilon \|f\| \mu(E)^{1/2}$ by Cauchy–Schwarz. □

Theorem 2.24. *The integral operator $\mathcal{K}$ of Definition 2.15 is compact.*

Proof. By the Stone–Weierstrass Theorem, for any $\epsilon > 0$ there are continuous $g_1, \dots, g_n, h_1, \dots, h_n$ with $\sup_{s,t} |\kappa(s,t) - \kappa_n(s,t)| < \epsilon$ for $\kappa_n(s,t) = \sum_{i=1}^n g_i(s) h_i(t)$. The integral operator $\mathcal{K}_n$ with kernel κ_n has range in $\operatorname{span}\{h_1, \dots, h_n\}$, hence is finite-rank (compact), and by Cauchy–Schwarz, $\|(\mathcal{K}_n - \mathcal{K})f\|^2 = \int_E \left(\int_E [\kappa_n(s,t) - \kappa(s,t)] f(s)\, d\mu(s) \right)^2 d\mu(t) \le \epsilon^2 \|f\|^2 \mu(E)^2$, so $\|\mathcal{K}_n - \mathcal{K}\| \to 0$ and $\mathcal{K}$ is compact by Theorem 2.6(4). □

If κ is symmetric, $\mathcal{K}$ is self-adjoint (Example 2.1 below), and Theorem 2.10 gives $\mathcal{K} = \sum_{j\ge1} \lambda_j e_j \otimes e_j$; by Lemma 2.1, the version of e_j given by $e_j(t) = \lambda_j^{-1} \int_E \kappa(s,t) e_j(s)\, d\mu(s)$ (for $\lambda_j \neq 0$) is continuous, and we always adopt this continuous version.

Example 2.1. *For $\mathcal{K}$ as in (2.14), $\langle \mathcal{K}f, g \rangle = \iint \kappa(t,s) f(s) g(t)\, ds\, dt = \langle f, \int_E \kappa(t, \cdot) g(t)\, dt \rangle$, so $\mathcal{K}^*$ has kernel $\kappa(t,s)$ (arguments reversed); $\mathcal{K}$ is self-adjoint exactly when κ is symmetric.*

Theorem 2.25. *The integral operator $\mathcal{K}$ is nonnegative definite if and only if its kernel κ is a nonnegative definite function, i.e. $\sum_{i,j} a_i a_j \kappa(t_i, t_j) \ge 0$ for all finite $\{a_i\} \subset \mathbb{R}$, $\{t_i\} \subset E$.*

Proof. By compactness of E and uniform continuity of κ, for any n there is a finite partition $\{E_{ni}\}$ of E of mesh $< \delta_n$ (with δ_n chosen so that κ varies by less than n^{-1} within a set of that diameter, via the Heine–Borel Theorem) and points $v_i \in E_{ni}$ such that the step kernel $\kappa_n(s,t) := \kappa(v_i, v_j)$ for $(s,t) \in E_{ni} \times E_{nj}$ satisfies $\sup_{s,t} |\kappa(s,t) - \kappa_n(s,t)| < n^{-1}$. Then $|\langle \mathcal{K}f, f \rangle - \langle \mathcal{K}_n f, f \rangle| \le n^{-1} \|f\|^2 \to 0$, and $\langle \mathcal{K}_n f, f \rangle = \sum_{i,j} \kappa(v_i, v_j) \int_{E_{ni}} f\, d\mu \int_{E_{nj}} f\, d\mu$, which is ≥ 0 for every f whenever κ is nonnegative definite; letting $n \to \infty$ gives $\langle \mathcal{K}f, f \rangle \ge 0$.

Conversely, suppose $\sum_{i,j=1}^m a_i a_j \kappa(v_i, v_j) < 0$ for some $\{a_i\}, \{v_i\}$. By continuity there are disjoint $E_1, \dots, E_m$ with $\mu(E_i) > 0$, $v_i \in E_i$, and $\sum_{i,j} a_i a_j \kappa(u_i, w_j) < 0$ uniformly for $u_i, w_i \in E_i$. Averaging over $E_i \times E_j$ (mean value theorem) gives $\sum_{i,j} a_i a_j \mu(E_i)^{-1} \mu(E_j)^{-1} \int_{E_i} \int_{E_j} \kappa\, d\mu\, d\mu$ 0, which is $\langle \mathcal{K}f, f \rangle$ for $f = \sum_i a_i \mu(E_i)^{-1} I_{E_i}$; so $\mathcal{K}$ is not nonnegative definite. □

We now come to Mercer's Theorem, which identifies the eigen-decomposition of a symmetric, nonnegative-definite integral operator with a convergent series expansion of its kernel — exactly the representation that, in Section 2.6, connects a covariance kernel $\gamma(s,t)$ to the eigenfunctions that FPCA estimates.

Lemma 2.2. *Let κ be continuous, symmetric, and nonnegative definite, with $\mathcal{K}$ its integral operator and eigenpairs $\{(\lambda_j, e_j)\}$. Then: (1) $\sum_j \lambda_j e_j^2(t) \le \kappa(t,t)$ for all t; (2) $\sum_j |\lambda_j e_j(s) e_j(t)| \le \{\kappa(t,t)\kappa(s,s)\}^{1/2}$ for all s,t; (3) $\lim_{n\to\infty} \sup_{s,t} \sum_{j>n} |\lambda_j e_j(s) e_j(t)| = 0$; (4) $\sum_j \lambda_j e_j(s) e_j(t)$ is well defined, and converges absolutely and uniformly to a (uniformly) continuous function of (s,t).*

Proof. (1) Let $\kappa_n(s,t) = \kappa(s,t) - \sum_{j=1}^n \lambda_j e_j(s) e_j(t)$, with integral operator $\mathcal{K}_n$. For any f, $\langle \mathcal{K}_n f, f\rangle = \langle \mathcal{K} f, f\rangle - \sum_{j=1}^n \lambda_j \langle f, e_j\rangle^2 = \sum_{j>n} \lambda_j \langle f, e_j\rangle^2 \ge 0$, so $\mathcal{K}_n$ is nonnegative definite, hence so is κ_n by Theorem 2.25; in particular $\kappa_n(t,t) \ge 0$, which is the claim.

(2) By Cauchy–Schwarz applied to part (1), for any finite index set J, $\sum_{j\in J} |\lambda_j e_j(s) e_j(t)| \le \Big(\sum_{j\in J} \lambda_j e_j^2(s)\Big)^{1/2} \Big(\sum_{j\in J} \lambda_j e_j^2(t)\Big)^{1/2} \le \{\kappa(s,s)\kappa(t,t)\}^{1/2}$.

(3) By the same Cauchy–Schwarz bound with $J = \{n{+}1, n{+}2, \dots\}$, $\sum_{j>n} |\lambda_j e_j(s) e_j(t)| \le \Big(\sum_{j>n} \lambda_j e_j^2(s)\Big)^{1/2} \Big(\sum_{j>n} \lambda_j e_j^2(t)\Big)^{1/2}$; by part (1) and dominated convergence, $\sum_{j>n} \lambda_j e_j^2(t) \downarrow 0$ for every t as $n \to \infty$, and by Dini's Theorem (monotone convergence of continuous functions to a continuous limit, on a compact space) this convergence is uniform in t; the claim follows.

(4) Given $\epsilon > 0$, part (3) gives n_ϵ with $\sup_{s,t} \sum_{j>n_\epsilon} |\lambda_j e_j(s) e_j(t)| < \epsilon$, and uniform continuity of $e_1, \dots, e_{n_\epsilon}$ gives $\delta > 0$ such that $\Big|\sum_{j\le n_\epsilon} \lambda_j e_j(s) e_j(t) - \sum_{j\le n_\epsilon} \lambda_j e_j(s') e_j(t')\Big| < \epsilon$ whenever $d((s,t),(s',t')) < \delta$; combining these two facts gives uniform continuity of the (absolutely, uniformly convergent, by (3)) limit. □

Theorem 2.26 (Mercer's Theorem)**.** *Let κ be continuous, symmetric, and nonnegative definite on $E \times E$, with integral operator $\mathcal{K}$ and eigenpairs $\{(\lambda_j, e_j)\}$. Then*

$$\kappa(s,t) = \sum_{j\ge 1} \lambda_j e_j(s) e_j(t), \qquad s,t \in E, \tag{2.15}$$

with the series converging absolutely and uniformly.

Proof. By Lemma 2.2(4), $\kappa'(s,t) := \sum_j \lambda_j e_j(s) e_j(t)$ is a well-defined continuous kernel, and the integral operator $\mathcal{K}'$ with kernel κ' satisfies $\mathcal{K}' e_j = \lambda_j e_j$ for every j and $\mathcal{K}' e = 0$ for $e \perp \overline{\text{span}}\{e_j\}$ (termwise integration, justified by uniform convergence), i.e. $\mathcal{K}'$ has the same eigen-decomposition as $\mathcal{K}$, so $\mathcal{K}' = \mathcal{K}$ by (2.3). Since distinct continuous kernels define distinct integral operators (if $\kappa_1 \ne \kappa_2$ on a set of positive $\mu\times\mu$-measure, some f separates $\int \kappa_1(s,\cdot) f(s) d\mu(s)$ from $\int \kappa_2(s,\cdot) f(s) d\mu(s)$), $\kappa = \kappa'$. □

Theorem 2.27. *Under the conditions of Theorem 2.26, $\mathcal{K}$ is trace class with*

$$\operatorname{trace}(\mathcal{K}) = \int_E \kappa(s,s)\, d\mu(s), \qquad \|\mathcal{K}\|_{HS}^2 = \iint_{E\times E} \kappa^2(s,t)\, d\mu(s) d\mu(t). \tag{2.16}$$

Proof. By (2.15) and Fubini's Theorem (justified by uniform convergence), $\operatorname{trace}(\mathcal{K}) = \sum_j \lambda_j = \sum_j \lambda_j \int_E e_j^2(t)\, d\mu(t) = \int_E \sum_j \lambda_j e_j^2(t)\, d\mu(t) = \int_E \kappa(t,t)\, d\mu(t)$, finite by Lemma 2.2(1); so $\mathcal{K}$ is trace class. For the HS norm, let $\kappa_n(s,t) = \sum_{j=1}^n \lambda_j e_j(s) e_j(t)$ with integral operator $\mathcal{K}_n$. Then $\operatorname{trace}(\mathcal{K}_n^2) = \sum_{j=1}^n \lambda_j^2 \uparrow \sum_j \lambda_j^2 = \|\mathcal{K}\|_{HS}^2$ (by (2.16)'s first identity applied to $\mathcal{K}_n^2$, which has kernel $\int_E \kappa_n(s,u)\kappa_n(u,t)\, d\mu(u)$, and equals $\iint \kappa_n^2\, d\mu\, d\mu$ by orthonormality of $\{e_j\}$), and $\iint \kappa_n^2\, d\mu\, d\mu \to \iint \kappa^2\, d\mu\, d\mu$ by Theorem 2.26 and dominated convergence. □

Combining Mercer's Theorem with the Eckart–Young Theorem (2.21) yields the fundamental optimality property of the truncated eigen-expansion of a covariance kernel that underlies FPCA dimension reduction: no rank-r kernel approximates κ better, in mean-square, than the truncated Mercer expansion.

Theorem 2.28. *Under the conditions of Theorem 2.26, for any r with $\lambda_r > 0$,*

$$\min_{\operatorname{rank}(W)=r} \iint_{E\times E} \{\kappa(s,t) - W(s,t)\}^2\, d\mu(s) d\mu(t) = \sum_{j>r} \lambda_j^2, \tag{2.17}$$

where the minimum, over kernels W of rank r (i.e. whose integral operator has rank r), is attained at $W(s,t) = \sum_{j=1}^r \lambda_j e_j(s) e_j(t)$.

Proof. By Theorem 2.21, the operator $\mathcal{W} = \sum_{j=1}^r \lambda_j e_j \otimes e_j$ minimizes $\|\mathcal{K} - \mathcal{W}\|_{HS}^2$ over rank-r operators $\mathcal{W}$, with minimal value $\sum_{j>r} \lambda_j^2$. Since $\mathcal{W}$ is an integral operator with (rank-r) kernel $W(s,t) = \sum_{j=1}^r \lambda_j e_j(s) e_j(t)$, and by (2.16) applied to $\mathcal{K} - \mathcal{W}$ (whose kernel is $\kappa - W$), $\|\mathcal{K} - \mathcal{W}\|_{HS}^2 = \iint \{\kappa - W\}^2\, d\mu\, d\mu$, the result follows. □

In Section 4.3, this theorem is the exact theoretical counterpart of the empirical observation that FPCA truncated to n_{80} components is the best rank-n_{80} mean-square summary of the estimated cross-country covariance structure available from any basis; it is what makes "variance explained" a meaningful optimality criterion rather than a merely descriptive one.

2.4 Perturbation Theory

Theorem 2.13 already showed that eigenvalues of a compact self-adjoint operator vary in a 1-Lipschitz way under perturbation of the operator, in operator norm. This section

develops the finer perturbation theory needed for the FPCA asymptotics of Section 2.8: expansions, to first order, of the eigenprojections, eigenvalues, and eigenvectors of a compact self-adjoint operator $\mathcal{T}$ under a perturbation $\Delta = \widetilde{\mathcal{T}} - \mathcal{T}$, together with explicit remainder bounds. Throughout, $\mathcal{T}$ plays the role of a population operator (e.g. a true covariance operator) and $\widetilde{\mathcal{T}}$ the role of an estimator; the results below are what let us convert operator-norm consistency of $\widetilde{\mathcal{T}}$ for $\mathcal{T}$ (Section 2.7) into consistency, and asymptotic normality, of the corresponding estimated eigenvalues and eigenfunctions. We follow [4], Section 5.1, restricting throughout to the self-adjoint case (the non-self-adjoint singular-value perturbation theory of that source's Section 5.2 is developed there for canonical correlation analysis and is not needed for FPCA).

2.4.1 The Resolvent and Contour Integration

The technical device underlying this section is contour integration of an operator-valued function, which requires the scalar field to be extended from $\mathbb{R}$ to $\mathbb{C}$; this complexification is a purely technical device (Definition 2.16 below is the only place it is used), and all operators, eigenvalues, and eigenvectors of interest remain real.

Definition 2.16. *A* complex *Hilbert space is defined as in Definition 2.4 (Hilbert space), with the scalar field $\mathbb{R}$ replaced by $\mathbb{C}$ and condition (iv) of the inner product (symmetry) replaced by conjugate symmetry, $\langle x, y\rangle = \overline{\langle y, x\rangle}$. Every real Hilbert space $\mathbb{H}$ embeds in its complexification in the natural way, and a (real) self-adjoint $\mathcal{T} \in \mathcal{B}(\mathbb{H})$ extends to a self-adjoint operator on the complexification.*

Definition 2.17. *For $\mathcal{T} \in \mathcal{B}(\mathbb{H})$ (on the complexification), the* spectrum $\sigma(\mathcal{T})$ *is the set of $z \in \mathbb{C}$ for which $\mathcal{T} - zI$ is not invertible in $\mathcal{B}(\mathbb{H})$; its complement $\rho(\mathcal{T})$ is the* resolvent set*. For a compact operator on an infinite-dimensional $\mathbb{H}$, $\sigma(\mathcal{T})$ consists exactly of the eigenvalues of $\mathcal{T}$ together with 0 (whether or not 0 is itself an eigenvalue). For $z \in \rho(\mathcal{T})$, using the spectral decomposition $\mathcal{T} = \sum_j \lambda_j \mathcal{P}_j$ (with $\mathcal{P}_j$ the projection onto the eigenspace of the nonrepeated eigenvalue λ_j, so $\mathcal{P}_j = \sum_{k=1}^{n_j} e_{jk} \otimes e_{jk}$ for an orthonormal basis $e_{j1}, \ldots, e_{jn_j}$ of that eigenspace of dimension n_j), the* resolvent *of $\mathcal{T}$ is*

$$\mathcal{R}(z) := (\mathcal{T} - zI)^{-1} = \sum_{j \geq 1} \frac{1}{\lambda_j - z} \mathcal{P}_j, \tag{2.18}$$

a bounded operator satisfying $\|\mathcal{R}(z)\| = 1/\min_j |z - \lambda_j|$.

The bound on $\|\mathcal{R}(z)\|$ follows since, for $f = \sum_j \mathcal{P}_j f$, $\|\mathcal{R}(z)f\|^2 = \sum_j |z-\lambda_j|^{-2}\|\mathcal{P}_j f\|^2 \le \max_j |z-\lambda_j|^{-2}\|f\|^2$, with equality approached by concentrating f near the eigenspace closest to z.

Definition 2.18. *Let Ω be a Jordan curve (simple closed curve) in $\mathbb{C}$ of length l_Ω, traversed counterclockwise, with $d_\Omega(z,z')$ the arc length from z to z' along Ω. For an indexed family $\{\mathcal{T}(z) : z \in \Omega\} \subset \mathcal{B}(\mathbb{H})$, and ordered points $z_0, \dots, z_n = z_0$ on Ω with ξ_j on the arc from z_j to z_{j+1}, define the Riemann-type sum $\mathcal{I}(\{z_j,\xi_j\}) := \sum_{j=0}^{n-1} \mathcal{T}(\xi_j)(z_{j+1}-z_j)$. If $\{z_{jn},\xi_{jn}\}$ is a sequence of such partitions with mesh $\to 0$ and $\sup_{d_\Omega(z,z')\le\delta} \|\mathcal{T}(z)-\mathcal{T}(z')\| \to 0$ as $\delta \downarrow 0$, the* contour integral *is*

$$\oint_\Omega \mathcal{T}(z)\,dz := \lim_{n\to\infty} \mathcal{I}(\{z_{jn},\xi_{jn}\}). \tag{2.19}$$

Lemma 2.3. *Under the modulus-of-continuity condition of Definition 2.18, the limit* (2.19) *exists and does not depend on the approximating sequence of partitions.*

Proof. For two partitions $\{z_j,\xi_j\}$, $\{\widetilde{z}_j,\widetilde{\xi}_j\}$ with mesh $\le \delta$, merge them into a single ordered partition $\{\check{z}_j,\check{\xi}_j\}$ refining both; writing the two Riemann sums as sums over the refined partition and subtracting term by term (each original arc is split into subarcs on which the corresponding ξ_j or $\widetilde{\xi}_j$ still lies), $\|\mathcal{I}(\{z_j,\xi_j\})-\mathcal{I}(\{\widetilde{z}_j,\widetilde{\xi}_j\})\| \le \sum_j \|\mathcal{T}(\check{\xi}_j)-\mathcal{T}(\check{\xi}'_j)\|\, d_\Omega(\check{z}_{j-1},\check{z}_j) \le \sup_{d_\Omega(z,z')\le\delta}\|\mathcal{T}(z)-\mathcal{T}(z')\|\, l_\Omega$ by the triangle inequality. Applying this to a sequence of partitions with mesh $\to 0$ shows $\{\mathcal{I}(\{z_{jn},\xi_{jn}\})\}$ is Cauchy in the Banach space $\mathcal{B}(\mathbb{H})$, hence convergent; applying it to two such sequences merged together shows the limit is independent of the sequence. □

Contour integration of an operator-valued function inherits linearity from (2.19) directly. Applying it to the resolvent yields a contour-integral representation of an eigenprojection, via the residue calculus fact $\frac{1}{2\pi i}\oint_\Omega (\lambda_k - z)^{-1}dz = -I(\lambda_k \text{ inside } \Omega)$ (Cauchy's integral formula, taking the contour to enclose no other singularity).

Theorem 2.29. *Let $\mathcal{T}$ be compact self-adjoint and let Ω be the boundary of a disk containing exactly the eigenvalues $\{\lambda_k : p \le k \le q\}$ (repeated eigenvalues, i.e. counted with multiplicity, all strictly interior to Ω) and no others. Then the projection $\mathcal{P}$ onto the union of the corresponding eigenspaces is*

$$\mathcal{P} = -\frac{1}{2\pi i}\oint_\Omega \mathcal{R}(z)\,dz. \tag{2.20}$$

Proof. Since $\|\mathcal{R}(z)-\mathcal{R}(z')\| = \max_j \left|\frac{1}{z-\lambda_j} - \frac{1}{z'-\lambda_j}\right| = \max_j \frac{|z-z'|}{|z-\lambda_j||z'-\lambda_j|} \le |z-z'|/\epsilon^2$ for $\epsilon := \inf_{z\in\Omega,j}|z-\lambda_j| > 0$, the modulus-of-continuity condition of Definition 2.18 holds and

(2.20) is well defined. By (2.18) and linearity of the contour integral, $-\frac{1}{2\pi i}\oint_\Omega \mathcal{R}(z)\,dz = -\frac{1}{2\pi i}\sum_k \mathcal{P}_k \oint_\Omega (\lambda_k - z)^{-1}\,dz = \sum_{p\le k\le q}\mathcal{P}_k = \mathcal{P}$, applying Cauchy's integral formula termwise (justified by the uniform bound $\|\mathcal{R}(z)\| \le 1/\epsilon$ on Ω, which makes the interchange of the (uniformly convergent) sum and the contour integral valid). □

2.4.2 Perturbation of Eigenprojections

We now compare the eigenprojections of a compact self-adjoint $\mathcal{T}$, with nonrepeated eigenvalues $\lambda_1 > \lambda_2 > \cdots$ and multiplicities n_j, to those of a perturbation $\widetilde{\mathcal{T}} = \mathcal{T} + \Delta$. When $\mathcal{T}$ has repeated eigenvalues, the eigenspaces of $\widetilde{\mathcal{T}}$ must be matched to those of $\mathcal{T}$ by dimension: writing $\lambda_1 \ge \lambda_2 \ge \cdots$ for the *repeated* eigenvalues of $\widetilde{\mathcal{T}}$ (i.e. listed with multiplicity) with eigenvectors $\{\widetilde{e}_k\}$ and $N_j := \sum_{k=1}^{j} n_k$, define

$$\widetilde{\mathcal{P}}_j := \sum_{k=N_{j-1}+1}^{N_j} \widetilde{e}_k \otimes \widetilde{e}_k, \tag{2.21}$$

i.e. the projection onto the span of exactly n_j eigenvectors of $\widetilde{\mathcal{T}}$, matching the dimension of $\mathcal{P}_j$; this is the pairing used throughout.

Theorem 2.30. *Let $\mathcal{T}$ be compact self-adjoint with nonrepeated eigenvalue λ_j, eigenprojection $\mathcal{P}_j$, and $\eta_j := \frac{1}{2}\min_{k\ne j}|\lambda_k - \lambda_j|$. Let $\widetilde{\mathcal{T}} = \mathcal{T} + \Delta$ be compact self-adjoint with $\|\Delta\| < \eta_j$, and let $\widetilde{\mathcal{P}}_j$ be as in* (2.21)*. Then*

$$\widetilde{\mathcal{P}}_j - \mathcal{P}_j = \mathcal{S}_j\Delta\mathcal{P}_j + \mathcal{P}_j\Delta\mathcal{S}_j + \frac{1}{2\pi i}\oint_{\Omega_j}\mathcal{M}(z)\,dz, \tag{2.22}$$

where Ω_j is the circle centered at λ_j of radius η_j, $\mathcal{S}_j := \sum_{k\ne j}(\lambda_k - \lambda_j)^{-1}\mathcal{P}_k$, and $\mathcal{M}(z) := \mathcal{R}(z)\sum_{k\ge 2}\{-\Delta\mathcal{R}(z)\}^k$. Consequently,

$$\left\|\widetilde{\mathcal{P}}_j - \mathcal{P}_j - (\mathcal{S}_j\Delta\mathcal{P}_j + \mathcal{P}_j\Delta\mathcal{S}_j)\right\| \le \frac{\delta_j^2}{1-\delta_j}, \qquad \delta_j := \|\Delta\|/\eta_j, \tag{2.23}$$

and, since $\|\mathcal{S}_j\| \le (2\eta_j)^{-1}$,

$$\|\mathcal{S}_j\Delta\mathcal{P}_j + \mathcal{P}_j\Delta\mathcal{S}_j\| \le \delta_j/\sqrt{2}, \qquad \textit{so that} \qquad \|\widetilde{\mathcal{P}}_j - \mathcal{P}_j\| \le \delta_j/\sqrt{2} + \delta_j^2/(1-\delta_j) \le \delta_j/(1-\delta_j). \tag{2.24}$$

Proof. Let $\widetilde{\mathcal{R}}$ denote the resolvent of $\widetilde{\mathcal{T}}$. By Theorem 2.13, $|\widetilde{\lambda} - \lambda_j| \le \|\Delta\| < \eta_j$ for every eigenvalue $\widetilde{\lambda}$ of $\widetilde{\mathcal{T}}$ matched to λ_j, so Ω_j encircles exactly the eigenvalues of $\widetilde{\mathcal{T}}$ corresponding

to λ_j and no others; Theorem 2.29 applies to both $\mathcal{T}$ and $\widetilde{\mathcal{T}}$ with this contour, giving

$$\widetilde{\mathcal{P}}_j - \mathcal{P}_j = -\frac{1}{2\pi i}\oint_{\Omega_j}\left(\widetilde{\mathcal{R}}(z) - \mathcal{R}(z)\right)dz. \tag{2.25}$$

Since $\widetilde{\mathcal{R}}(z) = (\Delta+\mathcal{T}-zI)^{-1} = \mathcal{R}(z)(\Delta\mathcal{R}(z)+I)^{-1}$ and, on Ω_j, $\|\Delta\mathcal{R}(z)\| \le \|\Delta\|/\eta_j = \delta_j < 1$ (using $\|\mathcal{R}(z)\| = 1/\eta_j$ on Ω_j by Definition 2.17), the Neumann series gives $\widetilde{\mathcal{R}}(z)-\mathcal{R}(z) = \mathcal{R}(z)\sum_{k\ge1}\{-\Delta\mathcal{R}(z)\}^k = -\mathcal{R}(z)\Delta\mathcal{R}(z) + \mathcal{M}(z)$. Substituting into (2.25),

$$\widetilde{\mathcal{P}}_j - \mathcal{P}_j = \frac{1}{2\pi i}\oint_{\Omega_j}\mathcal{R}(z)\Delta\mathcal{R}(z)\,dz + \frac{1}{2\pi i}\oint_{\Omega_j}\mathcal{M}(z)\,dz,$$

and, using $\mathcal{R}(z)\Delta\mathcal{R}(z) = \sum_{k,l}(\lambda_k - z)^{-1}(\lambda_l - z)^{-1}\mathcal{P}_k\Delta\mathcal{P}_l$ and the elementary contour integral $\frac{1}{2\pi i}\oint_{\Omega_j}(\lambda_k-z)^{-1}(\lambda_l-z)^{-1}dz$, which equals $-(\lambda_k-\lambda_j)^{-1}$ if $k\ne j, l=j$; $(\lambda_l-\lambda_j)^{-1}$ if $k=j, l\ne j$; and 0 otherwise (by Cauchy's integral formula, since exactly one of the two simple poles λ_k, λ_l lies inside Ω_j in these cases, and none or both do in the remaining cases, the latter contributing 0 by residues at infinity / direct partial-fraction cancellation), the first term equals $\sum_{k\ne j}(\lambda_k-\lambda_j)^{-1}(\mathcal{P}_k\Delta\mathcal{P}_j + \mathcal{P}_j\Delta\mathcal{P}_k) = \mathcal{S}_j\Delta\mathcal{P}_j + \mathcal{P}_j\Delta\mathcal{S}_j$, proving (2.22).

For (2.23): $\|\mathcal{M}(z)\| \le \|\mathcal{R}(z)\|\sum_{k\ge2}\|\Delta\mathcal{R}(z)\|^k \le \eta_j^{-1}\delta_j^2/(1-\delta_j)$ on Ω_j, so $\left\|\frac{1}{2\pi i}\oint_{\Omega_j}\mathcal{M}(z)\,dz\right\| \le \frac{l_{\Omega_j}}{2\pi}\eta_j^{-1}\delta_j^2/(1-\delta_j) = \delta_j^2/(1-\delta_j)$, since $l_{\Omega_j} = 2\pi\eta_j$.

For (2.24): $\|\mathcal{S}_j\Delta\mathcal{P}_j + \mathcal{P}_j\Delta\mathcal{S}_j\|^2 = \|\mathcal{S}_j\Delta\mathcal{P}_j\|^2 + \|\mathcal{P}_j\Delta\mathcal{S}_j\|^2 \le 2\|\mathcal{S}_j\|^2\|\Delta\|^2$ (orthogonality of the two terms' ranges/domains), and $\|\mathcal{S}_j\| = \max_{k\ne j}|\lambda_k-\lambda_j|^{-1} \le (2\eta_j)^{-1}$ gives the stated bound; combining with (2.23) via the triangle inequality gives the norm bound on $\widetilde{\mathcal{P}}_j - \mathcal{P}_j$. □

Theorem 2.30 shows that $\mathcal{S}_j\Delta\mathcal{P}_j+\mathcal{P}_j\Delta\mathcal{S}_j$ is a first-order (linear in Δ) approximation to $\widetilde{\mathcal{P}}_j - \mathcal{P}_j$, with remainder $O(\delta_j^2)$. This linearization is the technical heart of the asymptotic normality results in Section 2.8: it expresses the perturbation of an eigenprojection as an explicit *linear* functional of Δ (hence, when Δ is itself asymptotically Gaussian, so is the leading-order term of $\widetilde{\mathcal{P}}_j - \mathcal{P}_j$).

2.4.3 Perturbation of Eigenvalues

Theorem 2.31. *Under the conditions of Theorem 2.30, with $\widetilde{\lambda}_{N_{j-1}+1}, \dots, \widetilde{\lambda}_{N_j}$ the (repeated) eigenvalues of $\widetilde{\mathcal{T}}$ matched to λ_j and $\widetilde{e}_{N_{j-1}+1}, \dots, \widetilde{e}_{N_j}$ the corresponding orthonormal eigenvectors used in* (2.21)*,*

$$\widetilde{\mathcal{P}}_j\widetilde{\mathcal{T}}\widetilde{\mathcal{P}}_j - \lambda_j\widetilde{\mathcal{P}}_j = \widetilde{\mathcal{P}}_j\Delta\widetilde{\mathcal{P}}_j + (\widetilde{\mathcal{P}}_j - \mathcal{P}_j)(\mathcal{T}-\lambda_j I)(\widetilde{\mathcal{P}}_j - \mathcal{P}_j), \tag{2.26}$$

and, for each $k = N_{j-1} + 1, \dots, N_j$,

$$\widetilde{\lambda}_k - \lambda_j = \langle \Delta \widetilde{e}_k, \widetilde{e}_k \rangle + \Big\langle (\widetilde{\mathcal{P}}_j - \mathcal{P}_j)(\mathcal{T} - \lambda_j I)(\widetilde{\mathcal{P}}_j - \mathcal{P}_j)\widetilde{e}_k, \widetilde{e}_k \Big\rangle, \tag{2.27}$$

where the second term is $O(\|\widetilde{\mathcal{P}}_j - \mathcal{P}_j\|^2) = O(\delta_j^2)$ *by* (2.24)*, since it is bounded in norm by* $\|\mathcal{T}\| \, \|\widetilde{\mathcal{P}}_j - \mathcal{P}_j\|^2$.

Proof. Expanding $\widetilde{\mathcal{P}}_j \widetilde{\mathcal{T}} \widetilde{\mathcal{P}}_j - \lambda_j \widetilde{\mathcal{P}}_j = \widetilde{\mathcal{P}}_j \Delta \widetilde{\mathcal{P}}_j + \widetilde{\mathcal{P}}_j (\mathcal{T} - \lambda_j I) \widetilde{\mathcal{P}}_j$, and using $\mathcal{P}_j(\mathcal{T} - \lambda_j I) = 0$ (since $\mathcal{P}_j$ projects onto the λ_j-eigenspace) to write $\widetilde{\mathcal{P}}_j(\mathcal{T} - \lambda_j I)\widetilde{\mathcal{P}}_j = (\widetilde{\mathcal{P}}_j - \mathcal{P}_j)(\mathcal{T} - \lambda_j I)(\widetilde{\mathcal{P}}_j - \mathcal{P}_j)$ (the cross terms vanish since $(\mathcal{T} - \lambda_j I)\mathcal{P}_j = 0 = \mathcal{P}_j(\mathcal{T} - \lambda_j I)$), proves (2.26). Taking the inner product of both sides with $\widetilde{e}_k$ and using $\widetilde{\mathcal{P}}_j \widetilde{\mathcal{T}} \widetilde{\mathcal{P}}_j = \sum_{k=N_{j-1}+1}^{N_j} \widetilde{\lambda}_k \widetilde{e}_k \otimes \widetilde{e}_k$ and $\widetilde{\mathcal{P}}_j \widetilde{e}_k = \widetilde{e}_k$ gives (2.27). □

Identity (2.27) refines the crude bound of Theorem 2.13: to first order in Δ, $\widetilde{\lambda}_k - \lambda_j \approx \langle \Delta \widetilde{e}_k, \widetilde{e}_k \rangle$, with an explicit, quadratic-in-Δ remainder. In the leading case of interest for FPCA — λ_j of multiplicity one for every j, so $\mathcal{P}_j = e_j \otimes e_j$ and $\widetilde{\mathcal{P}}_j = \widetilde{e}_j \otimes \widetilde{e}_j$ for a single matched eigenvalue $\widetilde{\lambda}_j$ — this reduces to

$$\widetilde{\lambda}_j - \lambda_j = \langle \Delta e_j, e_j \rangle + \Big\langle (\widetilde{\mathcal{P}}_j - \mathcal{P}_j)(\mathcal{T} - \lambda_j I)(\widetilde{\mathcal{P}}_j - \mathcal{P}_j)e_j, e_j \Big\rangle, \tag{2.28}$$

obtained from (2.27) by symmetry of the argument in $e_j, \widetilde{e}_j$ (an identical derivation with the roles of $\mathcal{T}, \widetilde{\mathcal{T}}$ exchanged in the projection identity gives the version stated with e_j in place of $\widetilde{e}_j$ in the leading term, up to the same $O(\delta_j^2)$ remainder). This first-order expansion, $\widetilde{\lambda}_j - \lambda_j \approx \langle \Delta e_j, e_j \rangle$, is precisely what will drive the asymptotic normality of the estimated FPCA eigenvalues in Section 2.8: since $\Delta = \widetilde{\mathcal{T}} - \mathcal{T}$ is a linear functional of the estimated covariance operator, a central limit theorem for $\widetilde{\mathcal{T}}$ transfers, via this linearization, into one for $\widetilde{\lambda}_j$.

2.4.4 Perturbation of Eigenvectors

We first relate the norm of the difference between two unit eigenvectors to the norm of the difference between their (rank-one) projections — which is the quantity directly controlled by Theorem 2.30.

Lemma 2.4. *Let* $\mathcal{P} = e \otimes e$, $\widetilde{\mathcal{P}} = \widetilde{e} \otimes \widetilde{e}$ *for unit vectors* $e, \widetilde{e}$*. Then: (1) the eigenvalues of* $\mathcal{P} - \widetilde{\mathcal{P}}$ *are* $\pm(1 - \langle e, \widetilde{e} \rangle^2)^{1/2}$*; (2) if* $\langle e, \widetilde{e} \rangle \geq 0$, $\|e - \widetilde{e}\|^2 = 2\Big[1 - (1 - \|\mathcal{P} - \widetilde{\mathcal{P}}\|^2)^{1/2}\Big]$*; (3)* $\|\mathcal{P} - \widetilde{\mathcal{P}}\|_{HS}^2 = 2(1 - \langle e, \widetilde{e} \rangle^2) = 2\|\mathcal{P} - \widetilde{\mathcal{P}}\|^2$.

Proof. (1) The range of $\mathcal{P}-\widetilde{\mathcal{P}}$ is contained in $\operatorname{span}\{e,\widetilde{e}\}$, on which, in the basis $(e,\widetilde{e})$, $\mathcal{P}-\widetilde{\mathcal{P}}$ acts as the matrix $\begin{pmatrix} 1 & -\langle e,\widetilde{e}\rangle \\ \langle e,\widetilde{e}\rangle & -1 \end{pmatrix}$ (since $\mathcal{P}e=e$, $\mathcal{P}\widetilde{e}=\langle\widetilde{e},e\rangle e$, and similarly for $\widetilde{\mathcal{P}}$), whose characteristic polynomial is $\mu^2-(1-\langle e,\widetilde{e}\rangle^2)=0$, giving the stated eigenvalues.

(2) By part (1) and Theorem 2.15 (operator norm equals largest singular value, and $\mathcal{P}-\widetilde{\mathcal{P}}$ is self-adjoint so its singular values are |eigenvalues|), $\|\mathcal{P}-\widetilde{\mathcal{P}}\|^2=1-\langle e,\widetilde{e}\rangle^2$. Combined with $\|e-\widetilde{e}\|^2=2(1-\langle e,\widetilde{e}\rangle)$ and, for $\langle e,\widetilde{e}\rangle\geq 0$, $\langle e,\widetilde{e}\rangle=(1-\|\mathcal{P}-\widetilde{\mathcal{P}}\|^2)^{1/2}$, the claim follows.

(3) Immediate from part (1) (both nonzero eigenvalues of $\mathcal{P}-\widetilde{\mathcal{P}}$ contribute equally to the HS norm) and the operator-norm identity from part (2)'s proof. □

For $x\in[0,1]$, $1-(1-x)^{1/2}\leq x/2+x^2/(4(1-x))$ (Taylor expansion with remainder of $(1-x)^{1/2}$), so part (2) of Lemma 2.4 gives, whenever $\langle e,\widetilde{e}\rangle\geq 0$,

$$\|e-\widetilde{e}\|^2\leq\|\mathcal{P}-\widetilde{\mathcal{P}}\|^2+\frac{1}{2}\frac{\|\mathcal{P}-\widetilde{\mathcal{P}}\|^4}{1-\|\mathcal{P}-\widetilde{\mathcal{P}}\|^2},\tag{2.29}$$

and, since $(1+x)^{1/2}\leq 1+x/2$ for $x\in[0,1]$, taking square roots and simplifying,

$$\|e-\widetilde{e}\|\leq\|\mathcal{P}-\widetilde{\mathcal{P}}\|+\frac{1}{4}\frac{\|\mathcal{P}-\widetilde{\mathcal{P}}\|^3}{1-\|\mathcal{P}-\widetilde{\mathcal{P}}\|^2}.\tag{2.30}$$

We can now state the eigenvector perturbation expansion that, together with (2.28), will be the direct input to the FPCA asymptotic normality results of Section 2.8.

Theorem 2.32. *Let λ_j be a simple (multiplicity-one) eigenvalue of $\mathcal{T}$ with unit eigenvector e_j; the other eigenvalues of $\mathcal{T}$ need not be simple. Let $\widetilde{\mathcal{T}}=\mathcal{T}+\Delta$ be compact self-adjoint with matched eigenpair $(\widetilde{\lambda}_j,\widetilde{e}_j)$ chosen with $\langle e_j,\widetilde{e}_j\rangle\geq 0$, and suppose $\inf_{k\neq j}|\widetilde{\lambda}_j-\lambda_k|>0$. Then*

$$\widetilde{e}_j-e_j=\sum_{k\neq j}(\widetilde{\lambda}_j-\lambda_k)^{-1}\mathcal{P}_k\Delta\widetilde{e}_j+\mathcal{P}_j(\widetilde{e}_j-e_j).\tag{2.31}$$

Moreover, with $\eta_j=\frac{1}{2}\inf_{k\neq j}|\lambda_j-\lambda_k|$ and $\delta_j=\|\Delta\|/\eta_j$, if $\delta_j<1$ there is a finite function ψ with $\psi(\delta)\downarrow 1$ as $\delta\downarrow 0$ such that

$$\|e_j-\widetilde{e}_j-\mathcal{S}_j\Delta e_j\|\leq\psi(\delta_j)\delta_j^2,\qquad\mathcal{S}_j:=\sum_{k\neq j}(\lambda_k-\lambda_j)^{-1}\mathcal{P}_k.\tag{2.32}$$

Proof. For $k\neq j$, $(\widetilde{\lambda}_j-\lambda_k)\mathcal{P}_k(\widetilde{e}_j-e_j)=(\widetilde{\lambda}_j-\lambda_k)\mathcal{P}_k\widetilde{e}_j=\mathcal{P}_k\big(\widetilde{\mathcal{T}}\widetilde{e}_j-\mathcal{T}\mathcal{P}_k\widetilde{e}_j\big)=\mathcal{P}_k\Delta\widetilde{e}_j+\mathcal{P}_k(\mathcal{T}\widetilde{e}_j-\mathcal{T}\mathcal{P}_k\widetilde{e}_j)$; since $\mathcal{P}_k\mathcal{T}=\lambda_k\mathcal{P}_k$ this simplifies (writing $\mathcal{T}\widetilde{e}_j=\sum_l\lambda_l\mathcal{P}_l\widetilde{e}_j$) to $\mathcal{P}_k\Delta\widetilde{e}_j$, using $\widetilde{\mathcal{T}}\widetilde{e}_j=\widetilde{\lambda}_j\widetilde{e}_j$. Summing over $k\neq j$ and adding $\mathcal{P}_j(\widetilde{e}_j-e_j)$ recovers $\widetilde{e}_j-e_j=\sum_k\mathcal{P}_k(\widetilde{e}_j-e_j)$ by Theorem 2.10 (completeness of $\{\mathcal{P}_k\}$), proving (2.31).

For (2.32): Theorem 2.13 gives $|\widetilde{\lambda}_j - \lambda_j| \le \|\Delta\| < \eta_j \le \inf_{k\neq j}|\lambda_j - \lambda_k|$, so $\inf_{k\neq j}|\widetilde{\lambda}_j - \lambda_k| \ge \inf_{k\neq j}|\lambda_j - \lambda_k| - \|\Delta\| > 0$ and (2.31) applies. Expanding $(\widetilde{\lambda}_j - \lambda_k)^{-1}$ as a geometric series in $(\lambda_j - \widetilde{\lambda}_j)/(\lambda_j - \lambda_k)$ (convergent since $|\lambda_j - \widetilde{\lambda}_j| < |\lambda_j - \lambda_k|$) and substituting into (2.31) decomposes $\widetilde{e}_j - e_j$ into four terms: (i) $\sum_{k\neq j}(\lambda_j - \lambda_k)^{-1}\mathcal{P}_k\Delta e_j = \mathcal{S}_j\Delta e_j$; (ii) $\sum_{k\neq j}(\lambda_j - \lambda_k)^{-1}\mathcal{P}_k\Delta(\widetilde{e}_j - e_j)$; (iii) the $s \ge 1$ terms of the geometric expansion, $\sum_{k\neq j}\sum_{s\ge 1}(\lambda_j - \widetilde{\lambda}_j)^s(\lambda_j - \lambda_k)^{-s-1}\mathcal{P}_k\Delta\widetilde{e}_j$; (iv) $\mathcal{P}_j(\widetilde{e}_j - e_j)$.

Term (ii) is bounded, by Bessel's inequality, by $\|\Delta\|\|\widetilde{e}_j - e_j\|/(2\eta_j) = (\delta_j/2)\|\widetilde{e}_j - e_j\|$. Term (iii): by Parseval's relation (orthogonality of the $\mathcal{P}_k$ ranges), its squared norm is $\sum_{k\neq j}\left(\sum_{s\ge 1}(\lambda_j - \widetilde{\lambda}_j)^s(\lambda_j - \lambda_k)^{-s-1}\right)^2\|\mathcal{P}_k\Delta\widetilde{e}_j\|^2 \le \sum_{k\neq j}\frac{(\lambda_j - \widetilde{\lambda}_j)^2}{(\lambda_j - \lambda_k)^2(|\lambda_j - \lambda_k| - |\lambda_j - \widetilde{\lambda}_j|)^2}\|\mathcal{P}_k\Delta\widetilde{e}_j\|^2 \le \frac{\|\Delta\|^4}{4\eta_j^2(2\eta_j - \|\Delta\|)^2} \le \delta_j^4/4$ (bounding each denominator factor by $\eta_j \cdot \eta_j$, using $\|\Delta\| < \eta_j$, and $\sum_{k\neq j}\|\mathcal{P}_k\Delta\widetilde{e}_j\|^2 \le \|\Delta\|^2$ by Bessel). Term (iv): $\mathcal{P}_j(\widetilde{e}_j - e_j) = \langle\widetilde{e}_j - e_j, e_j\rangle e_j = (\langle\widetilde{e}_j, e_j\rangle - 1)e_j$, so $\|\mathcal{P}_j(\widetilde{e}_j - e_j)\| = 1 - \langle\widetilde{e}_j, e_j\rangle = \frac{1}{2}\|\widetilde{e}_j - e_j\|^2$.

Combining these four bounds with the triangle inequality, and using (2.30) together with (2.24) to control $\|\widetilde{e}_j - e_j\|$ (of order δ_j) in the terms (ii) and (iv) that are quadratic or higher in $\|\widetilde{e}_j - e_j\|$, yields $\|e_j - \widetilde{e}_j - \mathcal{S}_j\Delta e_j\| \le \psi(\delta_j)\delta_j^2$ for an explicit finite ψ with $\psi(\delta) \downarrow 1$ as $\delta \downarrow 0$ (each of the four terms other than $\mathcal{S}_j\Delta e_j$ contributes $O(\delta_j^2)$, with constants that remain bounded as $\delta_j \downarrow 0$). □

As with the eigenvalue expansion (2.28), Theorem 2.32 identifies $\mathcal{S}_j\Delta e_j$ as the first-order (linear-in-Δ) term of the eigenvector perturbation $\widetilde{e}_j - e_j$, with an explicit $O(\delta_j^2)$ remainder. Together, (2.28) and (2.32) reduce the asymptotic behavior of the estimated FPCA eigenvalue/eigenfunction pairs to the (much simpler) asymptotic behavior of the linear functionals $\langle\Delta e_j, e_j\rangle$ and $\mathcal{S}_j\Delta e_j$ of the covariance operator perturbation $\Delta = \widetilde{\Gamma} - \Gamma$ — which is exactly what the large-sample theory of Sections 2.6 and 2.7 establishes.

2.5 Smoothing and Regularization

Discretely sampled, possibly noisy functional data must be converted into genuine elements of a function space before the Hilbert-space machinery of the preceding sections applies. This section develops the general theory of penalized least-squares (regularization) estimators that accomplishes this, and specializes it to smoothing splines, which are the theoretical counterpart of the adaptive B-spline smoothing used in Section 4.2.1. We follow [4], Chapter 6.

2.5.1 The Functional Linear Model

Definition 2.19. *Let $\mathbb{Y}, \mathbb{H}$ be Hilbert spaces and $\mathcal{T} \in \mathcal{B}(\mathbb{H}, \mathbb{Y})$. The* functional linear model *is*

$$Y = \mathcal{T}m + \varepsilon, \qquad Y, \varepsilon \in \mathbb{Y}, \quad m \in \mathbb{H}, \tag{2.33}$$

where ε is a mean-zero error element and $\mathcal{T}$ is treated as known; the goal is to estimate m from an observation of Y.

The case directly relevant to Chapter 4 is the discretely sampled setting: $\mathbb{Y} = \mathbb{R}^n$ with $\|Y\|_{\mathbb{Y}}^2 = n^{-1}\sum_{i=1}^n Y_i^2$, $\mathbb{H}$ an RKHS of functions on $[0,1]$, and $\mathcal{T}g = (g(t_1), \ldots, g(t_n))^T$ for observation points $t_1, \ldots, t_n$; boundedness of $\mathcal{T}$ is precisely boundedness of the evaluation functionals, which characterizes an RKHS (Definition 2.7, Section 2.2.4). Model (2.33) also subsumes the case where the sample paths $x_1, \ldots, x_n$ of a mean-m process are themselves the object being smoothed: writing $Y_{ij} = x_i(t_{ij}) + \widetilde{\varepsilon}_{ij} = m(t_{ij}) + \{x_i(t_{ij}) - m(t_{ij})\} + \widetilde{\varepsilon}_{ij}$ and stacking, $\varepsilon_{ij} := x_i(t_{ij}) - m(t_{ij}) + \widetilde{\varepsilon}_{ij}$ recovers (2.33) with a (possibly correlated) error term.

2.5.2 Penalized Least-Squares Estimation

Ordinary least squares, $\widehat{m} = \operatorname{argmin}_{g\in\mathbb{H}}\|Y - \mathcal{T}g\|_{\mathbb{Y}}^2$, is ill-posed when $\mathbb{H}$ is infinite-dimensional: any g interpolating the data minimizes the criterion, so the estimator is undetermined. The remedy is to penalize roughness of g, measured by a nonnegative operator $\mathcal{W} \in \mathcal{B}(\mathbb{H})$ (e.g. $\langle g, \mathcal{W}g\rangle_{\mathbb{H}} = \int_0^1 |g^{(m)}(t)|^2\, dt$ for $\mathcal{W}$ the projection onto $\mathbb{H}_1$ of the Sobolev decomposition of Definition 2.8).

Theorem 2.33 (Representer theorem)**.** *Assume $\mathcal{T}^*\mathcal{T} + \eta\mathcal{W}$ is invertible. For $\eta \in (0, \infty)$, the unique minimizer over $g \in \mathbb{H}$ of*

$$f(g; \mathcal{T}, \mathcal{W}, Y, \eta) := \|Y - \mathcal{T}g\|_{\mathbb{Y}}^2 + \eta\langle g, \mathcal{W}g\rangle_{\mathbb{H}} \tag{2.34}$$

is

$$m_\eta := (\mathcal{T}^*\mathcal{T} + \eta\mathcal{W})^{-1}\mathcal{T}^*Y. \tag{2.35}$$

Proof. Write a candidate solution as $\widetilde{g} = m_\eta + g$. Expanding, $f(\widetilde{g}; \mathcal{T}, \mathcal{W}, Y, \eta) = \|Y\|_{\mathbb{Y}}^2 + \langle \mathcal{T}^*\mathcal{T}\widetilde{g}, \widetilde{g}\rangle_{\mathbb{H}} - 2\langle Y, \mathcal{T}\widetilde{g}\rangle_{\mathbb{Y}} + \eta\langle \mathcal{W}\widetilde{g}, \widetilde{g}\rangle_{\mathbb{H}}$. Since $\mathcal{T}^*Y = (\mathcal{T}^*\mathcal{T} + \eta\mathcal{W})m_\eta$ by (2.35), $\langle Y, \mathcal{T}\widetilde{g}\rangle_{\mathbb{Y}} = \langle (\mathcal{T}^*\mathcal{T} + \eta\mathcal{W})m_\eta, \widetilde{g}\rangle_{\mathbb{H}}$, so

$$f(\widetilde{g}; \mathcal{T}, \mathcal{W}, Y, \eta) = \|Y\|_{\mathbb{Y}}^2 + \left\langle (\mathcal{T}^*\mathcal{T} + \eta\mathcal{W})(\widetilde{g} - 2m_\eta), \widetilde{g}\right\rangle_{\mathbb{H}}.$$

Writing $\widetilde{g} - 2m_\eta = (\widetilde{g} - m_\eta) - m_\eta$ and expanding the inner product,

$$\Big\langle (\mathcal{T}^*\mathcal{T} + \eta\mathcal{W})(\widetilde{g} - 2m_\eta), \widetilde{g} \Big\rangle_{\mathbb{H}} = \Big\langle (\mathcal{T}^*\mathcal{T} + \eta\mathcal{W})g, g \Big\rangle_{\mathbb{H}} - \Big\langle (\mathcal{T}^*\mathcal{T} + \eta\mathcal{W})m_\eta, m_\eta \Big\rangle_{\mathbb{H}}$$

(using symmetry of $\mathcal{T}^*\mathcal{T} + \eta\mathcal{W}$ to combine cross terms). Since $\mathcal{T}^*\mathcal{T} + \eta\mathcal{W}$ is positive definite (it is invertible and nonnegative, being a sum of nonnegative operators), the first term, and hence $f(\widetilde{g}; \mathcal{T}, \mathcal{W}, Y, \eta)$, is uniquely minimized at $g = 0$, i.e. $\widetilde{g} = m_\eta$. □

Equation (2.35) is precisely the abstract form of the B-spline smoothing-spline criterion (4.2)–(4.3) used in Section 4.2.1: there, $\mathbb{Y} = \mathbb{R}^n$, $\mathbb{H} = \mathbb{W}^m(\mathcal{T})$, $\mathcal{W}$ penalizes $\int_{\mathcal{T}} [X^{(m)}(t)]^2\, dt$, and $\eta = \lambda$ is chosen data-adaptively per country via $\widehat{\sigma}$ rather than by the cross-validation methods of Section 2.5.5 below. The behavior of m_η at the extremes $\eta \to 0$ (which recovers the minimum-norm least-squares solution of $\mathcal{T}g = Y$, via the Moore–Penrose pseudoinverse $\mathcal{T}^\dagger$) and $\eta \to \infty$ (which forces m_η into $\mathrm{Ker}(\mathcal{W})$, the "ideal", unpenalized model class) is established in [4], Theorems 6.2.2–6.2.3; we omit the proofs here, as they rely on the theory of generalized (Moore–Penrose) operator inverses developed in that source's Section 3.5, which falls outside the scope of the preliminaries of Section 2.2 and is not needed for the fixed-η, data-adaptively-chosen smoothing spline actually used in this thesis.

2.5.3 Bias and Variance

Theorem 2.34. *Let m_η be as in* (2.35) *for the model* (2.33)*, and suppose a future, independent observation $Y_{\text{new}} = \mathcal{T}m + \varepsilon_{\text{new}}$ is to be predicted by $\mathcal{T}m_\eta$, with prediction risk* $\mathrm{Risk}(\eta) = \mathrm{Var}(\eta) + \mathrm{Bias}^2(\eta)$ *for*

$$\mathrm{Bias}^2(\eta) := \|\mathcal{T}(\mathbb{E}m_\eta - m)\|_{\mathbb{Y}}^2, \qquad \mathrm{Var}(\eta) := \mathbb{E}\|\mathcal{T}(m_\eta - \mathbb{E}m_\eta)\|_{\mathbb{Y}}^2. \tag{2.36}$$

Then $\mathrm{Bias}^2(\eta) \le \eta\langle m, \mathcal{W}m\rangle_{\mathbb{H}}$ *and* $\mathrm{Var}(\eta) = \mathbb{E}\|\mathcal{T}(\mathcal{T}^*\mathcal{T} + \eta\mathcal{W})^{-1}\mathcal{T}^*\varepsilon\|_{\mathbb{Y}}^2$.

Proof. $\mathbb{E}m_\eta = (\mathcal{T}^*\mathcal{T} + \eta\mathcal{W})^{-1}\mathcal{T}^*\mathcal{T}m$ is, by Theorem 2.33 applied with Y replaced by $\mathcal{T}m$, the minimizer over $g \in \mathbb{H}$ of $\|\mathcal{T}m - \mathcal{T}g\|_{\mathbb{Y}}^2 + \eta\langle g, \mathcal{W}g\rangle_{\mathbb{H}}$. Hence $\|\mathcal{T}(\mathbb{E}m_\eta - m)\|_{\mathbb{Y}}^2 \le \|\mathcal{T}(\mathbb{E}m_\eta - m)\|_{\mathbb{Y}}^2 + \eta\langle \mathbb{E}m_\eta, \mathcal{W}\mathbb{E}m_\eta\rangle_{\mathbb{H}} \le \|\mathcal{T}(g-m)\|_{\mathbb{Y}}^2 + \eta\langle g, \mathcal{W}g\rangle_{\mathbb{H}}$ for every g; taking $g = m$ gives the bias bound. The variance formula follows since $m_\eta - \mathbb{E}m_\eta = (\mathcal{T}^*\mathcal{T} + \eta\mathcal{W})^{-1}\mathcal{T}^*\varepsilon$ directly from (2.35) and linearity of expectation. □

Corollary 2.1. *Let $\mathbb{Y} = \mathbb{R}^n$ with $\|Y\|_{\mathbb{Y}}^2 = n^{-1}\sum_j Y_j^2$, and suppose ε has uncorrelated*

entries with mean 0 *and variance* σ^2*, and* $\mathcal{W}$ *is invertible. Then*

$$\mathrm{Var}(\eta) = \frac{\sigma^2}{n} \sum_{j=1}^{n} \left(\frac{\gamma_j}{\gamma_j + \eta} \right)^2, \tag{2.37}$$

where $\gamma_1 \geq \cdots \geq \gamma_n \geq 0$ *are the eigenvalues of the compact operator* $\mathcal{W}^{-1/2} \mathcal{T}^* \mathcal{T} \mathcal{W}^{-1/2}$*.*

Proof. Since $\mathrm{Im}(\mathcal{T})$ is finite-dimensional, $\mathcal{T}$, and hence $\widetilde{\mathcal{T}} := \mathcal{T}\mathcal{W}^{-1/2}$, is compact (Theorem 2.6(2)), with at most n singular values $\gamma_1 \geq \cdots \geq \gamma_n \geq 0$ (which are the eigenvalues of $\widetilde{\mathcal{T}}^* \widetilde{\mathcal{T}} = \mathcal{W}^{-1/2} \mathcal{T}^* \mathcal{T} \mathcal{W}^{-1/2}$, by Theorem 2.14). Writing $\mathcal{T}(\mathcal{T}^*\mathcal{T} + \eta \mathcal{W})^{-1} \mathcal{T}^* = \widetilde{\mathcal{T}}(\widetilde{\mathcal{T}}^* \widetilde{\mathcal{T}} + \eta I)^{-1} \widetilde{\mathcal{T}}^*$ (substituting $\mathcal{T} = \widetilde{\mathcal{T}} \mathcal{W}^{1/2}$) and using $\mathbb{E} \langle \varepsilon, e \rangle_{\mathbb{Y}}^2 = \sigma^2 e^T e / n^2$ for any $e \in \mathbb{R}^n$ together with the SVD of $\widetilde{\mathcal{T}}$ in Theorem 2.34's variance formula gives (2.37). □

2.5.4 A Computational Formula

Theorem 2.35. *Let* $\mathbb{Y} = \mathbb{R}^n$*,* $\mathbb{H} = \mathbb{H}_0 \oplus \mathbb{H}_1$ *with* $\dim(\mathbb{H}_0) = q < \infty$*, basis* $\{\phi_1, \ldots, \phi_q\}$ *of* $\mathbb{H}_0$*, and* $\mathcal{W} = \mathcal{P}_1$ *the projection onto* $\mathbb{H}_1$*. Write* $\mathcal{T} g = (\mathcal{T}_1 g, \ldots, \mathcal{T}_n g)^T$ *for bounded linear functionals* $\mathcal{T}_i$ *with representers* τ_i *(i.e.* $\mathcal{T}_i g = \langle \tau_i, g \rangle_{\mathbb{H}}$*), and let* $\xi_i := \mathcal{P}_1 \tau_i$*. Define* $\Phi := \{\mathcal{T}_i \phi_j\}_{i,j}$*,* $\Psi := \{\mathcal{T}_i \xi_j\}_{i,j} = \{\langle \xi_i, \xi_j \rangle_{\mathbb{H}}\}_{i,j}$*,* $\mathcal{U} := (\Phi, \Psi)$*, and* $\mathcal{V} := \left(\begin{smallmatrix} 0_{q \times q} & 0_{q \times n} \\ 0_{n \times q} & \Psi_{n \times n} \end{smallmatrix}\right)$*. If* $\mathcal{U}^T \mathcal{U} + \eta \mathcal{V}$ *is invertible, the minimizer of* (2.34) *is* $m_\eta = \sum_{j=1}^{q} \widehat{c}_j \phi_j + \sum_{i=1}^{n} \widehat{b}_i \xi_i$*, where*

$$\begin{pmatrix} \widehat{c} \\ \widehat{b} \end{pmatrix} = (\mathcal{U}^T \mathcal{U} + \eta \mathcal{V})^{-1} \mathcal{U}^T Y. \tag{2.38}$$

Proof. Write $g = \sum_j c_j \phi_j + \sum_i b_i \xi_i + h$ for $h \perp \mathrm{span}\{\phi_1, \ldots, \phi_q, \xi_1, \ldots, \xi_n\}$. Since $\tau_i - \xi_i \in \mathbb{H}_0 = \mathrm{span}\{\phi_j\}$, $\mathcal{T}_i h = \langle \tau_i, h \rangle_{\mathbb{H}} = \langle \tau_i - \xi_i, h \rangle_{\mathbb{H}} + \langle \xi_i, h \rangle_{\mathbb{H}} = 0$, so $\mathcal{T}_i g$ does not depend on h, while $\langle g, \mathcal{P}_1 g \rangle_{\mathbb{H}} = \|\mathcal{P}_1 g\|^2$ strictly increases with $\|h\|$ whenever $h \neq 0$ has a nonzero $\mathbb{H}_1$-component (since $h \perp \xi_i$ does not preclude $h \in \mathbb{H}_1$ in general, but any such component only adds to $\langle g, \mathcal{P}_1 g \rangle_{\mathbb{H}}$ without changing $\mathcal{T} g$); hence the minimizer has $h = 0$, i.e. $m_\eta \in \mathrm{span}\{\phi_j, \xi_i\}$. For such g, $\mathcal{T} g = \Phi c + \Psi b$ and $\langle g, \mathcal{P}_1 g \rangle_{\mathbb{H}} = b^T \Psi b$ (since $\mathcal{P}_1 \phi_j = 0$ and $\mathcal{P}_1 \xi_i = \xi_i$), so (2.34) becomes $\|Y - \mathcal{U}\binom{c}{b}\|_{\mathbb{Y}}^2 + \eta \binom{c}{b}^T \mathcal{V} \binom{c}{b}$, and Theorem 2.33 (applied in $\mathbb{R}^{q+n}$ with the identity map replaced by $\mathcal{U}$) gives (2.38). □

Theorem 2.36. *If* $\mathbb{H}_0 = \mathbb{H}(K_0)$*,* $\mathbb{H}_1 = \mathbb{H}(K_1)$*,* $\mathbb{H} = \mathbb{H}(K)$ *for reproducing kernels with* $K = K_0 + K_1$ *and* $\mathbb{H}_0 \cap \mathbb{H}_1 = \{0\}$*, then* $\mathbb{H} = \mathbb{H}_0 \oplus \mathbb{H}_1$ *and, in the notation of Theorem 2.35,* $\tau_i(t) = \mathcal{T}_i^{(\cdot)} K(t, \cdot)$ *and* $\xi_i(t) = \mathcal{T}_i^{(\cdot)} K_1(t, \cdot)$ *(applying* $\mathcal{T}_i$ *to the indicated argument).*

Proof. $\mathbb{H} = \mathbb{H}_0 \oplus \mathbb{H}_1$ is the kernel-sum decomposition of Definition 2.7. By the reproducing property, $\tau_i(t) = \langle \tau_i, K(t, \cdot) \rangle_{\mathbb{H}} = \mathcal{T}_i^{(\cdot)} K(t, \cdot)$. Similarly, $\xi_i(t) = \langle \xi_i, K(t, \cdot) \rangle_{\mathbb{H}} = \langle \mathcal{P}_1 \tau_i, K(t, \cdot) \rangle_{\mathbb{H}} = \langle \tau_i, \mathcal{P}_1 K(t, \cdot) \rangle_{\mathbb{H}} = \langle \tau_i, K_1(t, \cdot) \rangle_{\mathbb{H}} = \mathcal{T}_i^{(\cdot)} K_1(t, \cdot)$, using that $\mathcal{P}_1$ projects $K(t, \cdot) = K_0(t, \cdot) + K_1(t, \cdot)$ onto its $\mathbb{H}_1$-component. □

Theorem 2.36 applies directly to $\mathbb{H} = \mathbb{W}^q[0,1]$ with K_0, K_1 as in Definition 2.8 (i.e. Theorem 2.8.1 of [4]).

2.5.5 Regularization Parameter Selection

We now specialize to $\mathbb{Y} = \mathbb{R}^n$, $\mathbb{H} = \mathbb{H}_0 \oplus \mathbb{H}_1$ (finite-dimensional $\mathbb{H}_0$) of Theorem 2.35, and describe three data-adaptive methods for choosing η that do not require the closed-form, noise-level-based rule used in Section 4.2.1.

Let $m_\eta^{[k]}$ minimize $n^{-1}\sum_{i\neq k}(Y_i - \mathcal{T}_i g)^2 + \eta\langle g, \mathcal{P}_1 g\rangle_{\mathbb{H}}$ (leaving out observation k), and define ordinary cross-validation as $\mathrm{CV}(\eta) := n^{-1}\sum_{k=1}^n (Y_k - \mathcal{T}_k m_\eta^{[k]})^2$, chosen to minimize $\mathrm{CV}(\eta)$. Direct evaluation requires refitting n times per candidate η; the following device avoids this.

Lemma 2.5 (Wahba's leave-one-out lemma)**.** *For $y \in \mathbb{R}$, let $m_\eta[k, y]$ minimize $n^{-1}(y - \mathcal{T}_k g)^2 + n^{-1}\sum_{i\neq k}(Y_i - \mathcal{T}_i g)^2 + \eta\langle g, \mathcal{P}_1 g\rangle_{\mathbb{H}}$ (so $m_\eta = m_\eta[k, Y_k]$). Then $m_\eta[k, \mathcal{T}_k m_\eta^{[k]}] = m_\eta^{[k]}$.*

Proof. For any $g \in \mathbb{H}$, using that $m_\eta^{[k]}$ minimizes the leave-one-out criterion, and abbreviating $z := \mathcal{T}_k m_\eta^{[k]}$,

$$\begin{aligned} n^{-1}(\mathcal{T}_k m_\eta^{[k]} - z)^2 + n^{-1}\sum_{i\neq k}(Y_i - \mathcal{T}_i m_\eta^{[k]})^2 + \eta\langle m_\eta^{[k]}, \mathcal{P}_1 m_\eta^{[k]}\rangle_{\mathbb{H}} \\ = n^{-1}\sum_{i\neq k}(Y_i - \mathcal{T}_i m_\eta^{[k]})^2 + \eta\langle m_\eta^{[k]}, \mathcal{P}_1 m_\eta^{[k]}\rangle_{\mathbb{H}} \\ \leq n^{-1}\sum_{i\neq k}(Y_i - \mathcal{T}_i g)^2 + \eta\langle g, \mathcal{P}_1 g\rangle_{\mathbb{H}} \\ \leq n^{-1}(z - \mathcal{T}_k g)^2 + n^{-1}\sum_{i\neq k}(Y_i - \mathcal{T}_i g)^2 + \eta\langle g, \mathcal{P}_1 g\rangle_{\mathbb{H}}, \end{aligned}$$

i.e. $m_\eta^{[k]}$ minimizes the criterion defining $m_\eta[k, z]$, so $m_\eta[k, z] = m_\eta^{[k]}$. □

Define the "leverage" $a_{kk}(\eta) := (\mathcal{T}_k m_\eta - \mathcal{T}_k m_\eta^{[k]})/(Y_k - \mathcal{T}_k m_\eta^{[k]})$, so that $Y_k - \mathcal{T}_k m_\eta^{[k]} = (Y_k - \mathcal{T}_k m_\eta)/(1 - a_{kk}(\eta))$ and

$$\mathrm{CV}(\eta) = n^{-1}\sum_{k=1}^n \frac{(Y_k - \mathcal{T}_k m_\eta)^2}{[1 - a_{kk}(\eta)]^2}. \tag{2.39}$$

By Lemma 2.5, writing $Y_k' = \mathcal{T}_k m_\eta^{[k]}$, $a_{kk}(\eta) = \{\mathcal{T}_k m_\eta[k, Y_k] - \mathcal{T}_k m_\eta[k, Y_k']\}/(Y_k - Y_k')$; since $m_\eta[k, y]$ is, by Theorem 2.35 applied with the kth response replaced by y, linear in y, this ratio equals the derivative $\partial \mathcal{T}_k m_\eta[k, y]/\partial y|_{y=Y_k}$, i.e. the kth diagonal entry of the "hat

matrix" $\mathcal{A}(\eta) := \mathcal{U}(\mathcal{U}^T\mathcal{U} + \eta\mathcal{V})^{-1}\mathcal{U}^T$ satisfying $(\mathcal{T}_1 m_\eta, \ldots, \mathcal{T}_n m_\eta)^T = \mathcal{A}(\eta)Y$. So CV($\eta$) is computable directly from the residuals $Y - \mathcal{T}m_\eta$ and diag($\mathcal{A}(\eta)$), without refitting.

Generalized cross-validation replaces the individual leverages $a_{kk}(\eta)$ in (2.39) by their average, minimizing

$$\mathrm{GCV}(\eta) := \frac{\|Y - \mathcal{T}m_\eta\|_{\mathbb{Y}}^2}{\left(1 - n^{-1}\mathrm{trace}\,\mathcal{A}(\eta)\right)^2}. \tag{2.40}$$

Suppose ε has uncorrelated entries, mean 0, variance σ^2, and let $\mu := \mathbb{E}Y = (\mathcal{T}_1 m, \ldots, \mathcal{T}_n m)^T$. The prediction risk of $\mathcal{A}(\eta)Y$ for μ is $\mathrm{Risk}(\eta) := n^{-1}\mathbb{E}\|\mathcal{A}(\eta)Y - \mu\|_2^2 = n^{-1}\mu^T(I - \mathcal{A}(\eta))^T(I - \mathcal{A}(\eta))\mu + \sigma^2 n^{-1}\mathrm{trace}\,\mathcal{A}(\eta)\mathcal{A}(\eta)^T$ (bias-variance decomposition, as in Theorem 2.34 specialized to $\mathbb{R}^n$). Writing $\mathrm{MSE}(\eta) := \|(I - \mathcal{A}(\eta))Y\|_{\mathbb{Y}}^2$, so that $\mathrm{GCV}(\eta) = \mathrm{MSE}(\eta)/(1 - n^{-1}\mathrm{trace}\,\mathcal{A}(\eta))^2$, an identical bias-variance expansion of $\mathbb{E}[\mathrm{MSE}(\eta)]$ gives

$$\mathbb{E}[\mathrm{MSE}(\eta)] = \mathrm{Risk}(\eta) + \sigma^2 - \frac{2\sigma^2}{n}\mathrm{trace}\,\mathcal{A}(\eta). \tag{2.41}$$

Theorem 2.37 (GCV Theorem)**.** *Let* $\tau_j(\eta) := \mathrm{trace}\,\mathcal{A}^j(\eta)/n$, $j = 1, 2$, *and* $g(\eta) := \{2\tau_1(\eta) + \tau_1(\eta)^2/\tau_2(\eta)\}/(1 - \tau_1(\eta))^2$. *Then*

$$\frac{|\mathbb{E}\,\mathrm{GCV}(\eta) - \sigma^2 - \mathrm{Risk}(\eta)|}{\mathrm{Risk}(\eta)} \le g(\eta). \tag{2.42}$$

Proof. From (2.41) and $\mathrm{GCV}(\eta) = \mathbb{E}[\mathrm{MSE}(\eta)]/(1 - \tau_1(\eta))^2$ (in expectation), algebra gives

$$\mathbb{E}\,\mathrm{GCV}(\eta) - \sigma^2 - \mathrm{Risk}(\eta) = \frac{\tau_1(\eta)(2 - \tau_1(\eta))}{(1 - \tau_1(\eta))^2}\mathrm{Risk}(\eta) - \frac{\sigma^2\tau_1(\eta)^2}{(1 - \tau_1(\eta))^2}.$$

Dividing by $\mathrm{Risk}(\eta)$ and applying the triangle inequality,

$$\frac{|\mathbb{E}\,\mathrm{GCV}(\eta) - \sigma^2 - \mathrm{Risk}(\eta)|}{\mathrm{Risk}(\eta)} \le \frac{\tau_1(\eta)(2 - \tau_1(\eta))}{(1 - \tau_1(\eta))^2} + \frac{\tau_1(\eta)^2/\tau_2(\eta)}{(1 - \tau_1(\eta))^2} = g(\eta),$$

using $\sigma^2/\mathrm{Risk}(\eta) \le 1/\tau_2(\eta)$ (from (2.41) and nonnegativity of the bias term, since $\mathrm{trace}\,\mathcal{A}^2(\eta) \le n\sigma^2/\mathrm{Risk}(\eta)\cdot$ after rearrangement – more directly, $\sigma^2\mathrm{trace}\,\mathcal{A}^2(\eta)/n \le \mathrm{Risk}(\eta)$ is the variance component of $\mathrm{Risk}(\eta)$ in $\mathbb{R}^n$). □

When $g(\eta)$ is small, $\mathrm{GCV}(\eta)$ is thus a near-unbiased estimator of $\sigma^2 + \mathrm{Risk}(\eta)$. A related, exactly unbiased estimator of $\mathrm{Risk}(\eta)$, from (2.41), is Mallows' criterion $C_L(\eta) := \mathrm{MSE}(\eta) + 2\sigma^2 n^{-1}\mathrm{trace}\,\mathcal{A}(\eta)$, minimized in place of $\mathrm{GCV}(\eta)$ when an estimate of σ^2 is available; GCV avoids this requirement at the cost of a small bias governed by $g(\eta)$.

2.5.6 Splines

We close this section with the theory of polynomial splines, which resolves the penalized least-squares problem (2.34) explicitly for the roughness penalty $\langle g, \mathcal{W}g\rangle_{\mathbb{H}} = \int_0^1 |g^{(q)}(t)|^2\,dt$ on $\mathbb{H} = \mathbb{W}^q[0,1]$, and underlies the B-spline basis of Section 4.2.1.

Definition 2.20. *A* spline of order $q \geq 1$ *with knots* $0 < t_1 < \cdots < t_J < 1$ *is*

$$g(t) = \sum_{i=0}^{q-1} \theta_i t^i + \sum_{j=1}^{J} \delta_j (t - t_j)_+^{q-1}, \qquad \theta_i, \delta_j \in \mathbb{R}, \tag{2.43}$$

where $x_+ := \max(x, 0)$. *Equivalently,* g *is a piecewise polynomial of order* q *on each* $[t_i, t_{i+1})$, *with* $q-2$ *continuous derivatives (for* $q \geq 2$*) and* $(q-1)$*-st derivative a step function jumping at* $t_1, \ldots, t_J$. *The functions* $1, t, \ldots, t^{q-1}, (t-t_1)_+^{q-1}, \ldots, (t-t_J)_+^{q-1}$ *are linearly independent, so the space* $\mathcal{S}_q(t_1, \ldots, t_J)$ *of such splines has dimension* $q + J$.

For computation, an equivalent, nearly-orthogonal basis is preferable to the truncated-power basis of (2.43): the B-spline basis, constructed via divided differences.

Definition 2.21. *The* q*th-order* divided difference *of* g *at* $t_i, \ldots, t_{i+q}$ *is defined recursively by* $[t_i]g := g(t_i)$ *and*

$$[t_i, \ldots, t_{i+q}]g := \frac{[t_{i+1}, \ldots, t_{i+q}]g - [t_i, \ldots, t_{i+q-1}]g}{t_{i+q} - t_i}. \tag{2.44}$$

Theorem 2.38. *Let* $p_q(x) := \sum_{i=1}^{q} g(t_i)\ell_i(x)$ *for* $\ell_i(x) := \prod_{j \neq i}(x - t_j)/(t_i - t_j)$. *Then: (1)* p_q *is the unique polynomial of order* q *agreeing with* g *at* $t_1, \ldots, t_q$*; (2) for* $q = 1, 2, \ldots,$ *the coefficient of* x^q *in* p_{q+1} *(the interpolant of* g *at* $t_i, \ldots, t_{i+q}$*) is* $[t_i, \ldots, t_{i+q}]g$.

Proof. (1) ℓ_i has order q and vanishes at every t_j, $j \neq i$, except $\ell_i(t_i) = 1$; uniqueness of interpolating polynomials of order q through q points is standard.

(2) By induction on q; trivial for $q = 1$. Let $p_q, \widetilde{p}_q$ interpolate g at $t_i, \ldots, t_{i+q-1}$ and $t_{i+1}, \ldots, t_{i+q}$ respectively. Then $p(x) := \frac{x - t_i}{t_{i+q} - t_i}\widetilde{p}_q(x) + \frac{t_{i+q} - x}{t_{i+q} - t_i} p_q(x)$ has order $q+1$ and agrees with g at $t_i, \ldots, t_{i+q}$ (direct check at each node), so $p_{q+1} = p$ by uniqueness, and the coefficient of x^q in p_{q+1} is $\{[t_{i+1}, \ldots, t_{i+q}]g - [t_i, \ldots, t_{i+q-1}]g\}/(t_{i+q} - t_i) = [t_i, \ldots, t_{i+q}]g$ by the induction hypothesis and (2.44). □

Corollary 2.2. *If* g *is a polynomial of order* q *on* $[t_i, t_{i+q}]$, *then* $[t_i, \ldots, t_{i+q}]g = 0$.

Proof. With p_l as in Theorem 2.38, $p_l = g$ for $l \geq q$ (both interpolate g, which already has order $\leq q \leq l$), so $[t_i, \ldots, t_{i+q}]g$, the coefficient of x^q in $p_{q+1} = g$, is 0 since g has degree $< q$. □

Introduce phantom knots $t_{-(q-1)} < \cdots < t_{-1} \le 0 = t_0$ and $1 = t_{J+1} \le t_{J+2} < \cdots < t_{J+q}$ (arbitrary, e.g. all equal to 0 or 1 respectively), and define, for $i = -(q-1), \ldots, J$, the *B-spline*

$$N_{iq}(t) := (t_{i+q} - t_i)[t_i, \ldots, t_{i+q}](\cdot - t)_+^{q-1}. \tag{2.45}$$

Theorem 2.39. *The B-splines $\{N_{iq}\}_{i=-(q-1)}^{J}$ satisfy: (1) N_{iq} is a polynomial of order q on each (t_i, t_{i+1}); (2) $N_{iq}(t) = 0$ for $t \notin [t_i, t_{i+q}]$ (local support); (3) with $N_{i1} := I_{[t_i, t_{i+1})}$, $N_{iq}(t) = \frac{t-t_i}{t_{i+q-1}-t_i} N_{i(q-1)}(t) + \frac{t_{i+q}-t}{t_{i+q}-t_{i+1}} N_{(i+1)(q-1)}(t)$.*

Proof. (1) is immediate from (2.45). For (2): $N_{iq}(t) = 0$ for $t > t_{i+q}$ since every term $(t_{i+r} - t)_+^{q-1}$, $r = 0, \ldots, q$, then vanishes. For $t < t_i$, every $(t_{i+r} - t)_+^{q-1} = (t_{i+r} - t)^{q-1}$, so $N_{iq}(t) = (t_{i+q} - t_i)[t_i, \ldots, t_{i+q}](\cdot - t)^{q-1}$, and since $(\cdot - t)^{q-1}$ is a polynomial of order q on $[t_i, t_{i+q}]$, this vanishes by Corollary 2.2. (Part (3), the recursion, is a direct algebraic identity relating divided differences at nested knot sets; we omit the routine verification, cf. [4], Theorem 6.6.5.) □

By the local support property, N_{iq} and N_{jq} have disjoint supports whenever $|i - j| > q-1$, so $\langle N_{iq}, N_{jq} \rangle_2 = 0$ in that case ("near orthogonality"); this, together with the fact that the N_{iq} therefore have full rank on $\operatorname{span}\{N_{iq}\}$, shows they form a basis for $\mathcal{S}_q(t_1, \ldots, t_J)$, and this is the basis `scipy.interpolate.UnivariateSpline` constructs internally for the B-spline fits of Section 4.2.1.

Definition 2.22. *A spline g of order $2q$ with knots $t_1 < \cdots < t_J$ is a* natural spline *(notation $\mathcal{N}^{2q}(t_1, \ldots, t_J)$) if g reduces to a polynomial of order q outside $[t_1, t_J]$; equivalently, $g^{(j)}(0) = g^{(j)}(1) = 0$ for $j = q, \ldots, 2q-1$ (the* natural boundary conditions*), which impose $2q$ linear constraints, giving $\dim \mathcal{N}^{2q}(t_1, \ldots, t_J) = J$.*

Theorem 2.40. *For any $0 \le t_1 < \cdots < t_J \le 1$ and $z_1, \ldots, z_J \in \mathbb{R}$, there is a unique natural spline $g \in \mathcal{N}^{2q}(t_1, \ldots, t_J)$ with $g(t_j) = z_j$ for all j.*

Proof. This is a linear system: the J-dimensional space $\mathcal{N}^{2q}(t_1, \ldots, t_J)$ (Definition 2.22) is mapped to $\mathbb{R}^J$ by evaluation at $t_1, \ldots, t_J$; existence and uniqueness of the interpolant is equivalent to this linear map being a bijection, which holds since a natural spline vanishing at all t_j must (being a polynomial of order $q \le J$ on each subinterval, pieced together with $q-2$ continuous derivatives, and vanishing together with its top q derivatives at the boundary) vanish identically – the standard, purely linear-algebraic verification for natural spline interpolation ([4], Theorem 6.6.7; de Boor, 1978). □

The following optimality property of the natural interpolating spline is the key structural fact that reduces the (infinite-dimensional) smoothing problem in $\mathbb{W}^q[0,1]$ to a finite-dimensional one.

Theorem 2.41. *Let $\widetilde{g} \in \mathcal{N}^{2q}(t_1,\dots,t_J)$ interpolate $z_1,\dots,z_J$ (with $0 < t_1 < \cdots < t_J < 1$). Then for every $g \in \mathbb{W}^q[0,1]$ that also interpolates the z_j,*

$$\int_0^1 |\widetilde{g}^{(q)}(t)|^2\,dt \le \int_0^1 |g^{(q)}(t)|^2\,dt, \tag{2.46}$$

with equality iff $g = \widetilde{g} + h$ for h a qth-order polynomial vanishing at $t_1,\dots,t_J$ (so $h = 0$ identically if $J \ge q$).

Proof. Let $h := g - \widetilde{g}$, so $h(t_j) = 0$ for all j. Repeated integration by parts, using the natural boundary conditions on $\widetilde{g}$ (which kill the boundary terms at $0, 1$) and that $\widetilde{g}^{(2q-1)}$ is piecewise constant (being the $(q-1)$-st derivative of the order-q piecewise polynomial $\widetilde{g}^{(q)}$... more precisely, constant on each (t_j, t_{j+1}) since $\widetilde{g}$ has order $2q$), gives

$$\begin{aligned}\int_0^1 \widetilde{g}^{(q)} h^{(q)}\,dt &= -\int_0^1 \widetilde{g}^{(q+1)} h^{(q-1)}\,dt = \cdots = (-1)^{q-1}\int_0^1 \widetilde{g}^{(2q-1)} h^{(1)}\,dt \\ &= (-1)^{q-1}\sum_{j=1}^{J-1} \widetilde{g}^{(2q-1)}(t_j^+)\,[h(t_{j+1}) - h(t_j)] = 0,\end{aligned} \tag{2.47}$$

using $h(t_j) = 0$ for every j in the last step and that $\widetilde{g}$ is a polynomial of order q (so $\widetilde{g}^{(2q-1)} = 0$) outside $[t_1, t_J]$. Hence

$$\int_0^1 |g^{(q)}|^2\,dt = \int_0^1 |\widetilde{g}^{(q)}|^2\,dt + 2\int_0^1 \widetilde{g}^{(q)} h^{(q)}\,dt + \int_0^1 |h^{(q)}|^2\,dt = \int_0^1 |\widetilde{g}^{(q)}|^2\,dt + \int_0^1 |h^{(q)}|^2\,dt \ge \int_0^1 |\widetilde{g}^{(q)}|^2\,dt,$$

proving (2.46). Equality holds iff $h^{(q)} = 0$ a.e., i.e. (since $h \in \mathbb{W}^q[0,1]$, via the Taylor representation of Section 2.2) iff h is a polynomial of order q; as $h(t_j) = 0$ for $j = 1,\dots,J$ and a nonzero polynomial of order q has at most $q-1$ roots, $h = 0$ once $J \ge q$. □

Theorem 2.42. *Let $n > q$, $\eta \in (0,\infty)$, and $L : \mathbb{R}^n \to [0,\infty)$. If*

$$L(g(t_1),\dots,g(t_n)) + \eta\int_0^1 |g^{(q)}(t)|^2\,dt \tag{2.48}$$

has a minimizer over $g \in \mathbb{W}^q[0,1]$, that minimizer lies in $\mathcal{N}^{2q}(t_1,\dots,t_n)$.

Proof. Let $\widehat{g}$ minimize (2.48) and let $\widetilde{g} \in \mathcal{N}^{2q}(t_1,\dots,t_n)$ interpolate $\widehat{g}$ at $t_1,\dots,t_n$ (Theorem 2.40). Then $L(\widetilde{g}(t_1),\dots) = L(\widehat{g}(t_1),\dots)$ by construction, while Theorem 2.41 gives $\int|\widetilde{g}^{(q)}|^2 \le \int|\widehat{g}^{(q)}|^2$, so $\widetilde{g}$ attains a value of (2.48) no larger than $\widehat{g}$'s; by minimality of $\widehat{g}$,

equality holds and $\widetilde{g}$ is also a minimizer, necessarily equal to $\widehat{g}$ once $n \geq q$ by the equality case of Theorem 2.41. $\square$

Theorem 2.42 converts the smoothing spline problem (2.34) (with $\mathbb{H} = \mathbb{W}^q[0,1]$, $\mathcal{T}g = (g(t_1), \ldots, g(t_n))^T$, and L the residual sum of squares) from an infinite-dimensional to an n-dimensional problem: writing $g_j \in \mathcal{N}^{2q}(t_1, \ldots, t_n)$ for the natural spline interpolating the jth unit vector at $t_1, \ldots, t_n$, every $g \in \mathcal{N}^{2q}(t_1, \ldots, t_n)$ is $g = \sum_j g(t_j) g_j$, and minimizing (2.34) reduces to minimizing $\|Y - g\|_{\mathbb{Y}}^2 + \eta g^T \mathcal{W} g$ over $g = (g(t_1), \ldots, g(t_n))^T \in \mathbb{R}^n$, for $\mathcal{W} = \{\int_0^1 g_i^{(q)}(t) g_j^{(q)}(t)\, dt\}_{i,j}$, with explicit solution

$$m_\eta = (m_\eta(t_1), \ldots, m_\eta(t_n))^T = (I + \eta \mathcal{W})^{-1} Y, \qquad m_\eta(\cdot) = \sum_{j=1}^{n} m_\eta(t_j) g_j(\cdot). \tag{2.49}$$

This natural spline of order $2q$ is the *smoothing spline*; for $q = 2$ it is the cubic smoothing spline of (4.2), which is exactly the B-spline smoothing estimator of Section 4.2.1 (with $\eta = \lambda$).

We close with explicit rates for the bias and variance of the smoothing spline, under equally spaced design points; these are the theoretical counterpart of the RMSE–sparsity tradeoff explored empirically in Section 5.2.2.

Theorem 2.43. *Assume $\mathbb{E}[\varepsilon\varepsilon^T] = \sigma^2 I$, $t_j = (2j-1)/2n$, and let $m = (m(t_1), \ldots, m(t_n))^T$. For every n and $\eta > 0$,*

$$\|\mathbb{E} m_\eta - m\|_{\mathbb{Y}}^2 \leq \frac{\eta}{4}\big(m^T m + m^T \mathcal{W} m\big), \qquad \mathbb{E}\|m_\eta - \mathbb{E} m_\eta\|_{\mathbb{Y}}^2 \leq \frac{q\sigma^2}{n} + \frac{C\sigma^2}{n} \eta^{-1/(2q)}, \tag{2.50}$$

for a constant C depending only on q.

Proof. Let $0 = \gamma_1 = \cdots = \gamma_q < \gamma_{q+1} \leq \cdots \leq \gamma_n$ be the eigenvalues of $\mathcal{W}$ (known to satisfy $C_1(j-q)^{2q} \leq \gamma_j \leq C_2(j-q)^{2q}$ for $j > q$ and constants C_1, C_2 depending only on q, from the asymptotic theory of natural-spline Green's functions; cf. [4], citing Utreras, 1983) with eigenvectors $e_1, \ldots, e_n$. From (2.49), $\mathbb{E}(m_\eta - m) = (I + \eta\mathcal{W})^{-1} m - m$, so

$$\|\mathbb{E}(m_\eta - m)\|_{\mathbb{Y}}^2 = n^{-1} \sum_j \left(\frac{\eta\gamma_j}{1+\eta\gamma_j}\right)^2 (m^T e_j)^2 = n^{-1} \sum_j (1+\gamma_j)(m^T e_j)^2 \cdot (1+\gamma_j)^{-1} \left(\frac{\eta\gamma_j}{1+\eta\gamma_j}\right)^2.$$

Since $\sum_j (1+\gamma_j)(m^T e_j)^2 = m^T m + m^T \mathcal{W} m$ and, substituting $u = \gamma_j$, $\sup_{u \geq 0} (1+u)^{-1} (\eta u/(1+\eta u))^2 = \eta \sup_{u \geq 0} \frac{u}{1+u} \cdot \frac{u}{(1+u)(\eta+u)} \leq \eta \sup_{u \geq 0} \frac{u}{(1+u)^2} = \eta/4$ (bounding the second factor by its value at the maximizer of $u/(1+u)^2$), the bias bound follows.

For the variance, $\mathbb{E}\|m_\eta - \mathbb{E}m_\eta\|_{\mathbb{Y}}^2 = n^{-1}\mathbb{E}[\varepsilon^T(I+\eta\mathcal{W})^{-2}\varepsilon] = \sigma^2 n^{-1}\sum_j(1+\eta\gamma_j)^{-2}$ (using $\mathbb{E}[\varepsilon\varepsilon^T] = \sigma^2 I$ and the eigendecomposition of $\mathcal{W}$). The first q terms (with $\gamma_j = 0$) contribute $q\sigma^2/n$; the remaining terms are bounded, using $\gamma_j \geq C_1(j-q)^{2q}$ and approximating the sum by the integral $\int_0^\infty (1 + C_1\eta x^{2q})^{-2}\,dx = C\eta^{-1/(2q)}$ (substituting $u = C_1\eta x^{2q}$) for a constant $C = C(q)$, giving the stated rate. □

Theorem 2.43 makes explicit the bias-variance tradeoff invoked qualitatively in Section 4.2.1: the bias term grows linearly in η while the variance term decays like $\eta^{-1/(2q)}$, so that the mean-squared error $\text{Bias}^2(\eta) + \text{Var}(\eta)$ is minimized at an intermediate $\eta = O(n^{2q/(2q+1)})$ – the same $n^{-2q/(2q+1)}$ rate that appears in the classical nonparametric regression literature, and that motivates choosing λ proportional to the estimated noise level $\hat{\sigma}^2$ (rather than by a fixed rate in n) in the country-adaptive scheme of Equation (4.3).

2.6 Random Elements in a Hilbert Space

This section lays the probabilistic foundations of FDA: what it means for a Hilbert-space-valued object to be a random element, its mean and covariance operator, the Karhunen–Loève Theorem (the population-level statement of FPCA), and the large-sample theory (strong law of large numbers and central limit theorem) that Section 2.7 will use to study estimators of these quantities. We follow [4], Chapter 7, restricting to the material needed for the FPCA pipeline (the random-element and mean-square-continuous-process perspectives, and their large-sample theory); the material on reproducing-kernel-valued processes and canonical-correlation-oriented congruence relations in that source's Sections 7.5–7.6 is not needed here and is omitted.

2.6.1 Probability Measures on a Hilbert Space

Throughout, $\mathbb{H}$ is a separable Hilbert space with its norm-induced metric $d(f,g) = \|f-g\|$, and $\mathcal{B}(\mathbb{H})$ denotes its Borel σ-field (the smallest σ-field containing all norm-open subsets of $\mathbb{H}$) – an unfortunate notational clash with the space of bounded operators, resolved by context throughout. Let $\mathcal{M}$ denote the class of sets $\{x \in \mathbb{H} : \langle x, f\rangle \in B\}$ for $f \in \mathbb{H}$ and $B \subset \mathbb{R}$ open.

Theorem 2.44. $\sigma(\mathcal{M}) = \mathcal{B}(\mathbb{H})$.

Proof. Each set in $\mathcal{M}$ is open or closed (continuity of $x \mapsto \langle x, f\rangle$), so $\mathcal{M} \subset \mathcal{B}(\mathbb{H})$ and $\sigma(\mathcal{M}) \subset \mathcal{B}(\mathbb{H})$. For the reverse inclusion, it suffices to show every set $\{x : \|x - f\| > r\}$, $f \in \mathbb{H}$, $r > 0$, is in $\sigma(\mathcal{M})$. Fix a CONS $\{e_j\}$ (Theorem 2.2). Since $\|x\|^2 = \sum_k \langle x, e_k\rangle^2$,

$$\{x : \|x\| > r\} = \bigcup_{j=1}^{\infty} \Big\{x : \sum_{k=1}^{j} \langle x, e_k\rangle^2 > r^2\Big\}. \tag{2.51}$$

For $j = 2$: $\{x : \langle x, e_1\rangle^2 + \langle x, e_2\rangle^2 > r^2\} = \bigcup_{q\in\mathbb{Q}} \big(\{x : \langle x, e_1\rangle^2 > q\} \cap \{x : \langle x, e_2\rangle^2 > r^2 - q\}\big) \in \sigma(\mathcal{M})$, since both sets on the right are in $\mathcal{M}$; induction using the same device (splitting $\sum_{k\le j}\langle x, e_k\rangle^2 > r^2$ into the first coordinate versus the rest, over rational thresholds) extends this to every j, so (2.51) gives $\{x : \|x\| > r\} \in \sigma(\mathcal{M})$. For general f, $\{x : \|x - f\| > r\} = \{x : \|x\|^2 > r^2 + 2\langle x, f\rangle - \|f\|^2\} = \bigcup_{q\in\mathbb{Q}} \big(\{x : \|x\|^2 > q\} \cap \{x : 2\langle x, f\rangle < q - r^2 + \|f\|^2\}\big)$, a countable union of sets already shown (or immediately seen, for the second) to lie in $\sigma(\mathcal{M})$, so $\{x : \|x - f\| > r\} \in \sigma(\mathcal{M})$ and $\mathcal{B}(\mathbb{H}) \subset \sigma(\mathcal{M})$. □

Theorem 2.45. *Let $\chi : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{H}, \mathcal{B}(\mathbb{H}))$. Then: (1) χ is measurable if $\langle \chi, f\rangle$ is measurable for every $f \in \mathbb{H}$; (2) if χ is measurable, its distribution is uniquely determined by the marginal distributions of $\langle \chi, f\rangle$, $f \in \mathbb{H}$.*

Proof. (1) If $\langle \chi, f\rangle$ is measurable for all f, then for any $\{x : \langle x, f\rangle \in B\} \in \mathcal{M}$ (B open), $\chi^{-1}(\{x : \langle x, f\rangle \in B\}) = \{\omega : \langle \chi(\omega), f\rangle \in B\} \in \mathcal{F}$; Theorem 2.44 then gives measurability of χ (the converse direction is immediate from continuity of $x \mapsto \langle x, f\rangle$).

(2) Let $\check{\mathcal{M}}$ be the class of finite intersections of sets in $\mathcal{M}$; $\sigma(\check{\mathcal{M}}) = \sigma(\mathcal{M}) = \mathcal{B}(\mathbb{H})$. If probability measures μ_1, μ_2 agree on $\check{\mathcal{M}}$, then $\{A \in \mathcal{B}(\mathbb{H}) : \mu_1(A) = \mu_2(A)\}$ is a λ-system containing the π-system $\check{\mathcal{M}}$, so by the π-λ theorem it contains $\sigma(\check{\mathcal{M}}) = \mathcal{B}(\mathbb{H})$, i.e. $\check{\mathcal{M}}$ is a determining class. Since $\mathbb{P} \circ \chi^{-1}(\cap_{i=1}^{n}\{x : \langle x, f_i\rangle \in B_i\}) = \mathbb{P}(\langle \chi, f_i\rangle \in B_i,\ 1 \le i \le n)$, and the Cramér–Wold device determines a joint distribution in $\mathbb{R}^n$ from its one-dimensional projections, the distribution of χ is determined by the marginal distributions of $\langle \chi, f\rangle$, $f \in \mathbb{H}$. □

A measurable $\chi : \Omega \to \mathbb{H}$ is called a *random element* of $\mathbb{H}$; this is the natural infinite-dimensional generalization of a random vector, and is the object FPCA operates on – in Section 4's application, χ represents a country's (centered) GDP growth trajectory viewed as an element of $\mathbb{L}^2(\mathcal{T})$.

2.6.2 Mean and Covariance Operator of a Random Element

Definition 2.23. *If $\mathbb{E}\|\chi\| < \infty$, the* mean element *of a random element χ of $\mathbb{H}$ is the Bochner integral $m = \mathbb{E}(\chi) := \int_\Omega \chi \, d\mathbb{P}$ (well defined by Theorem 2.46 below); equivalently,*

since $f \mapsto \mathbb{E}[\langle \chi, f\rangle]$ is then a bounded linear functional, m is its representer (Theorem 2.4's Riesv-type argument, i.e. Theorem 3.2.1 of [4]), so that in either case

$$\langle m, f\rangle = \mathbb{E}[\langle \chi, f\rangle], \qquad f \in \mathbb{H}. \tag{2.52}$$

Theorem 2.46. *If $\mathbb{E}\|\chi\| < \infty$ for a random element χ of the separable Hilbert space $\mathbb{H}$, χ is Bochner integrable.*

This is a direct consequence of $\mathbb{H}$ being separable: taking $\mathbb{H}_n = \operatorname{span}\{e_1, \ldots, e_n\}$ for a CONS $\{e_j\}$ and g_n the projection of χ onto $\mathbb{H}_n$ satisfies the approximation condition for Bochner integrability by dominated convergence; we omit the routine verification ([4], Theorem 2.6.5).

Theorem 2.47. *If $\mathbb{E}\|\chi\|^2 < \infty$, $\mathbb{E}\|\chi - m\|^2 = \mathbb{E}\|\chi\|^2 - \|m\|^2$.*

Proof. Expand $\mathbb{E}\|\chi - m\|^2 = \mathbb{E}\|\chi\|^2 - 2\mathbb{E}\langle \chi, m\rangle + \|m\|^2$; by (2.52) with $f = m$, $\mathbb{E}\langle \chi, m\rangle = \langle m, m\rangle = \|m\|^2$. □

Definition 2.24. *If $\mathbb{E}\|\chi\|^2 < \infty$, the* covariance operator *of χ is the Bochner integral (in $\mathcal{B}_{HS}(\mathbb{H})$, well defined by Theorem 2.46 applied to $\mathcal{B}_{HS}(\mathbb{H})$ since $\|(\chi - m)\otimes(\chi - m)\|_{HS} = \|\chi - m\|^2$ has finite expectation by Theorem 2.47)*

$$\Gamma := \mathbb{E}[(\chi - m)\otimes(\chi - m)] := \int_\Omega (\chi - m)\otimes(\chi - m)\, d\mathbb{P}. \tag{2.53}$$

Theorem 2.48. $\mathbb{E}[(\chi - m)\otimes(\chi - m)] = \mathbb{E}(\chi\otimes\chi) - m\otimes m$.

Proof. $\chi\otimes m$, $m\otimes\chi$, $m\otimes m$ are all HS with $m\otimes m$ constant and $\mathbb{E}\|\chi\otimes m\|_{HS} = \mathbb{E}\|m\otimes\chi\|_{HS} = \|m\|\mathbb{E}\|\chi\| < \infty$. Since $(\mathbb{E}(m\otimes\chi))f = m\langle \mathbb{E}\chi, f\rangle$ (Theorem 2.46's linearity property, applied to the HS-valued Bochner integral) equals $\langle m, f\rangle m = (m\otimes m)f$ by (2.52), and likewise for $\mathbb{E}(\chi\otimes m)$, expanding $(\chi - m)\otimes(\chi - m) = \chi\otimes\chi - \chi\otimes m - m\otimes\chi + m\otimes m$ and taking expectations termwise gives the identity. □

We henceforth take $m = 0$ without loss of generality (replacing χ by $\chi - m$), so $\Gamma = \mathbb{E}(\chi\otimes\chi)$.

Theorem 2.49. *Suppose $m = 0$ and $\mathbb{E}\|\chi\|^2 < \infty$. For $f, g \in \mathbb{H}$: (1) $\langle \Gamma f, g\rangle = \mathbb{E}[\langle \chi, f\rangle\langle \chi, g\rangle]$; (2) Γ is nonnegative definite and trace class with $\|\Gamma\|_{TR} = \mathbb{E}\|\chi\|^2$; (3) $\mathbb{P}(\chi \in \operatorname{Im}(\Gamma)) = 1$.*

Proof. (1) By (2.53) and linearity of the Bochner integral (Theorem 2.46's linearity property applied to the bounded linear functional $\mathcal{T}f := \langle f, g\rangle$ on $\mathcal{B}_{HS}(\mathbb{H})$), $\Gamma f = \int_\Omega \chi\langle \chi, f\rangle\, d\mathbb{P}$, so $\langle \Gamma f, g\rangle = \mathbb{E}[\langle \chi, f\rangle\langle \chi, g\rangle]$.

(2) Nonnegative definiteness is immediate from (1) with $f = g$. For the trace: for any CONS $\{e_j\}$, $\|\Gamma\|_{TR} = \sum_j \langle \Gamma e_j, e_j\rangle = \sum_j \mathbb{E}\langle \chi, e_j\rangle^2 = \mathbb{E}\|\chi\|^2 < \infty$ (Parseval, Theorem 2.2), so Γ is trace class.

(3) By Theorem **??**(1) with self-adjointness of Γ, $(\mathrm{Im}(\Gamma))^\perp = \mathrm{Ker}(\Gamma)$. For $f \in (\mathrm{Im}(\Gamma))^\perp$, $\mathbb{E}[\langle \chi, f\rangle^2] = \langle \Gamma f, f\rangle = 0$ by part (1), so $\langle \chi, f\rangle = 0$ almost surely. Thus, with probability one, $\chi \perp (\mathrm{Im}(\Gamma))^\perp$, i.e. $\chi \in ((\mathrm{Im}(\Gamma))^\perp)^\perp = \mathrm{Im}(\Gamma)$ (closure, by Theorem 2.3). □

By Theorem 2.10 (applicable since, by Theorem 2.49(2), Γ is trace class, hence Hilbert–Schmidt, hence compact, and self-adjoint), Γ admits the eigendecomposition

$$\Gamma = \sum_{j\geq 1} \lambda_j e_j \otimes e_j, \tag{2.54}$$

with $\{e_j\}$ a CONS for $\mathrm{Im}(\Gamma)$, $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$ either finite in number or tending to 0, and each nonzero λ_j of finite multiplicity – extending the spectral decomposition of a covariance matrix to the functional setting.

Corollary 2.3 (Population Karhunen–Loève expansion)**.** *With probability one,* $\chi = \sum_{j\geq 1}\langle \chi, e_j\rangle e_j$, *where* $\{\langle \chi, e_j\rangle\}_{j\geq 1}$ *are uncorrelated, mean-zero random variables with* $\mathrm{Var}(\langle \chi, e_j\rangle) = \lambda_j$.

Proof. By Theorem 2.49(3), $\chi \in \mathrm{Im}(\Gamma) \subset \overline{\mathrm{span}}\{e_j\}$ almost surely; the pathwise Fourier expansion of Theorem 2.2 then gives $\chi = \sum_j \langle \chi, e_j\rangle e_j$ a.s. Uncorrelatedness and the stated variances follow from Theorem 2.49(1) applied to $f = e_i, g = e_j$: $\mathbb{E}[\langle \chi, e_i\rangle\langle \chi, e_j\rangle] = \langle \Gamma e_i, e_j\rangle = \lambda_j \delta_{ij}$. □

Corollary 2.3 *is* FPCA at the population level: the $\langle \chi, e_j\rangle$ are the population FPC scores, and the e_j the population FPCs. The following result gives the precise sense in which truncating this expansion at the leading n eigenfunctions is optimal – the population counterpart of the variance-explained criterion of Section 4.3.

Theorem 2.50. *For any CONS* $\{f_j\}$ *of* $\mathbb{H}$,

$$\mathbb{E}\left\|\chi - \sum_{j=1}^{n}\langle \chi, f_j\rangle f_j\right\|^2 = \mathbb{E}\|\chi\|^2 - \sum_{j=1}^{n}\langle \Gamma f_j, f_j\rangle, \tag{2.55}$$

which is minimized, over choices of orthonormal $f_1, \ldots, f_n$, *by* $f_j = e_j$, $1 \leq j \leq n$.

Proof. Expanding the square and using $\mathbb{E}\|\sum_{j\leq n}\langle \chi, f_j\rangle f_j\|^2 = \sum_{j\leq n}\mathbb{E}\langle \chi, f_j\rangle^2 = \sum_{j\leq n}\langle \Gamma f_j, f_j\rangle$ (Theorem 2.49(1)) and $\mathbb{E}\langle \chi, \sum_{j\leq n}\langle \chi, f_j\rangle f_j\rangle = \sum_{j\leq n}\mathbb{E}\langle \chi, f_j\rangle^2$ gives (2.55) directly. Since $\sum_{j=1}^n \langle \Gamma f_j, f_j\rangle \leq \sum_{j=1}^n \lambda_j$ for any orthonormal $f_1, \ldots, f_n$, with equality at $f_j = e_j$ (Theorem

2.11/Theorem 2.12, applied successively as in the proof of Theorem 2.21), the right side of (2.55) is minimized at $f_j = e_j$. □

2.6.3 Mean-Square Continuous Processes and the Karhunen–Loève Theorem

We now turn to the complementary, stochastic-process perspective: $X = \{X(t) : t \in E\}$ on $(\Omega, \mathcal{F}, \mathbb{P})$, for a compact metric space E (e.g. $E = \mathcal{T}$, the rescaled observation window), with mean function $m(t) := \mathbb{E}[X(t)]$ and covariance function $\gamma(s,t) := \mathrm{Cov}(X(s), X(t))$, provided these exist (a *second-order process*).

Theorem 2.51. *The covariance function γ of a second-order process is nonnegative definite.*

Proof. $\sum_{i,j} a_i a_j \gamma(t_i, t_j) = \mathrm{Var}\Big(\sum_j a_j X(t_j)\Big) \geq 0.$ □

By Theorem 2.51 and the Moore–Aronszajn Theorem (Definition 2.7), γ generates an RKHS $\mathbb{H}(\gamma)$ for which it is the reproducing kernel.

Definition 2.25. *A second-order process X is* mean-square continuous *if* $\lim_{n\to\infty} \mathbb{E}[X(t_n) - X(t)]^2 = 0$ *whenever $t_n \to t$ in E.*

Theorem 2.52. *A second-order process X is mean-square continuous if and only if m and γ are both continuous.*

Proof. Since $\mathbb{E}[X(s)-X(t)]^2 = \gamma(s,s)+\gamma(t,t)-2\gamma(s,t)+(m(s)-m(t))^2$, continuity of m, γ gives mean-square continuity. Conversely, mean-square continuity gives $|m(s) - m(t)| \leq \mathbb{E}[X(s)-X(t)]^{21/2} \to 0$ (continuity of m); assuming WLOG $m \equiv 0$, write $\gamma(s,t)-\gamma(s',t') = (\gamma(s,t) - \gamma(s',t)) + (\gamma(s',t) - \gamma(s',t'))$, and bound each difference by Cauchy–Schwarz: $|\gamma(s,t) - \gamma(s',t)| \leq \gamma(t,t)^{1/2}\mathbb{E}[X(s) - X(s')]^{21/2}$ and similarly for the second term, both of which $\to 0$ as $(s,t) \to (s',t')$ by mean-square continuity. □

For a mean-square continuous, mean-zero process, fix a finite measure μ on $(E, \mathcal{B}(E))$; the associated *covariance operator* $\mathcal{K}$ is the integral operator of Definition 2.15 with kernel γ, and by Mercer's Theorem (Theorem 2.26),

$$\gamma(s,t) = \sum_{j\geq 1} \lambda_j e_j(s) e_j(t), \qquad s,t \in E, \tag{2.56}$$

absolutely and uniformly convergent, with (λ_j, e_j) the eigenpairs of $\mathcal{K}$.

We next construct the $\mathbb{L}^2$ stochastic integral of X against $f \in \mathbb{L}^2(E, \mathcal{B}(E), \mu)$, which is what makes precise the otherwise formal expression $\int_E X(t)f(t)\,d\mu(t)$. For a partition $\{E_i\}_{i=1}^{m(n)}$ of E (with each $E_i \in \mathcal{B}(E)$ of diameter $< 1/n$, which exists by total boundedness of E) and arbitrary $t_i \in E_i$, let $I_X(f; \{E_i, t_i\}) := \sum_i X(t_i) \int_{E_i} f\,d\mu$. For two such partitions with parameters n, n', direct computation of the squared $\mathbb{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ difference between the two Riemann-type sums, expanded via $\mathbb{E}[X(t_i)X(t'_j)] = \gamma(t_i, t'_j)$, expresses it as a difference of double sums that (by uniform continuity of γ, on the compact $E \times E$) converges, as $n, n' \to \infty$, to 0 once each double sum is recognized as a Riemann-type approximation to $\iint_{E\times E} \gamma\, f\, f\, d\mu\, d\mu$ minus twice a cross term converging to the same double integral; this shows $\{I_X(f; \{E_i, t_i\})\}$ is Cauchy in $\mathbb{L}^2(\Omega, \mathcal{F}, \mathbb{P})$ and hence, by completeness (the Riesz–Fischer Theorem, Definition 2.4), converges to a limit $I_X(f) \in \mathbb{L}^2(\Omega, \mathcal{F}, \mathbb{P})$, independent of the approximating sequence of partitions.

Theorem 2.53. *Let X be mean-square continuous with mean zero. For $f, g \in \mathbb{L}^2(E, \mathcal{B}(E), \mu)$: (1) $\mathbb{E}[I_X(f)] = 0$; (2) $\mathbb{E}[I_X(f)X(t)] = \int_E \gamma(u, t)f(u)\,d\mu(u)$; (3) $\mathbb{E}[I_X(f)I_X(g)] = \iint_{E\times E} \gamma(u, v)f(u)g(v)$*

Proof. Each identity holds for the approximating Riemann-type sums by direct computation (finite linearity and $\mathbb{E}[X(t_i)X(t_j)] = \gamma(t_i, t_j)$), and passes to the limit $I_X(f)$ since $|\mathbb{E}[I_X(f)] - \mathbb{E}[I_X(f; \{E_i, t_i\})]| \le \mathbb{E}[(I_X(f) - I_X(f; \{E_i, t_i\}))^2]^{1/2} \to 0$ by construction of $I_X(f)$ as the $\mathbb{L}^2$ limit, and similarly (via Cauchy–Schwarz) for the other two identities. □

Corollary 2.4. $\mathbb{E}[I_X(e_i)I_X(e_j)] = \lambda_i \delta_{ij}$.

Proof. By Theorem 2.53(3) and (2.56) (Mercer expansion of γ), $\mathbb{E}[I_X(e_i)I_X(e_j)] = \iint \gamma(u, v)e_i(u)e_j(v)\,d$ $\langle \mathcal{K}e_j, e_i\rangle = \lambda_j \delta_{ij}$. □

Theorem 2.54 (Karhunen–Loève Theorem)**.** *Let X be mean-square continuous with mean zero. Then, with $X_n(t) := \sum_{j=1}^n I_X(e_j)e_j(t)$,*

$$\lim_{n\to\infty} \sup_{t\in E} \mathbb{E}[X(t) - X_n(t)]^2 = 0. \tag{2.57}$$

Proof. By Theorem 2.53(2) (with $f = e_j$) and Corollary 2.4,

$$\mathbb{E}[X_n(t)-X(t)]^2 = \mathbb{E}[X_n(t)]^2 - 2\mathbb{E}[X_n(t)X(t)] + \mathbb{E}[X(t)]^2 = \sum_{j=1}^n \lambda_j e_j^2(t) - 2\sum_{j=1}^n \lambda_j e_j^2(t) + \gamma(t,t) = \gamma(t,t) - \sum_{j=}$$

which tends to 0 uniformly in t by the uniform convergence in Mercer's Theorem ((2.56), Theorem 2.26). □

The $I_X(e_j)$ are the *scores* of X, X_n the n-term Karhunen–Loève expansion, and Theorem 2.54 states that this expansion converges to X uniformly in mean-square error – the exact process-perspective analog of Corollary 2.3, and the theoretical statement that the empirical FPCA of Chapter 5 approximates.

Theorem 2.55. *If X is jointly measurable (as a function of (t,ω), w.r.t. $\mathcal{B}(E)\times\mathcal{F}$) and $X(\cdot,\omega)\in\mathbb{H}:=\mathbb{L}^2(E,\mathcal{B}(E),\mu)$ for every ω, then $\omega\mapsto X(\cdot,\omega)$ is a random element of $\mathbb{H}$.*

Proof. Joint measurability implies, by the standard proof of Fubini's Theorem, that $\langle X(\cdot,\omega),f\rangle = \int_E X(t,\omega)f(t)\,d\mu(t)$ is $\mathcal{F}$-measurable in ω for every $f\in\mathbb{H}$; apply Theorem 2.45(1). □

Theorem 2.55 is what licenses treating a GDP growth trajectory – observed, and mean-square continuous, as a stochastic process $X(t)$ over the discretely sampled time domain – as a single random element $\chi = X(\cdot)$ of $\mathbb{L}^2(\mathcal{T})$ for the purposes of Sections 2.6.2 and 2.8: the two perspectives of this section coincide, and the covariance operator (2.53) of χ agrees with the integral operator (2.14) built from γ.

2.6.4 Large Sample Theory

We conclude with the strong law of large numbers and central limit theorem for i.i.d. random elements of a Hilbert space, which Section 2.7 applies to the sample mean and sample covariance operator.

Theorem 2.56. *If $\chi_1,\dots,\chi_n$ are pairwise independent, mean-zero random elements of $\mathbb{H}$ and $S_n:=\sum_{i=1}^n\chi_i$, then $\mathbb{E}\|S_n\|^2=\sum_{i=1}^n\mathbb{E}\|\chi_i\|^2$.*

Proof. Fix a CONS $\{e_k\}$. Pairwise independence gives $\mathbb{E}[\langle\chi_i,e_k\rangle\langle\chi_j,e_k\rangle]=\langle\mathbb{E}\chi_i,e_k\rangle\langle\mathbb{E}\chi_j,e_k\rangle=0$ for $i\neq j$. By Parseval (Theorem 2.2) and Tonelli's theorem (nonnegative summands), $\mathbb{E}\|S_n\|^2=\sum_k\mathbb{E}\langle S_n,e_k\rangle^2=\sum_k\sum_{i=1}^n\mathbb{E}\langle\chi_i,e_k\rangle^2=\sum_{i=1}^n\sum_k\mathbb{E}\langle\chi_i,e_k\rangle^2=\sum_{i=1}^n\mathbb{E}\|\chi_i\|^2$. □

Theorem 2.57 (Strong Law of Large Numbers)**.** *If $\chi_1,\chi_2,\dots$ are pairwise independent and identically distributed random elements of $\mathbb{H}$ with $\mathbb{E}\|\chi_1\|<\infty$, then $n^{-1}S_n\to\mathbb{E}(\chi_1)$ almost surely.*

Proof sketch. Following the truncation argument of Etemadi (1983), set $\chi_i':=\chi_i I_{\{\|\chi_i\|\le i\}}$, $S_n':=\sum_{i=1}^n\chi_i'$, and $k_n:=\lfloor\alpha^n\rfloor$ for fixed $\alpha>1$. Applying Theorem 2.47, Theorem 2.56, and Markov's inequality along the geometric subsequence $\{k_n\}$, together with the elementary bound $\sum_n k_n^{-2}I_{\{k_n\ge i\}}\le 4(1-\alpha^{-2})^{-1}i^{-2}$, gives $\sum_n\mathbb{P}(\|S_{k_n}'-\mathbb{E}S_{k_n}'\|/k_n>\epsilon)\le 4(1-\alpha^{-2})^{-1}\epsilon^{-2}\sum_i\mathbb{E}\|\chi_i'\|^2i^{-2}$, and this last sum is finite (bounded by a constant multiple

of $\mathbb{E}\|\chi_1\|$, via a layer-cake / dyadic decomposition of $\mathbb{E}\|\chi_1\|^2 I_{\{j\le\|\chi_1\|<j+1\}}$ summed against $\sum_{i>j} i^{-2} \asymp j^{-1}$); the Borel–Cantelli Lemma then gives $(S'_{k_n} - \mathbb{E}S'_{k_n})/k_n \to 0$ a.s. Dominated convergence gives $\mathbb{E}\chi'_n \to \mathbb{E}\chi_1$, hence (Cesàro) $\mathbb{E}S'_{k_n}/k_n \to \mathbb{E}\chi_1$, so $S'_{k_n}/k_n \to \mathbb{E}\chi_1$ a.s.; a further Borel–Cantelli argument shows $\chi'_i = \chi_i$ eventually a.s., upgrading this to $S_{k_n}/k_n \to \mathbb{E}\chi_1$ a.s. Finally, sandwiching a general n between consecutive $k_{m(n)-1} < n \le k_{m(n)}$ and bounding $\|S_n/n - S_{k_{m(n)}}/k_{m(n)}\|$ using the scalar strong law of large numbers applied to $\|\chi_i\|$ (real-valued) shows $\limsup_n \|S_n/n - S_{k_{m(n)}}/k_{m(n)}\| \le 2(\alpha-1)\mathbb{E}\|\chi_1\|$ a.s.; letting $\alpha \downarrow 1$ completes the proof. (Full details: [4], Theorem 7.7.2, adapting Etemadi, 1983.) □

For the central limit theorem, we need a suitable notion of weak convergence on $\mathbb{H}$: $\chi_n \xrightarrow{d} \chi$ if $\mathbb{E}[f(\chi_n)] \to \mathbb{E}[f(\chi)]$ for every bounded continuous $f : \mathbb{H} \to \mathbb{R}$. The key technical tool is a characterization of tightness adapted to Hilbert space.

Definition 2.26. *A family $\{\mu_\alpha\}_{\alpha\in I}$ of probability measures on $(\mathbb{H}, \mathcal{B}(\mathbb{H}))$ is* tight *if for every $\epsilon > 0$ there is a compact $W \subset \mathbb{H}$ with $\inf_\alpha \mu_\alpha(W) \ge 1-\epsilon$. For $S \subset \mathbb{H}$, $\epsilon > 0$, let $S^\epsilon := \{x \in \mathbb{H} : \inf_{z\in S} \|x - z\| \le \epsilon\}$.*

Theorem 2.58. *Let $\{\mu_\alpha\}_{\alpha\in I}$ be probability measures on $(\mathbb{H}, \mathcal{B}(\mathbb{H}))$. Suppose that for every $\epsilon, \delta > 0$ there is a finite $\{y_1, \dots, y_k\} \subset \mathbb{H}$ such that, with $S := \operatorname{span}\{y_1, \dots, y_k\}$: (1) $\inf_\alpha \mu_\alpha(S^\epsilon) \ge 1-\delta$; (2) $\inf_\alpha \mu_\alpha(\{x : |\langle x, y_j\rangle| \le r,\ j = 1, \dots, k\}) \ge 1-\delta$ for some $r > 0$. Then $\{\mu_\alpha\}$ is tight.*

Proof. Fix $\epsilon > 0$ and, applying the hypothesis with this $\epsilon = \delta$, obtain orthonormal (WLOG) $y_1, \dots, y_k$, $S = \operatorname{span}\{y_j\}$, and r satisfying (1)–(2); set $A_\epsilon := S^\epsilon \cap \{x : |\langle x, y_j\rangle| \le r,\ j \le k\}$, so $\inf_\alpha \mu_\alpha(A_\epsilon) \ge 1 - 2\epsilon$.

We show $A_\epsilon \subset W_\epsilon^\epsilon$ for a compact W_ϵ. For $x \in \mathbb{H}$ write $x = x_1 + x_2$ with x_1 the projection onto S. For $x \in A_\epsilon$: $\|x_2\| \le \epsilon$ (definition of S^ϵ), so for any $z = z_1 + z_2 \in \mathbb{H}$, $|\langle x_2, z_2\rangle| \le \epsilon\|z\|$; and $|\langle x_1, z_1\rangle| = |\sum_{j\le k}\langle x, y_j\rangle\langle z, y_j\rangle| \le kr\|z\|$ (Cauchy–Schwarz on the k-vector of coordinates, using $|\langle x, y_j\rangle| \le r$). Hence $\|x\| = \sup_{\|z\|\le 1} |\langle x, z\rangle| \le \epsilon + kr =: \nu$, so $A_\epsilon \subset B_\nu \cap S^\epsilon$ for $B_\nu := \{x : \|x\| \le \nu\}$.

If $x \in B_\nu \cap S^\epsilon$, there is $y_j \in S$ (finite-dimensional, so realizing the infimum, or within ϵ of it) with $\|x - y_j\| \le \epsilon$, so $\|y_j\| \le \nu + \epsilon$, giving $\inf_{z\in S\cap B_{\nu+\epsilon}} \|x - z\| \le \epsilon$; thus $B_\nu \cap S^\epsilon \subset (S \cap B_{\nu+\epsilon})^\epsilon =: W_\epsilon^\epsilon$ for the compact (finite-dimensional, closed, bounded) set $W_\epsilon := S \cap B_{\nu+\epsilon}$. So $A_\epsilon \subset W_\epsilon^\epsilon$ and $\inf_\alpha \mu_\alpha(W_\epsilon^\epsilon) \ge 1 - 2\epsilon$.

Repeating this construction at levels $\epsilon/2^j$, $j \ge 2$, and setting $W := \cap_{j\ge 2} W_{\epsilon/2^j}^{\epsilon/2^j}$ gives $\inf_\alpha \mu_\alpha(W) \ge 1 - \epsilon$ (union bound over the tail probabilities $2 \cdot \epsilon/2^j$). It remains to show W is compact: W is closed, and totally bounded because, writing $W = \cap_j (W_{\epsilon/2^j} \cap W) \cup$

(negligible remainder), each $W_{\epsilon/2^j} \cap W$ is compact (closed subset of the compact $W_{\epsilon/2^j}$), so for any $\nu' > 0$, choosing j with $\epsilon/2^j < \nu'/2$ and a finite $\nu'/2$-net F of $W_{\epsilon/2^j} \cap W$ gives $W \subset F^{\nu'}$ (any $x \in W$ is within $\epsilon/2^j < \nu'/2$ of $W_{\epsilon/2^j} \cap W$, which is within $\nu'/2$ of F). By the Heine–Borel Theorem (a complete, totally bounded metric space is compact; cf. Theorem 2.1.17 of [4]), W is compact. □

Theorem 2.59. *Let χ, χ_n be random elements of $\mathbb{H}$ with $\langle \chi_n, f\rangle \xrightarrow{d} \langle \chi, f\rangle$ in $\mathbb{R}$ for every $f \in \mathbb{H}$, and suppose that for every $\epsilon, \delta > 0$ there is a finite-dimensional S with $\inf_{n\geq 1} \mathbb{P}(\chi_n \in S^\epsilon) \geq 1 - \delta$. Then $\chi_n \xrightarrow{d} \chi$.*

Proof. The hypothesis verifies Theorem 2.58's conditions for $\{\mathbb{P}\circ\chi_n^{-1}\}$ (condition (2) follows from convergence in distribution of each $\langle \chi_n, y_j\rangle$, which is in particular bounded in probability), so this family is tight, hence (Prohorov's Theorem) relatively compact in the weak topology. Let $\mathbb{P}\circ\chi_{n'}^{-1} \xrightarrow{w} \nu_1$ and $\mathbb{P}\circ\chi_{n''}^{-1} \xrightarrow{w} \nu_2$ be two weakly convergent subsequences, with limiting random elements $\widetilde{\chi}, \check{\chi}$ (i.e. $\chi_{n'} \xrightarrow{d} \widetilde{\chi}$, $\chi_{n''} \xrightarrow{d} \check{\chi}$). By the continuous mapping theorem applied to $x \mapsto \langle x, f\rangle$, $\langle \chi_{n'}, f\rangle \xrightarrow{d} \langle \widetilde{\chi}, f\rangle$ and $\langle \chi_{n''}, f\rangle \xrightarrow{d} \langle \check{\chi}, f\rangle$; but both subsequences of $\langle \chi_n, f\rangle$ converge in distribution to $\langle \chi, f\rangle$ by hypothesis, so $\langle \widetilde{\chi}, f\rangle \overset{d}{=} \langle \check{\chi}, f\rangle \overset{d}{=} \langle \chi, f\rangle$ for every f; by Theorem 2.45(2), $\widetilde{\chi} \overset{d}{=} \check{\chi} \overset{d}{=} \chi$. As every weakly convergent subsequence of $\{\mathbb{P}\circ\chi_n^{-1}\}$ has the same limit $\mathbb{P}\circ\chi^{-1}$, the whole sequence converges weakly to it. □

Theorem 2.60 (Central Limit Theorem)**.** *Let $\chi_1, \chi_2, \ldots$ be i.i.d. random elements of $\mathbb{H}$ with mean 0 and $\mathbb{E}\|\chi_1\|^2 < \infty$. Then*

$$\xi_n := n^{-1/2} S_n \xrightarrow{d} \xi, \tag{2.58}$$

where ξ is a Gaussian random element of $\mathbb{H}$ with covariance operator $\mathbb{E}(\chi_1 \otimes \chi_1)$.

Proof. We verify the hypotheses of Theorem 2.59. For any $f \in \mathbb{H}$, $\langle \xi_n, f\rangle = n^{-1/2}\sum_i \langle \chi_i, f\rangle$ is a normalized sum of i.i.d. real random variables with mean 0 and variance $\langle \mathbb{E}(\chi_1 \otimes \chi_1)f, f\rangle$, so by the classical (scalar) CLT, $\langle \xi_n, f\rangle \xrightarrow{d} N(0, \langle \mathbb{E}(\chi_1 \otimes \chi_1)f, f\rangle)$, which is the distribution of $\langle \xi, f\rangle$.

For the tightness condition, fix a CONS $\{e_j\}$, let $S_J := \operatorname{span}\{e_1, \ldots, e_J\}$, and let ξ'_{nJ} be the projection of ξ_n onto $S_J^\perp$ (so $\mathbb{P}(\|\xi'_{nJ}\| \leq \epsilon) = \mathbb{P}(\xi_n \in S_J^\epsilon)$). Writing χ'_{iJ} for the projection of χ_i onto $S_J^\perp$, Chebyshev's inequality and Theorem 2.56 (applied to the i.i.d., hence pairwise independent, mean-zero elements χ'_{iJ}) give

$$\mathbb{P}(\|\xi'_{nJ}\| > \epsilon) \leq \epsilon^{-2}\mathbb{E}\|\xi'_{nJ}\|^2 = \epsilon^{-2} n^{-1} \sum_{i=1}^{n} \mathbb{E}\|\chi'_{iJ}\|^2 = \epsilon^{-2}\mathbb{E}\|\chi'_{1J}\|^2,$$

uniformly in n; since $\mathbb{E}\|\chi_1\|^2 < \infty$, $\mathbb{E}\|\chi'_{1J}\|^2 \to 0$ as $J \to \infty$ (dominated convergence, as

$\chi'_{1J} \to 0$ pointwise on Ω), so this can be made $< \delta$ for J large, verifying the hypothesis of Theorem 2.59 with $S = S_J$. □

Theorem 2.60 is the tool that Section 2.7 applies to $\chi_i = X_i - m$ (giving the asymptotic normality of the sample mean $\overline{X}_n$) and, via the linearization $\Delta = \widetilde{\Gamma} - \Gamma$ of Section 2.4, to the sample covariance operator and, in turn, to the estimated FPCA eigenvalues and eigenfunctions of Section 2.8.

2.7 Mean and Covariance Operator Estimation

This section studies the large-sample behavior of estimators of the mean element m and covariance operator Γ, under three progressively more realistic observation schemes: (i) completely observed functional data (the sample mean and sample covariance operator); (ii) discretely, noisily observed sample paths, estimated by local-linear smoothing; (iii) the same discretely observed data, estimated instead by penalized least squares (smoothing splines). We follow [4], Chapter 8. Case (i) is proved in full, since it follows directly and briefly from the large-sample theory of Section 2.6.4; cases (ii) and (iii) involve very long uniform-consistency arguments (empirical-process bounds via Bernstein's inequality over a discretized grid, for (ii); a Fréchet-derivative linearization of a penalized criterion in a Sobolev basis, for (iii)) that are not directly used by this thesis's estimation strategy — which individually smooths each country's trajectory with an adaptive B-spline smoothing spline (Section 4.2.1, justified in full by Section 2.5) and then applies the case-(i) sample covariance operator to the resulting smoothed curves. For (ii) and (iii) we therefore state the precise models and rate theorems, and give a proof sketch conveying the strategy and the source of each rate, rather than the complete technical derivation; full detail is in [4], Sections 8.2–8.3.

2.7.1 Sample Mean and Covariance Operator

Let χ be a random element of a separable Hilbert space $\mathbb{H}$ with $\mathbb{E}\|\chi\|^2 < \infty$, mean m, and covariance operator Γ (Section 2.6.2), and let $\chi_1, \ldots, \chi_n$ be an i.i.d. sample from χ. Define

the *sample mean* $m_n := n^{-1}\sum_{i=1}^n \chi_i$ and *sample covariance operator*

$$\Gamma_n := \frac{1}{n-1}\sum_{i=1}^{n}(\chi_i - m_n)\otimes(\chi_i - m_n). \tag{2.59}$$

Theorem 2.61. *If* $\mathbb{E}\|\chi_1\| < \infty$, $m_n \to m$ *almost surely. If* $\mathbb{E}\|\chi_1\|^2 < \infty$, $\sqrt{n}(m_n - m) \xrightarrow{d} \xi$ *in* $\mathbb{H}$*, where* ξ *is Gaussian with mean* 0 *and covariance operator* Γ*.*

Proof. Immediate from Theorem 2.57 (with χ_i in place of $\chi_i - m$, noting $n^{-1}S_n - m = n^{-1}\sum(\chi_i - m)$) and Theorem 2.60 (applied to $\chi_i - m$, whose covariance operator is Γ by definition). □

Theorem 2.62. *If* $\mathbb{E}\|\chi_1\|^2 < \infty$, $\Gamma_n \to \Gamma$ *almost surely. If* $\mathbb{E}\|\chi_1\|^4 < \infty$, $\sqrt{n}(\Gamma_n - \Gamma) \xrightarrow{d} \mathfrak{Z}$ *in* $\mathcal{B}_{HS}(\mathbb{H})$*, where* $\mathfrak{Z}$ *is Gaussian with mean* 0 *and covariance operator* $\mathbb{E}\Big[\{(\chi_1 - m)\otimes(\chi_1 - m) - \Gamma\}\otimes_{HS}\{(\chi_1 - m)\otimes(\chi_1 - m) - \Gamma\}\Big]$ *(an element of* $\mathcal{B}(\mathcal{B}_{HS}(\mathbb{H}))$*, via the HS-space tensor product of Definition 2.13, applied one level up to the Hilbert space* $\mathcal{B}_{HS}(\mathbb{H})$ *itself).*

Proof. Write $\Gamma_n = \frac{1}{n-1}\sum_{i=1}^n(\chi_i - m)\otimes(\chi_i - m) - \frac{n}{n-1}(m_n - m)\otimes(m_n - m)$ (direct algebraic expansion). The second term is $o(1)$ a.s. by Theorem 2.61 (so $m_n - m \to 0$ a.s., hence $(m_n - m)\otimes(m_n - m) \to 0$ in HS norm), and the first term $\to \Gamma$ a.s. by Theorem 2.57 applied to the i.i.d. $\mathcal{B}_{HS}(\mathbb{H})$-valued random elements $(\chi_i - m)\otimes(\chi_i - m)$, giving the strong consistency of Γ_n.

For the CLT, Theorem 2.60 applies to $(\chi_i - m)\otimes(\chi_i - m) - \Gamma$ (i.i.d., mean 0, valued in the Hilbert space $\mathcal{B}_{HS}(\mathbb{H})$) provided $\mathbb{E}\|(\chi_1 - m)\otimes(\chi_1 - m) - \Gamma\|_{HS}^2 < \infty$; by Theorem 2.47 applied in $\mathcal{B}_{HS}(\mathbb{H})$, this is bounded by $\mathbb{E}\|(\chi_1 - m)\otimes(\chi_1 - m)\|_{HS}^2 = \mathbb{E}\|\chi_1 - m\|^4 < \infty$. The $n/(n-1)$-weighted second term of Γ_n is $O_p(n^{-1})$ (since $\sqrt{n}(m_n - m)$ is asymptotically Gaussian by Theorem 2.61, so $n(m_n - m)\otimes(m_n - m) = O_p(1)$), hence contributes nothing to the limit of $\sqrt{n}(\Gamma_n - \Gamma)$, which therefore coincides with that of $\sqrt{n}\Big(n^{-1}\sum_i[(\chi_i - m)\otimes(\chi_i - m) - \Gamma]\Big)$. □

To interpret Theorem 2.62: for any $\mathcal{G} \in \mathcal{B}_{HS}(\mathbb{H})$, $\langle\sqrt{n}(\Gamma_n - \Gamma), \mathcal{G}\rangle_{HS} \xrightarrow{d} \langle\mathfrak{Z}, \mathcal{G}\rangle_{HS} \sim N(0, \mathrm{Var}(\langle(\chi_1 - m)\otimes(\chi_1 - m), \mathcal{G}\rangle_{HS}))$, and choosing a CONS $\{e_j\}$ with $e_1 = (\chi_1 - m)/\|\chi_1 - m\|$ shows $\langle(\chi_1 - m)\otimes(\chi_1 - m), \mathcal{G}\rangle_{HS} = \langle\chi_1 - m, \mathcal{G}(\chi_1 - m)\rangle$, so $\mathrm{Var}(\langle\sqrt{n}(\Gamma_n - \Gamma), \mathcal{G}\rangle_{HS}) \to \mathrm{Var}(\langle\chi_1 - m, \mathcal{G}(\chi_1 - m)\rangle)$ – an explicit, directly interpretable variance for any linear functional of the estimated covariance operator. This is the estimator, and the exact asymptotic theory, that Chapter 4 applies (with $\mathbb{H} = \mathbb{L}^2(\mathcal{T})$ and χ_i the B-spline-smoothed country trajectory) to obtain the empirical covariance operator whose spectral decomposition is the FPCA of Chapter 5.

2.7.2 Local Linear Estimation

Suppose instead that $X_1, \dots, X_n$ are i.i.d. copies of a mean-square continuous process X on $[0,1]$ with mean m and covariance γ, observed only at discrete, noisy points:

$$Y_{ij} = X_i(T_{ij}) + \varepsilon_{ij}, \qquad j = 1, \dots, r,\ i = 1, \dots, n, \tag{2.60}$$

with T_{ij} i.i.d. sampling points (density f, f' continuous) and ε_{ij} i.i.d. mean-zero noise, all mutually independent collections. A *local linear* estimator of $m(t)$ minimizes, in (a_0, a_1), the kernel-weighted criterion $\sum_{i,j} W_h(T_{ij}-t)(Y_{ij}-a_0-a_1(T_{ij}-t))^2$ for a symmetric density W on $[-1,1]$ (of bounded variation, with $\int u^j W(u)\,du$ equal to $1, 0, C \neq 0$ for $j = 0, 1, 2$) and bandwidth h, giving the closed-form estimator

$$m_h(t) = \frac{S_0(t)M_2(t) - S_1(t)M_1(t)}{M_0(t)M_2(t) - M_1(t)^2}, \qquad M_p(t) := \frac{1}{nr}\sum_{i,j} W_{h^p}(T_{ij}-t),\ \ S_p(t) := \frac{1}{nr}\sum_{i,j} W_{h^p}(T_{ij}-t)Y_{ij}, \tag{2.61}$$

where $W_{h^p}(u) := (u/h)^p W_h(u)$.

Theorem 2.63. *Suppose m'' is uniformly bounded, $\mathbb{E}|\varepsilon_{11}|^q < \infty$ and $\mathbb{E}\sup_t |X(t)|^q < \infty$ for some $q \in (2, \infty)$, and $h \to 0$ with $(h^2 + h/r)^{-1}(\log n/n)^{1-2/q} \to 0$. Then, with $\delta_{n1}(h) := (\{1 + (hr)^{-1}\}\log n/n)^{1/2}$,*

$$\sup_{t\in[0,1]} |m_h(t) - m(t)| = O(h^2 + \delta_{n1}(h)) \quad \text{a.s.} \tag{2.62}$$

Proof sketch. Write $m_h(t) - m(t) = \{\widetilde{S}_0(t)M_2(t) - \widetilde{S}_1(t)M_1(t)\}/\{M_0(t)M_2(t) - M_1(t)^2\}$ for $\widetilde{S}_p(t) := S_p(t) - m(t)M_p(t) - hm'(t)M_{p+1}(t)$. The denominator and the M_p converge a.s. uniformly to nonzero limits by a general uniform strong law for kernel-weighted sums of bounded triangular arrays (stated below as Lemma 2.6), so the rate is governed by $\widetilde{S}_p$. Splitting $\widetilde{S}_p(t) = \{$a deterministic Taylor-remainder term, $O(h^2)$ a.s. by boundedness of $m''\}$ $+ U_p(t)$ for $U_p(t) := (nr)^{-1}\sum_{i,j} W_{h^p}(T_{ij} - t)[\varepsilon_{ij} + X_i(T_{ij}) - m(T_{ij})]$, a further application of Lemma 2.6 (to the two zero-mean triangular arrays comprising U_p) gives $\sup_t |U_p(t)| = O(\delta_{n1}(h))$ a.s. □

Lemma 2.6. *Let $\{Z_{ij}\}$ be a triangular array with $\sup_{i,j}\mathbb{E}|Z_{ij}|^q < \infty$, $(nr)^{-1}\sum_{i,j}|Z_{ij}|^q = O(1)$ a.s., and $\sup_{i,j,k}\mathbb{E}[|Z_{ij}Z_{ik}| \mid T_{ij}, T_{ik}] < C$ a.s., for some $q \in (2,\infty)$; let $\overline{Z}_p(t) := (nr)^{-1}\sum_{i,j} W_{h^p}(T_{ij}-t)Z_{ij}$ and $\beta_n := h^2 + h/r$ with $\beta_n^{-1}(\log n/n)^{1-2/q} \to 0$. Then $\sup_t |\overline{Z}_p(t) - \mathbb{E}\overline{Z}_p(t)| = O(\{\beta_n \log n/n\}^{1/2})$ a.s.*

This is [4], Lemma 8.2.2; its proof is a uniform law of large numbers for kernel-weighted triangular arrays, obtained by discretizing $[0,1]$ into a grid of $O(1/h)$ points, truncating Z_{ij} at a slowly growing threshold Q_n, and applying Bernstein's inequality plus Borel–Cantelli at each grid point (a two-page empirical-process argument we do not reproduce, as it is generic nonparametric-regression technology unrelated to the FDA-specific content of this thesis).

Corollary 2.5. *Under the conditions of Theorem 2.63: if r is bounded, $\sup_t |m_h(t) - m(t)| = O(h^2 + [\log n/(nh)]^{1/2})$ a.s. (the classical univariate nonparametric rate); if $r = r_n$ with $r_n^{-1} \lesssim h \lesssim (\log n/n)^{1/4}$, $\sup_t |m_h(t) - m(t)| = O([\log n/n]^{1/2})$ a.s. (a parametric rate, attained once $r \gtrsim n^{1/4}$).*

An analogous, more involved construction (weighting products $Y_{ij}Y_{ik}$, $j \neq k$, by $W_h(T_{ij} - s)W_h(T_{ik} - t)$ to estimate $R(s,t) := \mathbb{E}[X(s)X(t)]$, then setting $\gamma_h(s,t) := R_{h_R}(s,t) - m_{h_m}(s)m_{h_m}(t)$) yields a local-linear covariance estimator γ_h with rate $\sup_{s,t} |\gamma_h(s,t) - \gamma(s,t)| = O(h_m^2 + h_R^2 + \delta_{n1}(h_m) + \delta_{n2}(h_R))$ a.s. (with $\delta_{n2}(h) := (\{1 + (hr)^{-1} + (hr)^{-2}\} \log n/n)^{1/2}$), under fourth-moment analogs of the hypotheses above ([4], Theorem 8.2.4); the sparse and dense rates parallel Corollary 2.5, with the sparse rate $O(n^{-1/3})$ optimized at bandwidth $\widehat{h}_R \asymp n^{-1/6}$.

The rate that will actually matter for FPCA is not the pointwise kernel rate for $\gamma_h(s,t) - \gamma(s,t)$, but the rate at which the induced *integral operator* converges – and these turn out to differ.

Theorem 2.64. *Under the conditions of the local-linear covariance rate above, for any bounded measurable e on $[0,1]$, with $(\Delta_h e)(t) := \int_0^1 \{\gamma_h(s,t) - \gamma(s,t)\} e(s)\, ds$,*

$$\sup_{t \in [0,1]} |(\Delta_h e)(t)| = O(h_m^2 + h_R^2 + \delta_{n1}(h_m) + \delta_{n1}(h_R)) \quad a.s. \tag{2.63}$$

Note $\delta_{n1}(h_R)$, not the (slower) $\delta_{n2}(h_R)$ of the pointwise rate: integrating against a fixed e before taking the supremum in t averages out one order of the kernel's stochastic fluctuation. This gain is exactly the mechanism Section 2.8 needs to transfer local-linear covariance estimation into a rate on the estimated covariance *operator* in $\mathcal{B}(\mathbb{L}^2[0,1])$, hence (via Section 2.4) into a rate for the estimated eigenvalues and eigenfunctions; we omit the proof ([4], Theorem 8.2.7), which decomposes $\Delta_h e$ analogously to Theorem 2.63 and reapplies Lemma 2.6-type bounds to the resulting integrated stochastic terms.

2.7.3 Penalized Least-Squares Estimation

We return to model (2.60) (with T_{ij} now taken uniform on $[0,1]$, for simplicity), but assume instead that X is a random element of the Sobolev space $\mathbb{W}^q[0,1]$ of Definition 2.8, $q>1$, with CONS $\{e_j\}$ (the eigenfunctions of $(-1)^q D^{2q}$ from Theorem 2.8.3 of [4], satisfying $\langle e_j,e_k\rangle_2=\delta_{jk}$, $\langle e_j^{(q)},e_k^{(q)}\rangle_2=\gamma_j\delta_{jk}$ with $\gamma_1=\cdots=\gamma_q=0$ and $C_1 j^{2q}\le\gamma_{j+q}\le C_2 j^{2q}$), so that $\|v\|^2_{\mathbb{W}^q[0,1]}=\sum_j(1+\gamma_j)v_j^2$ for $v=\sum_j v_j e_j$. This is a more restrictive but sharper setting than Section 2.7.2's, and the resulting smoothing-spline estimators are rate-optimal.

Theorem 2.65 (Minimax lower bound; Cai and Yuan, 2011)**.** *Let $\mathbb{P}(q,C_1)$ be the probability laws of $\mathbb{W}^q[0,1]$-valued processes X with $\mathbb{E}\|X^{(q)}\|_2^2\le C_1$. There is $C_2>0$, depending only on $C_1,\sigma^2=\mathbb{E}\varepsilon_{11}^2$, such that every estimator $\widehat{m}$ of m satisfies*

$$\limsup_{n\to\infty}\sup_{\mathbb{P}_X\in\mathbb{P}(q,C_1)}\mathbb{P}\Big(\|\widehat{m}-m\|_2^2>C_2\{(nr)^{-2q/(2q+1)}+n^{-1}\}\Big)>0. \tag{2.64}$$

The smoothing-spline estimator $m_\eta:=\operatorname{argmin}_{v\in\mathbb{W}^q[0,1]}f_{rn,\eta}(v)$ for $f_{rn,\eta}(v):=(nr)^{-1}\sum_{i,j}(Y_{ij}-v(T_{ij}))^2+\eta\|v^{(q)}\|_2^2$ (the case of Theorem 2.33 with $\mathcal{T}$ the discrete evaluation map on $[0,1]$ into $\mathbb{R}^{nr}$) attains this lower bound.

Theorem 2.66. *If $\eta\asymp(rn)^{-2q/(2q+1)}$, $\|m_\eta-m\|_2^2=O_p((nr)^{-2q/(2q+1)}+n^{-1})$.*

Proof sketch. Following Cai and Yuan (2011), let $f_{\infty,\eta}(v):=\mathbb{E}f_{rn,\eta}(v)=\mathbb{E}[(Y_{11}-m(T_{11}))^2]+\|m-v\|_2^2+\eta\|v^{(q)}\|_2^2$, $\overline{m}_\eta:=\operatorname{argmin}_v f_{\infty,\eta}(v)$, and the one-step Newton-type intermediate $\widetilde{m}_\eta:=m_\eta-(f''_{\infty,\eta})^{-1}f'_{rn,\eta}(m_\eta)$ (Fréchet derivatives, in the sense of Section 3.6 of [4]), so that $m_\eta-m=(m_\eta-\widetilde{m}_\eta)+(\widetilde{m}_\eta-\overline{m}_\eta)+(\overline{m}_\eta-m)$.

The last term is the easiest: since m_η minimizes $f_{\infty,\eta}$ with m itself a competitor, $\|\overline{m}_\eta-m\|_2^2\le f_{\infty,\eta}(\overline{m}_\eta)-\mathbb{E}[(Y_{11}-m(T_{11}))^2]\le f_{\infty,\eta}(m)-\mathbb{E}[(Y_{11}-m(T_{11}))^2]=\eta\|m^{(q)}\|_2^2=O((rn)^{-2q/(2q+1)})$ under the assumed rate on η – an instance of the bias bound Theorem 2.34 specialized to this criterion.

The remaining two terms, $\|m_\eta-\widetilde{m}_\eta\|_2^2$ and $\|\widetilde{m}_\eta-\overline{m}_\eta\|_2^2$, are shown to be $O_p((nr)^{-2q/(2q+1)}+n^{-1})$ by expressing both in the interpolation-scale norms $\|v\|_\alpha^2:=\sum_j(1+\gamma_j)^\alpha v_j^2$, $\alpha\in[0,1]$ (which satisfy $\|v\|_0=\|v\|_2\le\|v\|_\alpha\le\|v\|_1=\|v\|_{\mathbb{W}^q[0,1]}$), using the explicit representation $(\widetilde{f}''_{\infty,\eta})^{-1}v=\frac12\sum_j\frac{1+\gamma_j}{1+\eta\gamma_j}v_j e_j$ of the (Riesz-representer form of the) second Fréchet derivative's inverse. This reduces $\|m_\eta-\widetilde{m}_\eta\|_\alpha^2$ and $\|\widetilde{m}_\eta-\overline{m}_\eta\|_\alpha^2$ to explicit weighted sums, $\frac14\sum_k\frac{(1+\gamma_k)^\alpha}{(1+\eta\gamma_k)^2}(f'_{rn,\eta}(\overline{m}_\eta)e_k)^2$ and an analogous expression, whose expectations are bounded via a variance computation on $f'_{rn,\eta}(\overline{m}_\eta)e_k$ (using $\mathbb{E}[f'_{rn,\eta}(\overline{m}_\eta)e_k]=0$, since $f'_{\infty,\eta}(\overline{m}_\eta)=0$) together with the polynomial growth rate $\gamma_j\asymp j^{2q}$, giving $O_p(n^{-1}+(nr)^{-1}\eta^{-\alpha-1/(2q)})$-type

bounds that, at $\alpha = 0$ and the stated rate for η, both reduce to $O_p((nr)^{-2q/(2q+1)} + n^{-1})$. We omit the (lengthy, but routine) algebraic details; full derivation in [4], Theorem 8.3.2. □

An entirely parallel construction — penalizing a bivariate smoothing criterion $f_{rn,\eta}(K) := \{r(r-1)n\}^{-1} \sum_i \sum_{j \neq k} (Y_{ij}Y_{ik} - K(T_{ij}, T_{ik}))^2 + \eta \|K\|^2_{\mathbb{H}}$ over the tensor-product Sobolev space $\mathbb{H} = \mathbb{W}^q[0,1] \otimes \mathbb{W}^q[0,1]$ (Definition 2.7's product-kernel construction) — gives a smoothing-spline covariance estimator K_η.

Theorem 2.67. *At the rate $\eta \asymp (rn)^{-2q/(2q+1)}$, with $\delta_n^2 := (nr)^{-2q/(2q+1)} + n^{-1}$,*

$$\|K_\eta - \gamma\|^2_{\mathbb{L}^2([0,1]^2)} = O_p(\delta_n^2). \tag{2.65}$$

Consequently, if $\Delta_\eta \in \mathcal{B}(\mathbb{L}^2[0,1])$ is the integral operator with kernel $K_\eta - \gamma$, $\|\Delta_\eta\|^2 = O_p(\delta_n^2)$.

Proof sketch. By the same three-term decomposition and interpolation-norm argument as Theorem 2.66, now applied to the bivariate Fourier coefficients $\{h_{\nu\tau}\}$ of $K_\eta - \overline{K}_\eta$ in the tensor basis $\{e_\nu \otimes e_\tau\}$ ([4], culminating result of Section 8.3); the operator-norm bound follows from (2.65) via the Cauchy–Schwarz bound of Definition 2.15, $\|\Delta_\eta\| \leq \left(\iint (K_\eta - \gamma)^2 \right)^{1/2} = \|K_\eta - \gamma\|_{\mathbb{L}^2([0,1]^2)}$. □

Comparing Sections 2.7.2 and 2.7.3: at $q = 2$ the local-linear and penalized-least-squares rates essentially coincide, but the penalized least-squares (smoothing-spline) approach requires the stronger assumption that sample paths be differentiable (as opposed to merely bounded), while offering better computational efficiency – the trade-off underlying the choice, in Section 4.2.1, of an adaptive smoothing-spline (rather than local-linear kernel) estimator for each country's GDP growth trajectory.

2.8 Functional Principal Component Analysis

This closing section is the capstone of the chapter: it combines the spectral decomposition of Section 2.3 (Mercer's Theorem identifying a covariance kernel with its eigen-expansion), the Karhunen–Loève Theorem of Section 2.6 (FPCA at the population level), the covariance operator estimators of Section 2.7, and the eigenprojection/eigenvalue/ eigenvector perturbation theory of Section 2.4, to establish the large-sample distribution of the *estimated* FPCA eigenvalues and eigenfunctions – exactly the objects Chapter 5 reports

(Table 5.2, the cumulative variance curves of Figure 5.10, and the FPC score plots of Figure 5.13). We follow [4], Chapter 9, again treating the two discretely-observed-data variants (Sections 2.8.2, 2.8.3) with proof sketches, since Section 2.8.1's completely-observed-data theory is what applies once each country's trajectory has already been individually smoothed (Section 4.2.1).

Throughout, χ (or X) denotes the (mean-zero, WLOG) functional observation with covariance operator $\Gamma = \sum_j \lambda_j \mathcal{P}_j$, nonrepeated eigenvalues $\lambda_1 > \lambda_2 > \cdots$ of multiplicity $r_j \geq 1$ and eigenprojections $\mathcal{P}_j$ (Theorem 2.10); by the Karhunen–Loève Theorem (Corollary 2.3/Theorem 2.54), $\chi = \sum_j Z_j e_j$ a.s. for uncorrelated scores Z_j with $\mathrm{Var}(Z_j) = \lambda_j$ – functional PCA *is* the empirical version of this decomposition.

2.8.1 Estimation via the Sample Covariance Operator

Let $\chi_1, \ldots, \chi_n$ be i.i.d. copies of χ with sample covariance operator Γ_n (Definition (2.59)), (possibly repeated) eigenvalues $\{\lambda_{jn}\}$, and matched eigenprojections $\mathcal{P}_{jn}$ (Equation (2.21)). Set $\Delta_n := \Gamma_n - \Gamma$.

Theorem 2.68. *If* $\mathbb{E}\|\chi\|^4 < \infty$,

$$n^{1/2}(\mathcal{P}_{jn} - \mathcal{P}_j) \xrightarrow{d} \mathcal{S}_j \mathfrak{Z} \mathcal{P}_j + \mathcal{P}_j \mathfrak{Z} \mathcal{S}_j, \tag{2.66}$$

where $\mathfrak{Z}$ *is the distributional limit of* $n^{1/2}(\Gamma_n - \Gamma)$ *from Theorem 2.62 and* $\mathcal{S}_j = \sum_{k \neq j} (\lambda_j - \lambda_k)^{-1} \mathcal{P}_k$ *as in Theorem 2.30.*

Proof. By Theorem 2.62, $\Delta_n \to 0$ a.s., so $\mathbb{P}(\|\Delta_n\| < \eta_j \text{ eventually}) = 1$ for $\eta_j = \frac{1}{2}\min_{k \neq j} |\lambda_k - \lambda_j|$; Theorem 2.30 then applies eventually a.s., giving $\mathcal{P}_{jn} - \mathcal{P}_j = \mathcal{S}_j \Delta_n \mathcal{P}_j + \mathcal{P}_j \Delta_n \mathcal{S}_j + O(\|\Delta_n\|^2)$. Since $\|\Delta_n\| = O_p(n^{-1/2})$ (Theorem 2.62), $n^{1/2} \cdot O(\|\Delta_n\|^2) = O_p(n^{-1/2}) = o_p(1)$, so $n^{1/2}(\mathcal{P}_{jn} - \mathcal{P}_j)$ is asymptotically equivalent to $\mathcal{S}_j(n^{1/2}\Delta_n)\mathcal{P}_j + \mathcal{P}_j(n^{1/2}\Delta_n)\mathcal{S}_j$, a continuous (bounded bilinear) functional of $n^{1/2}\Delta_n$; the continuous mapping theorem and $n^{1/2}\Delta_n \xrightarrow{d} \mathfrak{Z}$ (Theorem 2.62) give (2.66). □

Theorem 2.69. *If* $\mathbb{E}\|\chi\|^4 < \infty$, *for any fixed* j,

$$\left(n^{1/2}(\lambda_{kn} - \lambda_j)\right)_{k = N_{j-1}+1, \ldots, N_j} \xrightarrow{d} \mathrm{eig}(\mathcal{P}_j \mathfrak{Z} \mathcal{P}_j), \tag{2.67}$$

the (random) eigenvalues of the $r_j \times r_j$ *self-adjoint operator* $\mathcal{P}_j \mathfrak{Z} \mathcal{P}_j$ *(restricted to the eigenspace of* $\mathcal{P}_j$*).*

Proof. The $n^{1/2}(\lambda_{kn} - \lambda_j)$, $k = N_{j-1} + 1, \ldots, N_j$, are the eigenvalues of $n^{1/2}(\mathcal{P}_{jn}\Gamma_n\mathcal{P}_{jn} - \lambda_j\mathcal{P}_{jn})$. By Theorem 2.31 (with $\widetilde{\mathcal{T}} = \Gamma_n$),

$$\mathcal{P}_{jn}\Gamma_n\mathcal{P}_{jn}-\lambda_j\mathcal{P}_{jn} = \mathcal{P}_j\Delta_n\mathcal{P}_j+(\mathcal{P}_{jn}-\mathcal{P}_j)\Delta_n\mathcal{P}_{jn}+\mathcal{P}_j\Delta_n(\mathcal{P}_{jn}-\mathcal{P}_j)+(\mathcal{P}_{jn}-\mathcal{P}_j)(\Gamma-\lambda_j I)(\mathcal{P}_{jn}-\mathcal{P}_j),$$

using $\mathcal{P}_{jn}\Delta_n\mathcal{P}_{jn} = \mathcal{P}_j\Delta_n\mathcal{P}_j+(\mathcal{P}_{jn}-\mathcal{P}_j)\Delta_n\mathcal{P}_{jn}+\mathcal{P}_j\Delta_n(\mathcal{P}_{jn}-\mathcal{P}_j)$ (algebraic expansion). By Theorem 2.68, $\mathcal{P}_{jn}-\mathcal{P}_j = O_p(n^{-1/2})$, and $\Delta_n = O_p(n^{-1/2})$, so each of the last three terms is $O_p(n^{-1})$; multiplying by $n^{1/2}$, these vanish in probability, leaving $n^{1/2}(\mathcal{P}_{jn}\Gamma_n\mathcal{P}_{jn} - \lambda_j\mathcal{P}_{jn})$ asymptotically equivalent to $\mathcal{P}_j(n^{1/2}\Delta_n)\mathcal{P}_j$. Since eigenvalues of a self-adjoint operator restricted to a fixed finite-dimensional subspace are a continuous function of the operator (Theorem 2.13, applied within the r_j-dimensional range of $\mathcal{P}_j$), the continuous mapping theorem and $n^{1/2}\Delta_n \xrightarrow{d} \mathfrak{Z}$ give (2.67). □

The case $r_j = 1$, of primary interest (and the generic case for a covariance operator with simple spectrum), admits an explicit limit for both the eigenvalue and the eigenfunction.

Theorem 2.70. *Suppose $\mathbb{E}\|\chi\|^4 < \infty$ and λ_j has unit multiplicity with eigenfunction e_j; let (λ_{jn}, e_{jn}) be the matched eigenpair of Γ_n (with $\langle e_{jn}, e_j\rangle \geq 0$). Then*

$$n^{1/2}(\lambda_{jn} - \lambda_j) \xrightarrow{d} \langle \mathfrak{Z}e_j, e_j\rangle, \qquad n^{1/2}(e_{jn} - e_j) \xrightarrow{d} \sum_{k\neq j}(\lambda_j - \lambda_k)^{-1}\mathcal{P}_k\mathfrak{Z}e_j. \tag{2.68}$$

Proof. By Theorem 2.31 (Equation (2.28)) with $\Delta = \Delta_n$, $\lambda_{jn}-\lambda_j = \langle\Delta_n e_j, e_j\rangle+O_p(\|\Delta_n\|^2)$, and $n^{1/2}\cdot O_p(\|\Delta_n\|^2) = O_p(n^{-1/2}) = o_p(1)$; the first identity follows from $n^{1/2}\Delta_n \xrightarrow{d} \mathfrak{Z}$ and the continuous mapping theorem applied to $\Delta \mapsto \langle\Delta e_j, e_j\rangle$. Similarly, by Theorem 2.32 (Equation (2.32)), $e_{jn}-e_j = \mathcal{S}_j\Delta_n e_j+O_p(\|\Delta_n\|^2)$, and the same argument, with continuous mapping applied to $\Delta \mapsto \mathcal{S}_j\Delta e_j$, gives the second identity. □

Theorem 2.70 is the precise asymptotic justification for treating the eigenvalues and eigenfunctions reported in Table 5.2 as consistent, asymptotically Gaussian estimates of the population FPCA of the (B-spline-smoothed) GDP growth process: $\sqrt{n}$-consistency holds under only a fourth-moment condition on the smoothed trajectories, with no smoothing-parameter-dependent bias term, precisely because the input to Γ_n is already-smoothed, effectively fully observed functional data.

2.8.2 Estimation via Local Linear Smoothing

We return to the discretely, noisily observed model (2.60) and the local-linear covariance kernel estimator $\gamma_h(s,t) = \sum_j \widehat{\lambda}_j\widehat{e}_j(s)\widehat{e}_j(t)$ of Section 2.7.2, with Δ_h the integral operator

with kernel $\gamma_h - \gamma$ (rate given by Theorem 2.64). Assume the λ_j are distinct, and resolve the sign ambiguity of $\widehat{e}_j$ by choosing $\langle \widehat{e}_j, e_j \rangle \geq 0$.

Since $\|\Delta_h\| \leq \|\Delta_h e\|$ is not generally true, but $\|\mathcal{P}_j - \widehat{\mathcal{P}}_j\| = O_p(\|\Delta_h\|/\gamma_j)$ follows directly from (2.24) (with $\gamma_j := \frac{1}{2}\min_{\lambda_k \neq \lambda_j} |\lambda_k - \lambda_j|$) whenever $\|\Delta_h\|/\gamma_j \to 0$, Theorem 2.64 already gives the rate of $\widehat{\mathcal{P}}_j - \mathcal{P}_j$ (simultaneously over j). We now refine this to $\widehat{\lambda}_j$ and $\widehat{e}_j$ individually.

Theorem 2.71. *Under the conditions of Theorem 2.64, for any j with $\lambda_j > 0$,*

$$\widehat{\lambda}_j - \lambda_j = O\Big((\log n/n)^{1/2} + h_m^2 + h_R^2 + \delta_{n1}^2(h_m) + \delta_{n2}^2(h_R)\Big) \quad a.s. \tag{2.69}$$

Proof sketch. By Equations (2.28) and (2.24), $\widehat{\lambda}_j - \lambda_j = \iint (\gamma_h - \gamma)(s,t) e_j(s) e_j(t)\, ds\, dt + O(\|\Delta_h\|^2) =: A_{n1} - A_{n2} + O(\|\Delta_h\|^2)$, splitting $\gamma_h = R_{h_R} - m_{h_m} \otimes m_{h_m}$ into its two constituent local-linear estimators (Section 2.7.2). A_{n2}, the m_{h_m}-driven term, is $O((\log n/n)^{1/2} + h_m^2)$ a.s. by Theorem 2.63 applied twice. For A_{n1}, substituting the decomposition of $R_{h_R} - R$ from the (sketched) proof of the local-linear covariance rate reduces A_{n1} to an average of the form $n^{-1}\sum_i Z_{ni}$ for bounded-moment, independent-across-i Z_{ni} (pairing sampling points within each curve i), to which the truncation argument of Lemma 2.7 below applies, giving $A_{n1} = O((\log n/n)^{1/2} + h_R^2)$ a.s. Combining the two bounds with $\|\Delta_h\|^2 = O(h_m^4 + h_R^4 + \delta_{n1}^2(h_m) + \delta_{n2}^2(h_R))$ a.s. (Theorems 2.64–2.63 combined, as in the source) and simplifying (since $(\log n/n)^{1/2} = o(\delta_{n1}(h_m))$) gives (2.69). □

Lemma 2.7. *Let $Z_{n1}, \ldots, Z_{nn}$ be independent, mean-zero, with $\sup_{i,n} \mathbb{E}|Z_{ni}|^\delta < \infty$ and $n^{-1}\sum_i |Z_{ni}|^\delta = O(1)$ a.s., for some $\delta \in (2, \infty)$. Then $n^{-1}\sum_i Z_{ni} = O((\log n/n)^{1/2})$ a.s.*

Proof. Let $a_n = (\log n/n)^{1/2}$ and split $Z_{ni} = Z_{ni}^{\succ} + Z_{ni}^{\prec}$ for $Z_{ni}^{\succ} := Z_{ni} I(|Z_{ni}| > a_n^{-1})$, $Z_{ni}^{\prec} := Z_{ni} I(|Z_{ni}| \leq a_n^{-1})$. By Hölder's inequality, $a_n^{-1} n^{-1} \sum_i |Z_{ni}^{\succ}| \leq a_n^{-1} n^{-1} \sum_i |Z_{ni}|^{1-\delta} |Z_{ni}|^\delta \leq a_n^{1-\delta} n^{-1} \sum_i |Z_{ni}|^\delta \to 0$ a.s. (since $a_n^{1-\delta} \to 0$ and the sum is $O(1)$ a.s.), and likewise in expectation, so $n^{-1}\sum_i (Z_{ni}^{\succ} - \mathbb{E}Z_{ni}^{\succ}) = o(a_n)$ a.s. For the truncated part, Bernstein's inequality gives $\mathbb{P}\Big(|n^{-1}\sum_i (Z_{ni}^{\prec} - \mathbb{E}Z_{ni}^{\prec})| > B a_n\Big) \leq \exp\{-B^2 \log n/(2\sigma^2 + (2/3)B)\}$, summable in n for B large; Borel–Cantelli completes the proof. □

Theorem 2.72. *Under the conditions of Theorem 2.64, for $\lambda_j > 0$,*

$$\|\widehat{e}_j - e_j\| = O(h_m^2 + h_R^2 + \delta_{n1}(h_m) + \delta_{n1}(h_R)) \quad a.s., \qquad \sup_t |\widehat{e}_j(t) - e_j(t)| = O(h_m^2 + h_R^2 + \delta_{n1}(h_m) + \delta_{n1}(h_R) - \tag{2.70}$$

Proof. By Theorem 2.32 (Equation (2.31)), $\widehat{e}_j - e_j = \sum_{k \neq j} (\lambda_j - \lambda_k)^{-1} \langle \Delta_h e_j, e_k \rangle e_k + O(\|\Delta_h\|^2)$, so by Bessel's inequality $\|\widehat{e}_j - e_j\| \leq C(\|\Delta_h e_j\| + \|\Delta_h\|^2)$ for a finite C depending

on γ_j. By Theorems 2.64 and (as invoked in the proof of Theorem 2.71) the local-linear rates, $\|\Delta_h\|^2 = O(h_m^4+h_R^4+\delta_{n1}^2(h_m)+\delta_{n2}^2(h_R))$ and $\|\Delta_h e_j\| = O(h_m^2+h_R^2+\delta_{n1}(h_m)+\delta_{n1}(h_R))$ a.s., giving the first bound.

For the uniform bound, for any t, $\widehat{\lambda}_j\widehat{e}_j(t) - \lambda_j e_j(t) = \int_0^1(\gamma_h(s,t) - \gamma(s,t))e_j(s)\,ds + \int_0^1 \gamma_h(s,t)(\widehat{e}_j(s) - e_j(s))\,ds =: (\Delta_h e_j)(t) + R_n(t)$, where by Cauchy–Schwarz, $|R_n(t)| \le \sup_{s,t}|\gamma_h(s,t)| \cdot \|\widehat{e}_j - e_j\| = O(\|\widehat{e}_j - e_j\|)$ a.s. (using uniform boundedness of γ_h, itself $O(1)$ a.s. by consistency); so, using the already-established rate for $\|\widehat{e}_j - e_j\|$, $\widehat{\lambda}_j\widehat{e}_j(t) - \lambda_j e_j(t) = O(h_m^2 + h_R^2 + \delta_{n1}(h_m) + \delta_{n1}(h_R))$ a.s., uniformly in t. Writing $\lambda_j(\widehat{e}_j(t) - e_j(t)) = \{\widehat{\lambda}_j\widehat{e}_j(t) - \lambda_j e_j(t)\} - (\widehat{\lambda}_j - \lambda_j)\widehat{e}_j(t)$ and combining this bound with Theorem 2.71's rate for $\widehat{\lambda}_j - \lambda_j$ (and uniform boundedness of $\widehat{e}_j$) via the triangle inequality gives the second bound (the $(\log n/n)^{1/2}$ term from Theorem 2.71 being dominated by $\delta_{n1}(h_m)$). □

Corollary 2.6. *Under the conditions of Theorem 2.64, with $\lambda_j > 0$: if $r \le M$ (sparse) and bandwidths chosen optimally, $\widehat{\lambda}_j - \lambda_j = O(\{\log n/n\}^{1/2})$ a.s., while $\|\widehat{e}_j - e_j\|$ and $\sup_t|\widehat{e}_j(t) - e_j(t)|$ attain the classical nonparametric rate $O(h_R^2 + \{\log n/(nh_R)\}^{1/2})$ a.s. (at $h_R \asymp n^{-1/5}$); if $r = r_n \to \infty$ with $r_n^{-1} \lesssim h_m, h_R \lesssim (\log n/n)^{1/4}$ (dense), all three quantities attain the parametric rate $O(\{\log n/n\}^{1/2})$ a.s.*

So, unlike the mean and covariance *kernel* estimators themselves (Section 2.7.2), the eigenvalues of the local-linear covariance operator are always $\sqrt{n}$-consistent, sparse or dense; only the eigenfunctions pay the nonparametric-rate penalty in the sparse regime – a reflection of the operator-level rate gain already noted after Theorem 2.64.

2.8.3 Estimation via Penalized Least Squares

Finally, we obtain FPCA from the smoothing-spline covariance estimator K_η of Theorem 2.67. Recall $K_\eta(s,t) = \sum_{i=1}^n R_i(s)^T\mathcal{A}_i R_i(t)$ for symmetric, zero-diagonal matrices $\mathcal{A}_i$ and $R_i(s) := (R_q(s,T_{i1}),\dots,R_q(s,T_{ir}))^T$, R_q the reproducing kernel of $\mathbb{W}^q[0,1]$ (Definition 2.8). Every eigenfunction of K_η is thus of the form $\widehat{e}(\cdot) = v^T R(\cdot)$ for $R(\cdot) := (R_1(\cdot)^T,\dots,R_n(\cdot)^T)^T$ and some vector v.

Theorem 2.73. *Let $\mathcal{A} := \operatorname{diag}(\mathcal{A}_1,\dots,\mathcal{A}_n)$ and $\mathcal{G} := \{\int_0^1 R_q(s,T_{i\nu})R_q(s,T_{j\tau})\,ds\}_{i,j;\nu,\tau=1:r}$ (the Gram matrix of the $\{R_i\}$). If u is an eigenvector of $\mathcal{B} := \mathcal{G}^{1/2}\mathcal{A}\mathcal{G}^{1/2}$ with eigenvalue $\widehat{\lambda}$, then $\widehat{e}(\cdot) := u^T\mathcal{G}^{-1/2}R(\cdot)$ is an eigenfunction of K_η with eigenvalue $\widehat{\lambda}$, and distinct eigenvectors of $\mathcal{B}$ give $\mathbb{L}^2[0,1]$-orthonormal eigenfunctions of K_η.*

Proof. For eigenvectors u_i, u_j of $\mathcal{B}$ with $v_i := \mathcal{G}^{-1/2}u_i$, $\widehat{e}_i(\cdot) := v_i^T R(\cdot)$, orthonormality holds since $\int_0^1 \widehat{e}_i(s)\widehat{e}_j(s)\,ds = v_i^T\mathcal{G}v_j = u_i^T u_j = \delta_{ij}$ (using the definition of $\mathcal{G}$ and $v_i = \mathcal{G}^{-1/2}u_i$).

And, writing $\{\widehat{\lambda}_j\}$ for the eigenvalues of $\mathcal{B}$ with eigenvectors $\{u_j\}$, $\sum_j \widehat{\lambda}_j \widehat{e}_j(s)\widehat{e}_j(t) = R(s)^T\mathcal{G}^{-1/2}\Big(\sum_j \widehat{\lambda}_j u_j u_j^T\Big)\mathcal{G}^{-1/2}R(t) = R(s)^T\mathcal{G}^{-1/2}\mathcal{B}\mathcal{G}^{-1/2}R(t) = R(s)^T\mathcal{A}R(t) = K_\eta(s,t)$, identifying $\{(\widehat{\lambda}_j, \widehat{e}_j)\}$ as the eigen-decomposition of K_η. □

We estimate λ_j (assumed simple) by the jth largest eigenvalue $\widehat{\lambda}_j$ of $\mathcal{B}$, and e_j by $\widehat{e}_j(\cdot) = u_j^T\mathcal{G}^{-1/2}R(\cdot)$ (sign resolved by $\langle \widehat{e}_j, e_j\rangle \geq 0$).

Theorem 2.74. *Suppose λ_j has unit multiplicity and $\eta \asymp (\log n/(rn))^{2q/(2q+1)}$. Under the conditions of Theorem 2.67, with $\delta_n^2 := (\log n/(rn))^{2q/(2q+1)} + n^{-1}$,*

$$\widehat{\lambda}_j - \lambda_j = O_p(\delta_n), \qquad \|\widehat{e}_j - e_j\|^2 = O_p(\delta_n^2). \tag{2.71}$$

Proof. By Theorem 2.67, $\|\Delta_\eta\|^2 = O_p(\delta_n^2)$ for Δ_η the operator with kernel $K_\eta - \gamma$; by (2.24), $\|\widehat{\mathcal{P}}_j - \mathcal{P}_j\| = \|\widehat{e}_j \otimes \widehat{e}_j - e_j \otimes e_j\| = O_p(\delta_n)$. By Lemma 2.4(2) and $(1-x)^{1/2} = 1 - x/2 + o(x)$ as $x \to 0$, $\|\widehat{e}_j - e_j\|^2 = 2\Big[1 - (1 - \|\widehat{\mathcal{P}}_j - \mathcal{P}_j\|^2)^{1/2}\Big] = \|\widehat{\mathcal{P}}_j - \mathcal{P}_j\|^2 + o_p(\delta_n^2) = O_p(\delta_n^2)$. The eigenvalue rate follows from Equation (2.28) exactly as in the proof of Theorem 2.70, using $\|\Delta_\eta\| = O_p(\delta_n)$. □

Theorems 2.71–2.74 show that all three estimation schemes considered in this chapter – sample covariance operator, local-linear smoothing, and penalized least squares – yield eigenvalue and eigenfunction estimators whose consistency and (in the completely-observed case) asymptotic normality are direct, mechanical consequences of two facts established earlier in this chapter: operator-norm (or Hilbert–Schmidt-norm) consistency of the estimated covariance operator for Γ (Section 2.7), and the linearization of eigenprojections, eigenvalues, and eigenfunctions around this operator-norm perturbation (Section 2.4). This is the theoretical justification, in full, for treating the FPCA eigenvalues and eigenfunctions estimated in Chapter 5 – via the sample covariance operator of the B-spline-smoothed country trajectories – as consistent, asymptotically normal estimates of the population functional principal components of Latin American GDP growth.

2.9 Chapter Summary and Scope for the Remaining Bases

This chapter has developed, from first principles, the complete rigorous justification for the B-spline-based FPCA pipeline of Chapter 4: Sections 2.2– 2.3 built the Hilbert-space

and operator-theoretic machinery culminating in Mercer's Theorem, which identifies a covariance kernel with the eigen-decomposition FPCA estimates; Section 2.4 showed that this eigen-decomposition depends continuously, and to first order linearly, on the covariance operator itself; Section 2.5 established that the smoothing spline underlying the adaptive B-spline basis of Section 4.2.1 is both well defined (Theorem 2.33) and, via the natural-spline optimality theorem, exactly computable in finite dimensions; Section 2.6 gave the Karhunen–Loève Theorem, the population-level statement of FPCA; and Sections 2.7 – 2.8 combined all of the above into consistency and asymptotic normality of the estimated FPCA eigenvalues and eigenfunctions themselves (Theorem 2.70), the precise sense in which the empirical results of Chapter 5 are statistically meaningful rather than merely descriptive.

As flagged in Section 2.1, [4] develops this pipeline for a general basis system and, concretely, for smoothing splines (B-splines with a roughness penalty) – matching Section 4.2.1 of this thesis – but does not treat the Fourier or wavelet basis systems that Chapter 4 compares against the B-spline baseline. The theory that would complete this chapter's coverage of those two bases includes: for the **Fourier basis**, classical approximation-theoretic results on the rate of trigonometric approximation for functions of bounded variation, and a rigorous account of the Gibbs phenomenon near jump discontinuities, explaining the artefacts documented empirically in Section 5.2.3; for the **wavelet basis**, the characterization of Besov spaces $B^s_{p,q}$ as the natural regularity classes for spatially inhomogeneous functions, the minimax theory of wavelet shrinkage of [3] already cited in Section 4.2.3, and nonlinear n-term approximation theory establishing why a sparse wavelet representation can outperform any linear (fixed-basis) method, including B-splines and Fourier truncation, on functions with localized irregularities such as the crisis episodes analyzed in Chapter 5. This material will be incorporated from a dedicated approximation-theory source in a subsequent revision of this chapter.

Chapter 3

State of the Art

3.1 Overview

The representation of a discretely sampled trajectory as an element of a function space, obtained by expanding it in a finite system of basis functions, is the operation that everything else in functional data analysis is built on; [1] remains the reference work that establishes this operation and the analyses constructed on top of it, including FPCA. Chapter 1 argued that, in applications to volatile economic series, this operation is rarely treated as an object of empirical scrutiny in its own right, and Chapter 2 developed the operator-theoretic machinery that any basis expansion feeds into. This chapter reviews the literature that bears on that argument. A systematic search of the bibliographic databases, screened against explicit eligibility criteria, yielded 91 sources retained for full analysis, of which 44 qualify as direct antecedents: work that both justifies its choice of basis and evaluates the quality of the resulting representation.

Section 3.2 reviews how functional basis families are chosen and justified in practice. Section 3.3 examines how that choice propagates into functional principal component analysis. Section 3.4 turns to smoothing and penalization, and to the extent to which their effect on the fitted representation is disentangled from the effect of the basis family itself. Section 3.5 reviews the use of wavelet bases for non-smooth or non-stationary processes. Section 3.6 narrows the scope to economic and financial applications, with particular attention to Latin American and Ecuadorian antecedents. Section 3.7 consolidates these five threads into a single methodological gap, and Section 3.8 positions the present thesis

relative to it.

3.2 Basis choice: how functional basis families are justified in practice

The reviewed literature converges markedly on B-spline bases as the default choice for representing functional data: among the 91 reviewed works, 30 of those that adopt a single basis family choose B-splines, against 5 that choose Fourier and only 2 that choose wavelets. This numerical convergence does not reflect a methodological consensus; authors within the field state the underlying tension explicitly. [8] describe the prevailing practice as "a rather ad hoc choice" among Fourier, wavelet, or spline bases, and [9] report that "a notable gap persists in the literature regarding comprehensive comparisons between these methodologies" – two works in this same corpus, two years apart, state nearly the same objection: the choice of basis responds to convention more often than to an explicit justification.

When a theoretical justification is offered, it typically rests on a structural property attributed to the data, not on an empirical comparison. [10] are explicit on this point: they prefer B-splines over Fourier "due to the aperiodicity of the data," while the general literature reserves Fourier bases precisely for the opposite case, periodic signals [11]. Wavelet bases, in turn, are justified not by periodicity but by locality: [12] and [13] adopt them specifically to capture discontinuities or localized features that a global basis oversmooths. This correspondence – periodicity to Fourier, global smoothness to B-splines, locality or discontinuity to wavelets – is the theoretical argument available in the field, but it is rarely applied systematically: most of the 91 reviewed works simply adopt B-splines without discussing whether the series at hand satisfies the condition that would, in theory, favor that choice.

The minority of works that directly compare basis families (25 of 91) shows that this choice is not cosmetic: results depend strongly on the application domain and do not generalize from one setting to another. [14] find that a regularized Gaussian radial basis outperforms B-splines in mean squared error for unbalanced data; [15] find the opposite for GPS geodetic curves, where P-splines outperform Fourier in two of three components; [13]

report an even larger gap in spectral classification, where a wavelet-based method reaches an adjusted Rand index of 0.901 against only 0.320 for k-means on B-spline bases. This heterogeneity of outcomes – no single family dominates universally – is precisely what makes it untenable to extrapolate the suitability of B-splines, or of any other family, from one domain to another without domain-specific evidence: if the choice of basis were immaterial, these 25 comparative works would not report different winners across applications.

A second, more recent line of work (2016–2024) responds to the same problem not by choosing more carefully among the classical families but by avoiding the choice altogether: the matrix-completion methods of [16, 17] and the orthonormal "splinet" bases of [11, 18, 19] estimate an orthonormal basis directly from the data instead of fixing one in advance. [20], in a register closer to that of this thesis, show that penalizing the basis (P-splines) matters as much as, or more than, the family chosen, reducing the mean squared error substantially relative to unpenalized regression splines within the same B-spline family – a result that already anticipates Section 3.4: the effect of the basis family and the effect of the smoothing regime are not the same question, and the general FDA literature is beginning to separate them. What it does not do, as Section 3.6 shows, is separate them for economic or financial series.

3.3 FPCA: how basis choice propagates into estimation and truncation

The most direct evidence that basis choice conditions the outcome of FPCA does not come from a comparison of basis families in itself, but from a controlled experiment that precedes it: [21] fits the same curves with B-spline, Fourier, and power bases, fixing the same dimension $p = 5$ in all three cases to isolate the effect of the family, and finds that "k-means clustering results vary depending on how the curves are fit to the data." This is the cleanest empirical check in the corpus that PR2 has an affirmative answer: the basis family not only affects the fit, it affects the downstream analysis built on top of that fit. A second source of variation, orthogonal to the family, lies in the formulation of FPCA itself: [22] show that a dynamic FPCA, computed in the frequency domain, explains more variance than a static FPCA under the same underlying basis (80% against 73% in their

application to PM10 pollution data) – how the FPCA itself is computed competes with the basis in which the curve is represented as a source of differences in the results.

Where Section 3.2 found a partial convergence toward B-splines in the choice of family, the criterion used to truncate that basis – how many basis functions to retain – shows no comparable convergence. The corpus documents at least eight distinct approaches in active use: GCV, AIC/BIC and their variants, generalized information criteria (GIC, MAIC, GBIC), automatic Bayesian selection (Bernoulli-type latent variables in [23]; sparse or variational priors in [24]), an eigenvalue-ratio criterion, as used by [25], cumulative explained-variance thresholds, and purpose-built bootstrap procedures [26]. [27] directly compare four of these criteria (GCV, GIC, MAIC, GBIC) within the same study and treat the choice of criterion as a design decision as consequential as the basis family itself – a reminder that choosing the basis well and choosing how many functions of that basis to use are two independent decisions that are rarely studied separately.

One antecedent isolates that independence explicitly, in a way that directly anticipates the central argument of this thesis: [28] deliberately fix a large dimension K in their B-spline basis and control smoothness exclusively through the penalization parameter, instead of selecting K by a criterion such as GCV. It is, within the corpus, the methodological design closest to separating the effect of basis dimension from the effect of the smoothing regime – but it does so within a single family (B-splines), not as a comparison across families, and not on volatile economic series (Section 3.6). This same tension, between selecting the dimension automatically or decoupling it from smoothing, appears again, in a different form, in Section 3.4.

3.4 Smoothing, penalization, and parameter selection

The separation between the effect of the basis family and the effect of the smoothing regime, which PR3 poses as an open question, does have antecedents within a single family. [20] hold the B-spline family fixed and vary only the type of penalization – none (regression splines), continuous (smoothing splines), and discrete (P-splines) – finding that both penalized approaches reduce the mean squared error substantially relative to the unpenalized case. [28] take this separation further: they fix a large dimension K and control smooth-

ness exclusively through the penalization parameter, instead of selecting K by a criterion such as GCV. In both cases the basis family is held constant; what varies is the smoothing regime, and the effect is measurable and substantial. This is exactly the type of design that PR3 calls for, but applied within a single family, not as a comparison across distinct families, and not on economic or financial series.

When the practical selection of the smoothing parameter λ is examined, the most commonly cited automatic criterion, generalized cross-validation, does not always perform as expected. [29] document a case in which neither GCV nor ordinary cross-validation produced a reasonable value of λ, and the authors had to fall back on a subjective assessment of the fit. This result is relevant to PR3 because it marks a practical limit: if the automatic criterion for selecting λ can fail, any comparison between basis families that relies on an automatic, unverified selection of the smoothing parameter risks confounding the effect of the family with an artifact of a poor choice of λ. [27] come closest, within the corpus, to controlling both sources of variation at once: they compare three basis families and, for each, four distinct selection criteria (GCV, GIC, MAIC, GBIC), which allows the analysis to separate how much of the observed difference comes from the family and how much from the selection criterion. The application domain is climate, not economic.

A more recent group of works avoids the problem rather than solving it: instead of first selecting the basis dimension and then, separately, the smoothing parameter, they formulate both decisions as a single Bayesian inference problem. [23], [30], [31], and [24] use shrinkage priors (horseshoe, spike-and-slab, sparse priors via variational inference) that select which basis functions to retain and how much to penalize them within the same estimation step. This unification solves the practical problem of having to tune two parameters separately, but it does not answer PR3 in the sense that matters to this thesis: by merging both decisions into a single mechanism, it does not allow isolating how much of the result is explained by the chosen basis family and how much by the smoothing regime applied to it.

Of the 21 direct antecedents of PR3, only one works with economic or financial series: [32], on the functional volatility of crude-oil futures. But its form of regularization is not the roughness penalization discussed by the other works in this section; it is a non-negativity constraint on data-driven bases. No antecedent in this corpus applies to economic or

financial series the kind of explicit separation between basis family and smoothing regime that does exist, within a single family, in [20] and [28]. Section 3.6 returns to this point.

3.5 Wavelet bases and non-smooth processes

PR4 asks how wavelet bases perform relative to polynomial or trigonometric bases for non-smooth or non-stationary processes, but among the 14 direct antecedents of this axis, applications that actually use a wavelet basis are a minority: only [13] and [12] adopt one as the central component of their method. Most of the works that address irregular processes respond to that irregularity without resorting to wavelets at all, which is itself relevant to PR4: within the recent practice documented by this corpus, wavelets are not the default response to the lack of smoothness that theory would suggest.

When the direct comparison is made, the result does not favor wavelets systematically. [13] find a considerable advantage of wavelet-based clustering over B-spline bases (an adjusted Rand index of 0.901 against 0.320), but [33] report the opposite: a wavelet-PCA analysis revealed no temporal changes beyond those already captured by a Fourier-based FPCA with smoothing. [9] describe splines and wavelets as "complementary," with similar mean-squared-error performance in one of its case studies, and explicitly notes that the literature lacks comprehensive comparisons between the two families. The pattern is the same as in Section 3.2: no single family dominates universally, and the outcome depends on the type of irregularity present in the data.

The most frequent strategy among the antecedents of PR4 is not to replace the smooth basis with a wavelet, but to adapt the smooth basis to the irregularity. [34] align the location of the basis functions with the change point they aim to detect, instead of fixing their position in advance, and show that this alignment substantially improves detection power relative to a standard FPCA. [35] replace B-spline interpolation with cubic Hermite polynomials for commodity forward curves, which exhibit breaks at contract expiry points, and find a better fit with this alternative, also non-wavelet. [20] and [28], already discussed in Section 3.4, absorb roughness through the penalization parameter within the same B-spline family, rather than switching families. None of these three strategies uses wavelets; all three modify a smooth basis to accommodate irregularity.

None of the PR4 antecedents that directly compare wavelets against other families works with economic or financial series. The two antecedents of this axis that do – [36], on intraday returns of the SPY ETF, and [32], on the volatility of crude-oil futures – do not offer that comparison: the former mentions wavelets only as a typical alternative, untested in the study itself, and the latter regularizes through non-negativity constraints, not through a wavelet basis. The question posed by PR4 therefore has no direct empirical answer for the economic-financial domain within the reviewed literature – another point that Section 3.6 takes up.

3.6 FDA in economics and finance; Latin American and Ecuadorian antecedents

Functional data analysis is well established as an applied tool in economics and finance: the 35 items in this axis cover electricity demand, stock indices, exchange rates, commodities, cryptocurrencies, and income distribution. But this breadth of domains contrasts with methodological depth: only 11 of the 35 (31%) reach the level of a direct antecedent ($Q3 \geq 1$ and $Q4 \geq 1$), against 48% of direct antecedents in the full corpus. Most of the remaining 24 applications use FDA as an analysis tool without discussing the choice of basis or evaluating the quality of the resulting representation – the same pattern of incidental use already documented as a reason for exclusion (E1) during Phase 5 screening – with scores concentrated between 0 and 4 out of 8.

Among the 11 direct antecedents, none applies to economic or financial series the separation between the effect of the basis family and the effect of smoothing that does exist, within a single family, in [20] and [28] (Section 3.4). [10] theoretically justify their choice of B-splines by the aperiodicity of the realized volatility they study, but do not compare it against another family. [37], on the Colombian stock market, combine P-splines with FPCA and clustering to characterize the functional correlation between 26 companies and the price of Brent crude oil, but likewise do not compare basis families against each other. Each economic antecedent satisfies some subset of the four valuation criteria – theoretical justification, reported parameters, evaluation of the representation – but none satisfies all four at once the way [27] or [20] do outside the economic domain.

Regional coverage is even more limited. Of the 91 items in the corpus, only three have any connection to Latin America: [38], on electricity demand in Colombia; [37], already cited; and [2] itself, on demographic and economic indicators for the region. None of the 91 items retrieved by the systematic search works specifically with Ecuadorian data. This scarcity is not an artifact of the search strings – Phase 4 had already anticipated a low volume for the regional search string – but a pattern that is now confirmed within the set already screened and scored for methodological quality: Latin American literature on FDA exists, but it is scarce, and its intersection with the comparison of basis families is, within this corpus, empty.

3.7 Critical synthesis and the research gap

Before restating it as an actionable gap, the candidate gap must be checked against the extraction matrix itself rather than asserted from an impression of the five axes just reviewed. Cross-referencing the 25 items that explicitly compare two or more basis families against the questions they address and their application domain narrows the picture considerably: only 5 of those 25 touch PR5, applied to electricity prices [39], wastewater epidemiology (a marginal connection to PR5) [33], natural gas flows [40], intraday returns of the SPY exchange-traded fund [36], and commodity forward curves [35]. None of these five isolates the effect of the basis family from the effect of the smoothing or penalization regime as an explicit methodological contribution; each reports model comparisons or extensions rather than a design that controls both factors separately. The only item in the corpus that does separate basis family from estimation method explicitly, [27], applies to weekly climate data, not to economic or financial series. And, consistent with the regional scarcity documented in Section 3.6, none of the 91 items applies to Latin American or Ecuadorian data. The candidate gap therefore holds, but it must be stated more precisely: the issue is not that systematic basis comparison is absent from FDA in general – 25 of 91 items (27.5%) perform one, and at least one isolates family from estimation method – but that this comparison has not been made for volatile economic or financial series, still less with the effect of smoothing isolated or with Latin American data.

Three observations follow from the five axes reviewed above. **What is established.**

Representing functional data through basis expansion is well established (B-splines dominate with 30 of 91 single-family uses, followed by empirical/FPCA bases with 12 of 91), and the criteria for selecting basis dimension are mature and in widespread use (GCV, AIC/BIC, and cross-validation appear in most items that report a criterion). **What is fragmentary.** Systematic comparison across basis families is not uncommon in the general FDA literature – 25 of 91 items (27.5%) explicitly compare two or more families, and at least one ([27]) formally isolates the effect of the basis family from the effect of the estimation method. But this comparative literature concentrates on domains outside economics and finance (climate, genetics, spectroscopy, geodesy, EEG): only 5 of those 25 works touch economic or financial series, and none of them separates the effect of the family from the effect of the smoothing regime. **What is absent.** No work in the reviewed literature offers a systematic, controlled comparison of functional basis families for FPCA representation of volatile economic or financial series that isolates the effect of basis choice from the effect of penalization or the smoothing regime, and no application of this kind exists for Latin American or Ecuadorian data. This is precisely where [2] – the article that motivates this thesis – stops: it uses B-splines exclusively, without comparison and without penalization, and its own proposed future work explicitly calls for "considering different approximation basis, specifically for the case of GDP-GR" – textual confirmation, from within the field's own literature, of the exact gap this thesis addresses.

3.8 Positioning of this thesis

This thesis is designed to close the gap identified above on its narrowest and most defensible terms: a controlled comparison of functional basis families for volatile economic time series that isolates the effect of basis choice from the effect of the smoothing regime, applied to Latin American data. Chapter 4 extends the dataset of [2] – the study that motivates this work – to an annual panel of GDP growth rates for 20 Latin American countries from 1960 to 2024, incorporating six years absent from the original study, among them the 2020 COVID-19 contraction and the region's heterogeneous post-pandemic recovery. Unlike the 25 comparative works identified in Section 3.7, none of which applies to economic or financial data with the effect of smoothing controlled, the comparison here

fixes reconstruction quality across families before comparing them: the wavelet threshold is calibrated, per country, to match the RMSE achieved by the B-spline fit, so that differences in sparsity and in the resulting FPCA are attributable to the basis family rather than to an uncontrolled difference in smoothing level. This design answers, for the economic-financial domain that Section 3.6 found empty, the same question that [20] and [28] answer only within a single basis family (Section 3.4): what is the effect of the basis family once the smoothing regime is no longer allowed to vary freely across it.

The scope of this comparison is deliberately narrower than some of the alternatives reviewed in this chapter. It does not estimate a data-driven orthonormal basis, as the matrix-completion and splinet methods of Section 3.2 do, nor does it unify basis selection and penalization into a single Bayesian mechanism, as the shrinkage-prior methods of Section 3.4 do; both choices would reintroduce, in a different form, the confound this thesis is designed to remove. It restricts the comparison to three classical, non-data-driven families – B-spline, Fourier, and wavelet (Haar, DB2, Sym4) – and evaluates them jointly on reconstruction fidelity, coefficient economy, and the interpretability of the resulting functional principal component analysis, directly extending the single-basis analysis of [2] in the three directions its own proposed future work calls for. Within the literature reviewed in this chapter, it is, to the best of this review's knowledge, the first systematic comparison of this kind applied to Latin American economic data.

Chapter 4

Methodology

4.1 Data

The dataset consists of annual GDP growth rates (expressed as percentage points) for 20 Latin American countries over the period 1960–2024, sourced from the World Bank Open Data repository. The countries included are: Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Cuba, Dominican Republic, Ecuador, El Salvador, Guatemala, Haiti, Honduras, Mexico, Nicaragua, Panama, Paraguay, Peru, Uruguay, and Venezuela. This extends the dataset of [2] by six additional years (2019–2024), incorporating both the COVID-19 contraction of 2020 and the heterogeneous post-pandemic recovery trajectories across the region.

Missing values are present for the year 1960 across most countries, and additionally for El Salvador (1960–1965) and Cuba (1960–1971). All missing values are handled prior to basis expansion via linear interpolation between adjacent observed values, followed by forward-fill and backward-fill to address leading and trailing gaps. This imputation is applied exclusively for the purpose of constructing the functional representation; all reconstruction quality metrics are evaluated only on observed, non-imputed data points.

4.2 Basis Expansion Methods

Functional Data Analysis requires that discrete observations $\{(t_j,\, y_i(t_j))\}_{j=1}^{T}$ for each country i be converted into a continuous functional object $X_i(t)$ defined over $\mathcal{T} = [1960,\, 2024]$.

This is achieved by expressing each function as a linear combination of basis elements:

$$X_i(t) \approx \sum_{k=1}^{K} c_{ik}\,\phi_k(t), \tag{4.1}$$

where $\{\phi_k\}_{k=1}^{K}$ is the chosen basis system and c_{ik} are the estimated coefficients. The choice of basis is the central object of study of this thesis. Three families are compared.

4.2.1 B-spline Basis (Baseline)

B-splines are piecewise polynomial functions of degree p joined at interior points called knots, with global continuity up to order $p-1$. They are the standard basis in FDA for non-periodic data [1] and serve as the baseline for comparison, replicating the approach of [2].

GDP growth trajectories are assumed to belong to the Sobolev space $W_2^m(\mathcal{T})$, which requires that derivatives up to order m be square-integrable. The smoothing spline is obtained by minimizing:

$$\sum_{j=1}^{T}\Big(y_i(t_j) - X_i(t_j)\Big)^2 + \lambda \int_{\mathcal{T}} \Big[X_i^{(m)}(t)\Big]^2\,dt, \tag{4.2}$$

where the smoothing parameter λ controls the bias-variance tradeoff. In this work λ is chosen adaptively per country via a robust noise estimate $\hat{\sigma}$, computed from the vector of first differences Δy as:

$$\hat{\sigma} = \frac{\text{median}(|\Delta y - \text{median}(\Delta y)|)}{0.6745\,\sqrt{2}}, \tag{4.3}$$

and the smoothing parameter is set to $\lambda = n\,\hat{\sigma}^2$, where n is the number of observed points. This data-driven choice ensures that the smoothing level is proportional to each country's intrinsic volatility rather than fixed globally.

4.2.2 Fourier Basis

The Fourier basis consists of sinusoidal functions of increasing frequency and is optimal for functions in $L^2(\mathcal{T})$ that exhibit periodic or quasi-periodic behaviour. A Fourier approxi-

mation retaining K frequency components is:

$$X_i(t) \;=\; a_0 \;+\; \sum_{k=1}^{K}\Big[a_k \cos\Big(\tfrac{2\pi k\, t}{T}\Big) \;+\; b_k \sin\Big(\tfrac{2\pi k\, t}{T}\Big)\Big], \tag{4.4}$$

implemented via the Discrete Fourier Transform (DFT) by retaining the K lowest-frequency components and zeroing the remainder. Three levels are compared: $K \in \{5,\, 10,\, 15\}$.

A known limitation of the Fourier basis for economic data is the *Gibbs phenomenon*: near discontinuities or structural breaks—such as the 1982 Latin American Debt Crisis or the 2020 COVID-19 contraction—a finite Fourier truncation produces spurious oscillations in the neighbourhood of the jump. This is examined empirically in Chapter 5.

4.2.3 Wavelet Basis

Wavelets provide an unconditional basis for Besov spaces $B^s_{p,q}(\mathcal{T})$, which measure regularity in a way that tolerates spatial inhomogeneity—that is, functions that are smooth over some time intervals yet exhibit sharp transitions in others [3]. A function with a sparse wavelet representation belongs to a Besov ball of high regularity, whereas a function requiring many wavelet coefficients to represent accurately does not admit efficient approximation in either Sobolev or Fourier bases.

A multi-level discrete wavelet transform (DWT) decomposes a signal y into approximation and detail coefficients:

$$y \;=\; \sum_{k} a_{J,k}\, \phi_{J,k}(t) \;+\; \sum_{j=1}^{J}\sum_{k} d_{j,k}\, \psi_{j,k}(t), \tag{4.5}$$

where $\phi_{J,k}$ and $\psi_{j,k}$ are translates of the scaling function and the mother wavelet at decomposition level j and translation k, $J = 4$ is the number of decomposition levels used throughout, $a_{J,k}$ are the approximation coefficients at the coarsest scale, and $d_{j,k}$ are the detail coefficients at scale j.

The DWT is implemented as an explicit orthogonal matrix $\mathbf{W}$ constructed with circular (periodic) boundary conditions, so that $\mathbf{W}\mathbf{W}^\top = \mathbf{I}$ exactly. The inverse transform is then $\mathbf{W}^\top$, guaranteeing machine-precision round-trip fidelity independent of signal length. Three wavelet families of increasing smoothness order are compared:

- **Haar**: the simplest orthogonal wavelet, piecewise constant, with one vanishing moment. Suitable for detecting level shifts.
- **Daubechies 2 (DB2)**: the shortest smooth orthogonal wavelet, filter length 4, with one vanishing moment. Balances time and frequency localisation.
- **Symlet 4 (Sym4)**: a nearly symmetric variant of Daubechies wavelets, filter length 8, with three vanishing moments. Provides smoother reconstructions while maintaining compact support.

Denoising is achieved via *soft thresholding* of the detail coefficients following [3]:

$$\hat{d}_{j,k} \;=\; \operatorname{sgn}(d_{j,k})\,\max\Big(|d_{j,k}| - \lambda,\; 0\Big), \tag{4.6}$$

where $\lambda \geq 0$ is the threshold parameter. The approximation coefficients $a_{J,k}$ are never thresholded. Setting $\lambda = 0$ yields lossless reconstruction; increasing λ progressively removes fine-scale detail, interpolating continuously between the raw-data representation and a smooth approximation.

4.3 Comparison Metrics

Three metrics are used to compare basis performance, each addressing a distinct aspect of approximation quality.

4.3.1 Reconstruction Error

The root mean squared error (RMSE) between the observed values and the functional reconstruction, evaluated exclusively on non-imputed observed time points:

$$\mathrm{RMSE}_i \;=\; \sqrt{\frac{1}{|\mathcal{O}_i|}\sum_{t\,\in\,\mathcal{O}_i}\Big(y_i(t) - \hat{X}_i(t)\Big)^2}, \tag{4.7}$$

where $\mathcal{O}_i$ denotes the set of observed (non-missing) time points for country i.

Because wavelets admit lossless reconstruction at $\lambda = 0$ (giving RMSE = 0), a direct RMSE comparison across bases is only meaningful at equivalent smoothing levels. The fair comparison strategy adopted here is: for each country, find the wavelet threshold λ^*

such that $\mathrm{RMSE}_i^{\mathrm{wav}}(\lambda^*) \approx \mathrm{RMSE}_i^{\mathrm{bsp}}$, and then count the number of nonzero coefficients required. This directly tests the Besov space hypothesis: at equivalent reconstruction quality, a sparser basis requires fewer nonzero coefficients.

4.3.2 Sparsity

The compression ratio under soft thresholding, defined as the fraction of wavelet coefficients set to zero:

$$\mathrm{Sparsity}(\lambda) \;=\; 1 - \frac{\|\hat{\mathbf{d}}(\lambda)\|_0}{|\mathbf{d}|}, \tag{4.8}$$

where $|\mathbf{d}|$ denotes the total number of wavelet coefficients across all decomposition levels.

4.3.3 FPCA Variance Explained

Functional Principal Component Analysis (FPCA) is applied to the matrix of functional reconstructions $\mathbf{X} \in \mathbb{R}^{N \times T}$ (where $N = 20$ countries and $T = 65$ time points). The centred matrix $\widetilde{\mathbf{X}} = \mathbf{X} - \mathbf{1}\bar{X}^\top$ is decomposed via the Singular Value Decomposition (SVD):

$$\widetilde{\mathbf{X}} \;=\; \mathbf{U}\,\mathbf{S}\,\mathbf{V}^\top, \tag{4.9}$$

where the columns of $\mathbf{V}$ are the functional principal components (eigenfunctions of the empirical covariance operator), the diagonal entries of $\mathbf{S}^2$ are proportional to the eigenvalues, and the rows of $\mathbf{US}$ are the PC scores per country.

The percentage of variance explained by the k-th FPC is:

$$\mathrm{VE}_k \;=\; \frac{s_k^2}{\sum_j s_j^2} \;\times\; 100\%, \tag{4.10}$$

and the number of FPCs required to reach 80% cumulative variance is recorded as n_{80}. A smaller n_{80} indicates a more parsimonious functional representation of the cross-country variation structure.

4.4 Computational Implementation

All analyses are implemented in Python 3.12. B-spline fitting uses `scipy.interpolate.UnivariateSp` Fourier expansions use `scipy.fft`; wavelet transforms are implemented from first principles using explicit orthogonal matrices, without relying on external wavelet libraries, to ensure full transparency and reproducibility of the reconstruction pipeline. Visualisations are produced with `matplotlib` and `seaborn`. Complete source code is provided in Appendix A.

Chapter 5

Results and Discussion

5.1 Exploratory Analysis

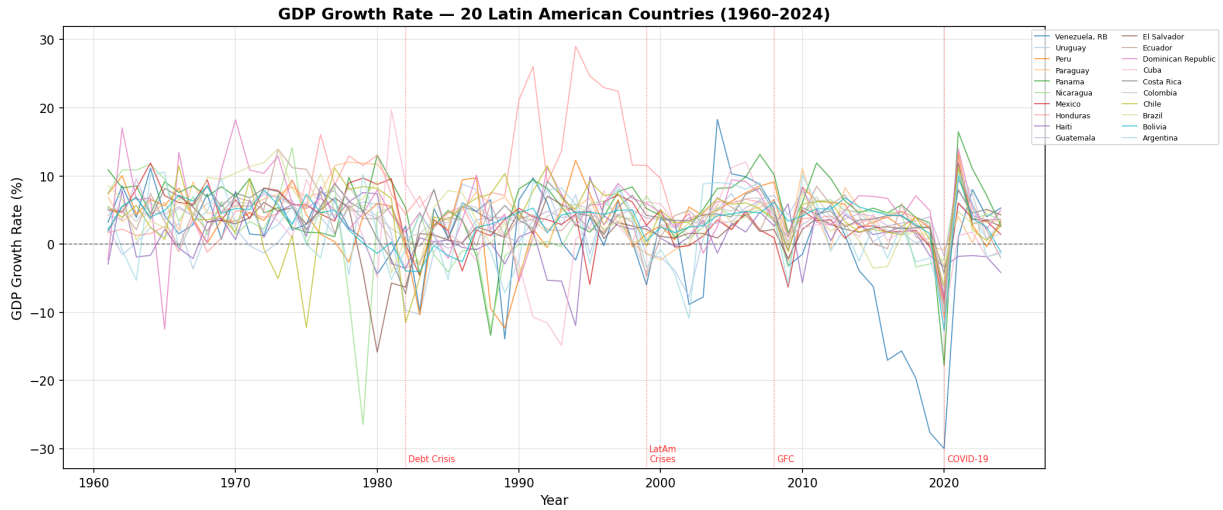


Figure 5.1: GDP growth rate trajectories for 20 Latin American countries, 1960–2024. Red dotted lines mark major regional crises.

Figure 5.1 presents the GDP growth trajectories for all 20 countries over the period 1960–2024. The data exhibits three defining characteristics that motivate the methodological choices of this thesis.

First, the series is *heteroskedastic*: the cross-sectional standard deviation of growth rates varies substantially over time.

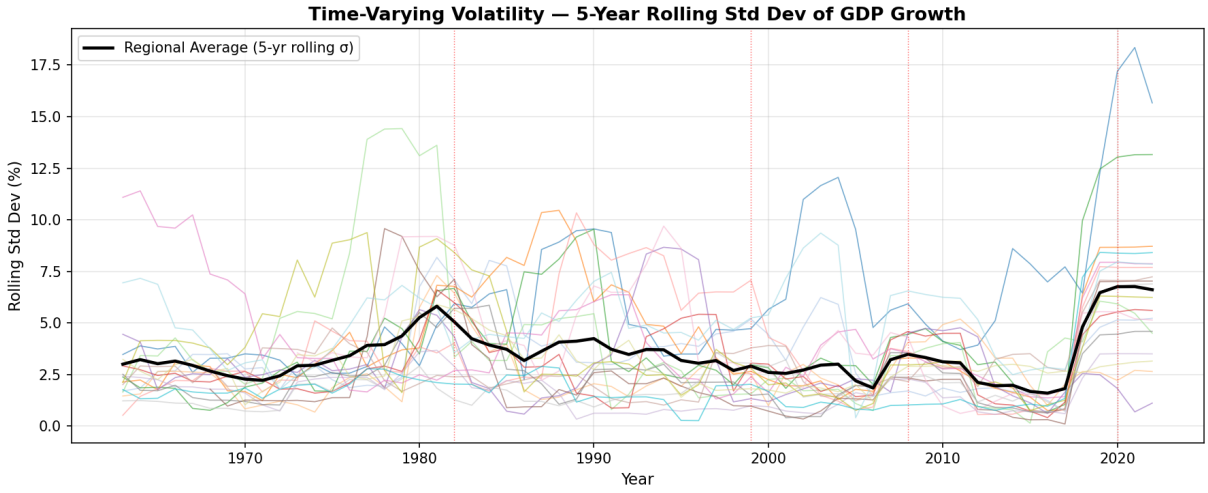


Figure 5.2: Rolling std (time-varying volatility).

Figure 5.2 shows the 5-year rolling standard deviation, which peaks near the 1982 Latin American Debt Crisis and again after the 2020 COVID-19 contraction, with a relatively calm period from approximately 1995 to 2015. This non-stationarity of variance violates the global smoothness assumptions underlying both B-spline and Fourier bases.

Second, *country-level volatility is highly heterogeneous.* Venezuela records the highest standard deviation ($\hat{\sigma} = 8.76\%$), more than three times that of Guatemala ($\hat{\sigma} = 2.40\%$), the most stable economy in the sample (Figure 5.3). This heterogeneity implies that a single smoothing parameterisation cannot be appropriate for all countries simultaneously.

Third, the GDP growth heatmap (Figure 5.4) reveals that the major regional crises—the 1982 Debt Crisis, the 1999 LatAm crises, the 2008 Global Financial Crisis, and the 2020 COVID-19 contraction—affect countries with different timing, magnitude, and duration. This *temporal localisation* of shocks is the key property motivating the wavelet basis: unlike Fourier methods, which describe variation globally across the entire time domain, wavelets localise simultaneously in time and frequency.

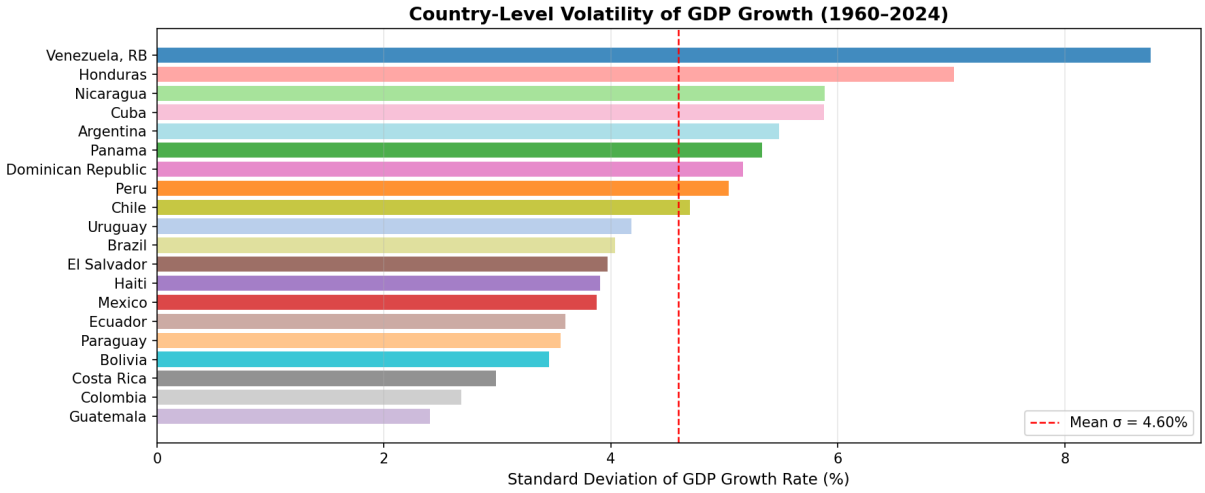


Figure 5.3: Country volatility bar chart

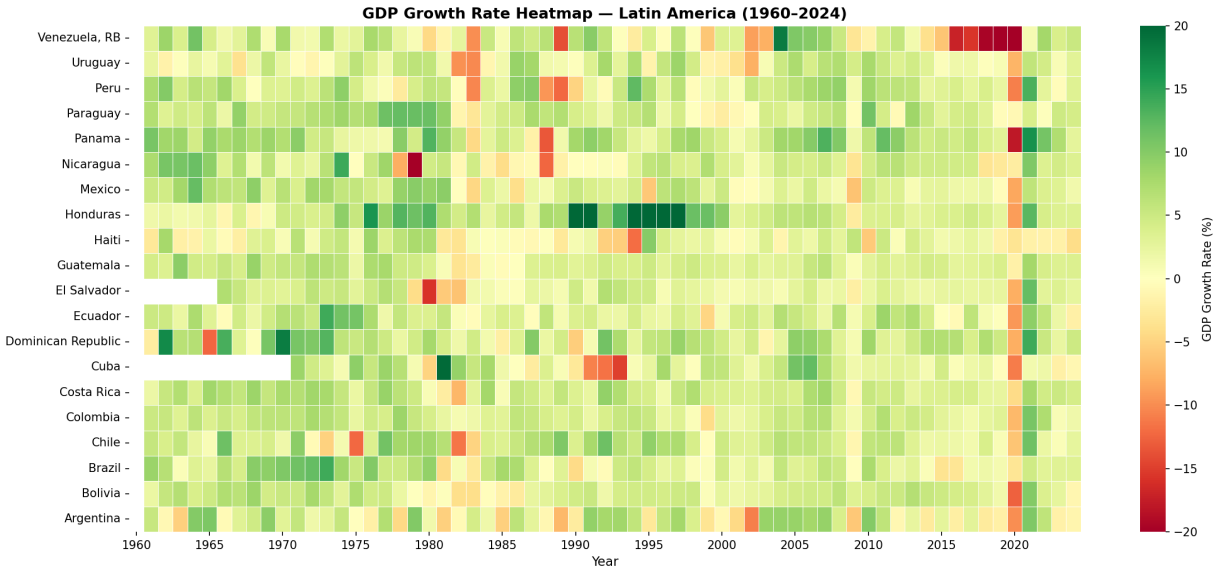


Figure 5.4: Heatmap of GDP growth over time

5.2 Basis Expansion Results

5.2.1 Reconstruction Quality and Coefficient Economy

Table 5.1 reports the mean RMSE across all 20 countries for each basis configuration at matched smoothing levels. As explained in Section 4.3.1, wavelets are evaluated at the per-country threshold λ^* that matches the B-spline RMSE, so that the comparison is made at equivalent reconstruction quality.

Table 5.1: Mean RMSE and coefficient use across all 20 countries. Wavelet columns report results at the threshold λ^* matched to the B-spline RMSE of each country individually.

Method	**Mean RMSE**	**Mean coeff. used**	**Total coeff.**
B-spline (adaptive)	2.51	19.5 knots	—
Fourier ($K = 15$)	2.65	29	65
Wavelet DB2 (λ^*)	2.52	38.8	128
Wavelet Sym4 (λ^*)	2.53	38.0	128

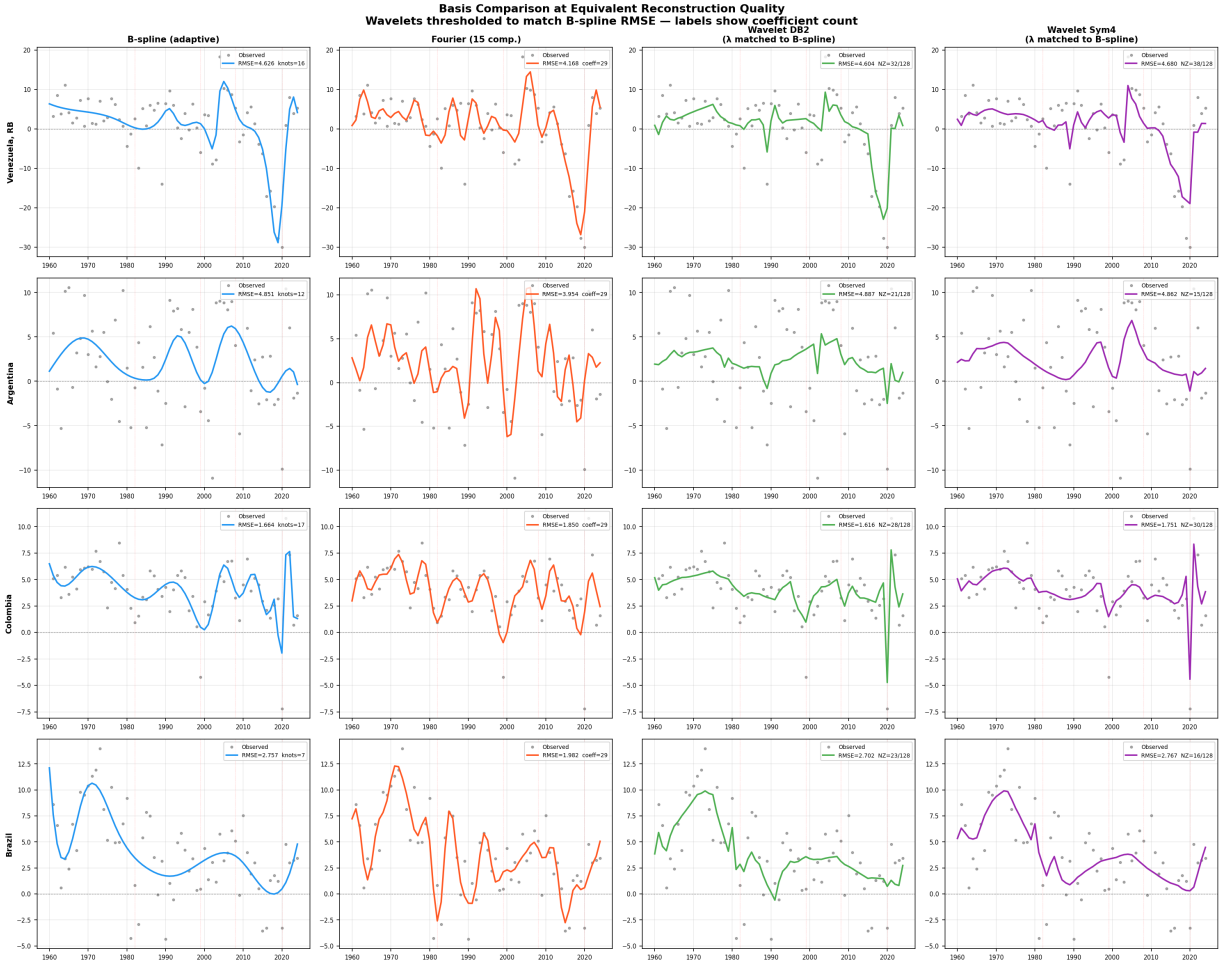


Figure 5.5: Basis comparison grid.

The B-spline achieves the lowest mean RMSE with the fewest knots (19.5 on average). However, as Figure 5.5 demonstrates for Venezuela, Argentina, Colombia, and Brazil, this apparent parsimony is misleading: the B-spline reconstruction completely suppresses the crisis episodes that define the economic histories of the most volatile countries. The smooth trend it recovers discards signal rather than compressing it.

At matched RMSE, the Sym4 wavelet requires on average 38 nonzero coefficients out

of a total of 128 (70% compression), compared to the B-spline's 19.5 knots. These 38 coefficients, however, preserve the crisis episodes. The remaining 90 zeroed coefficients correspond to genuinely small detail components whose omission causes no perceptible distortion of the economic signal.

Figure 5.6 presents the coefficient count comparison for all 20 countries. The wavelet coefficient count correlates strongly with country volatility: Venezuela, Honduras, and Nicaragua—the three most volatile economies—require the most nonzero wavelet coefficients to match B-spline smoothing quality. This is direct empirical confirmation that their GDP trajectories are less regular in the Besov sense.

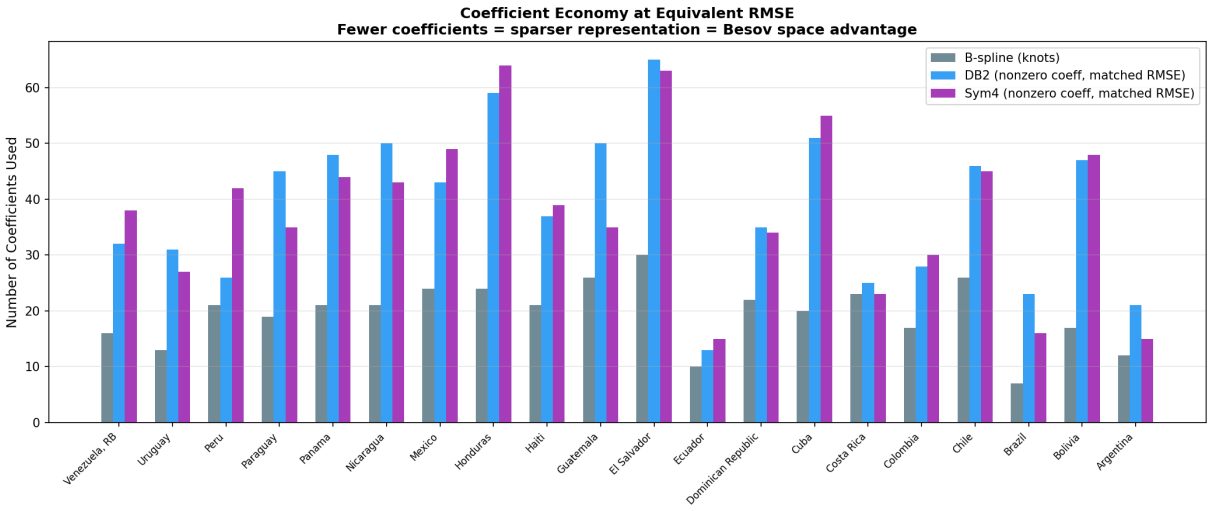


Figure 5.6: RMSE vs sparsity curves.

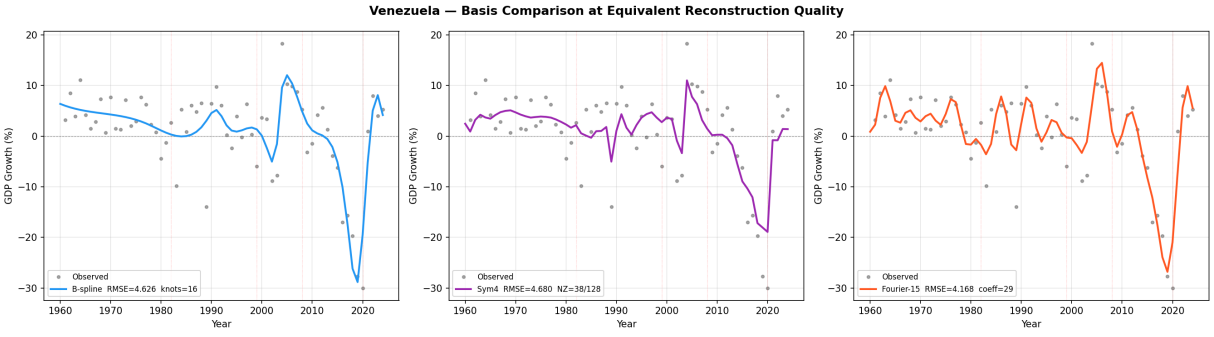


Figure 5.7: Venezuela: B-spline vs Sym4 at matched RMSE side by side.

The Venezuela case (Figure 5.7) illustrates the qualitative differences most sharply. The B-spline (RMSE = 4.63, 16 knots) produces a monotone declining trend from 2000 onward

that misses both the 2004–2008 oil boom and the catastrophic 2015–2020 contraction. The Sym4 wavelet at matched RMSE (38/128 nonzero coefficients) correctly identifies both episodes while remaining appropriately smooth during the stable 1960–1990 period. The Fourier reconstruction (15 components, RMSE = 4.17) introduces visible spurious oscillations during the stable 1960–1990 period, consistent with the Gibbs phenomenon discussed in Section 5.2.3 below.

5.2.2 Sparsity-Fidelity Tradeoff and the Besov Space Argument

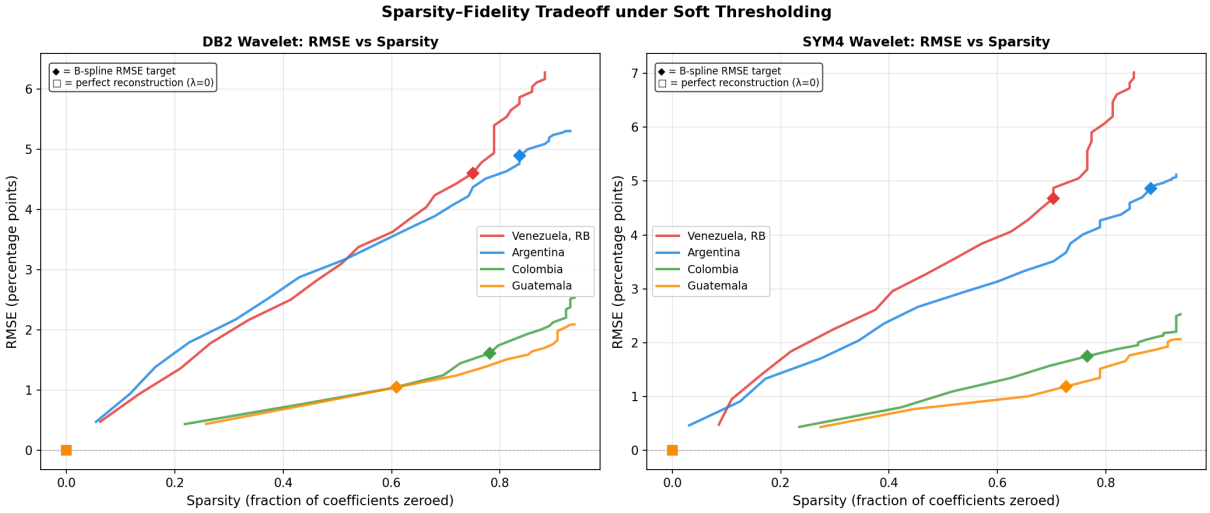


Figure 5.8: Wavelet energy distribution.

Figure 5.8 presents the sparsity-fidelity tradeoff curves for four countries of contrasting volatility under Sym4 and DB2 wavelet thresholding. Each curve traces RMSE as a function of the fraction of coefficients zeroed as λ increases from 0 to 15.

The shape of these curves provides direct empirical evidence for the Besov space characterisation. For Guatemala and Colombia—the two least volatile economies in the sample—the curves are relatively flat: RMSE increases slowly as coefficients are zeroed, meaning the GDP trajectories of these countries are genuinely sparse in the wavelet basis and live near a Besov ball of high regularity. For Venezuela and Argentina, the curves are steep: significant RMSE accumulates rapidly as coefficients are zeroed, consistent with lower Besov regularity arising from the multiscale shocks that characterise these economies.

This country-specific regularity structure is invisible to B-splines or Fourier bases, which

apply a single global smoothness criterion to all countries. The wavelet basis, by contrast, adapts to the local regularity of each trajectory, providing a richer characterisation of cross-country heterogeneity in economic volatility. The marked diamond on each curve in Figure 5.8 indicates the B-spline RMSE target: stable economies reach this target at higher sparsity levels (more efficient compression), while volatile economies reach it at lower sparsity levels.

5.2.3 Fourier Basis and the Gibbs Phenomenon

The power spectrum of Venezuelan GDP growth (Figure 5.9) shows no dominant peak at any specific period. Power is broadly distributed across 15–35 year periods, inconsistent with a regular business cycle and suggesting that the growth process is driven by aperiodic commodity and political shocks rather than a deterministic cycle. This undermines the theoretical motivation for the Fourier basis on this dataset.

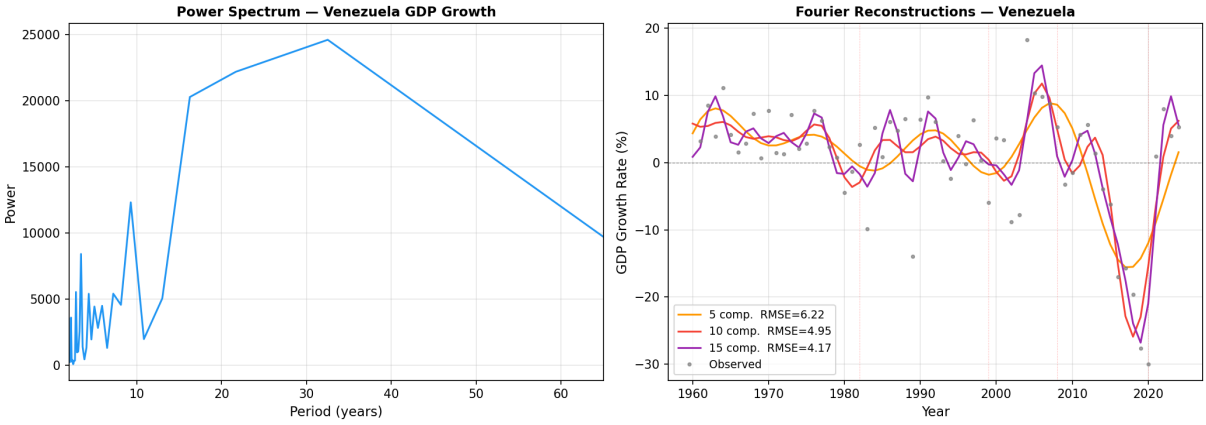


Figure 5.9: Fourier spectrum — Venezuela.

Empirically, increasing the number of Fourier components from 5 to 15 reduces RMSE from 6.22 to 4.17 for Venezuela, but introduces visible spurious oscillations during the stable 1960–1995 period (Figure 5.7, right panel). This is consistent with the Gibbs phenomenon: the finite-term Fourier approximation oscillates near the structural breaks at 1983 and 2020, corrupting the representation of surrounding smooth periods. Fourier-15 requires 29 coefficients to achieve mean RMSE = 2.65 across all countries, comparable to B-splines, but at the cost of these Gibbs-type artefacts near crisis years.

5.3 FPCA Results

5.3.1 Replication of the Original Result

Table 5.2 extends Table 2 of [2] to include all basis configurations tested in this thesis.

Table 5.2: FPCA variance explanation results. VE_k denotes the percentage of variance explained by the k-th FPC; n_{80} is the number of FPCs required to explain at least 80% of total variance. The original paper result is included for direct comparison.

Configuration	**VE_1**	**VE_2**	**VE_3**	**3-FPC**	**n_{80}**
Raw data (baseline)	21.3%	15.9%	10.0%	47.2%	8
B-spline (smooth $\sigma\times2$)	53.5%	25.0%	10.8%	89.2%	3
Fourier ($K = 5$)	40.4%	22.5%	12.5%	75.4%	4
Fourier ($K = 15$)	26.9%	17.6%	11.4%	55.9%	7
Wavelet Sym4 ($\lambda = 0$, lossless)	21.3%	15.9%	10.0%	47.2%	8
Wavelet Sym4 ($\lambda = 5$)	33.8%	22.9%	10.8%	67.6%	5
Wavelet Sym4 ($\lambda = 12$)	45.8%	27.0%	13.4%	86.2%	3
Wavelet DB2 ($\lambda = 5$)	34.2%	23.6%	11.4%	69.1%	5
Wavelet Haar ($\lambda = 5$)	32.6%	23.2%	11.3%	67.2%	5
Original paper [2]:					
B-spline order 5, dim 48	17.7%	16.5%	12.9%	47.1%	> 8

The raw data FPCA (equivalent to an interpolating B-spline, $\lambda = 0$) yields $VE_1 = 21.3\%$, $VE_2 = 15.9\%$, $VE_3 = 10.0\%$, with a 3-FPC total of 47.2%—nearly identical to the original paper's 47.1%. This confirms that the extended dataset (1960–2024 vs. 1970–2018) and revised country coverage preserve the essential variance structure identified by [2], and validates the full pipeline before proceeding to the alternative bases.

5.3.2 Effect of Basis Choice on FPCA Parsimony

The cumulative variance curves in Figure 5.10 reveal a clear and interpretable ordering among all configurations. The basis choice has a dramatic effect on FPCA parsimony, but this effect must be interpreted carefully rather than equated with quality.

B-spline smoothing ($\sigma\times2$) requires only 3 FPCs for 80% variance explained. However, Figure 5.11 shows that the B-spline FPC1 is a monotone declining trend, capturing the post-1980 growth slowdown and explaining 53.5% of variance. This high parsimony arises because smoothing has removed the crisis variation and left only the trend. It is a poorer description of the data—information has been discarded rather than organised.

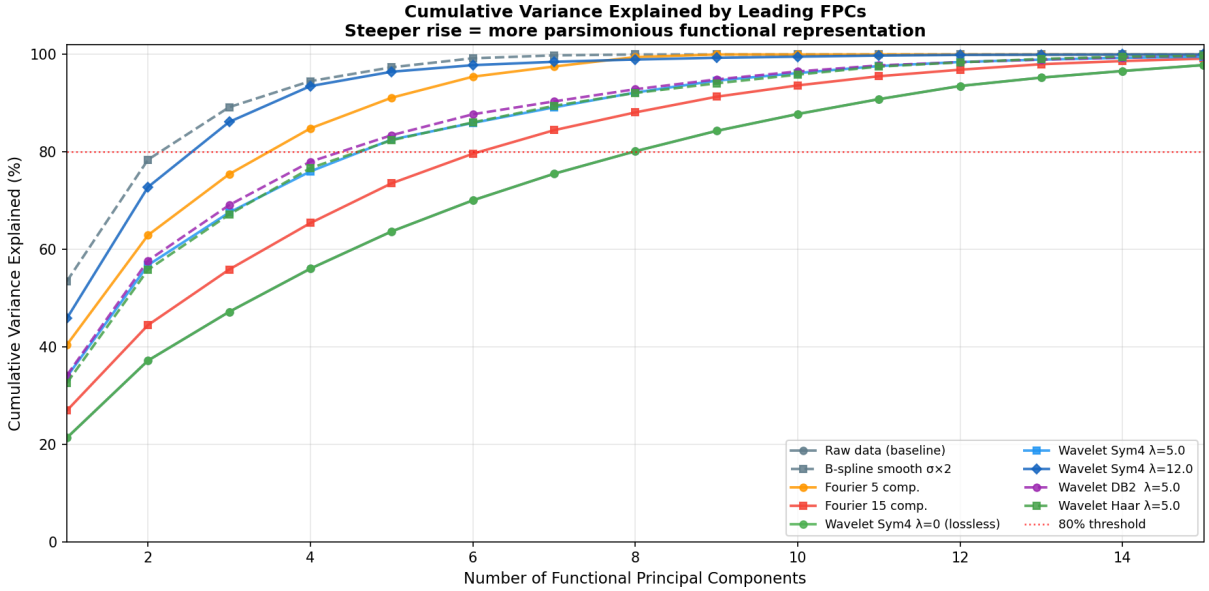


Figure 5.10: Cumulative variance curves.

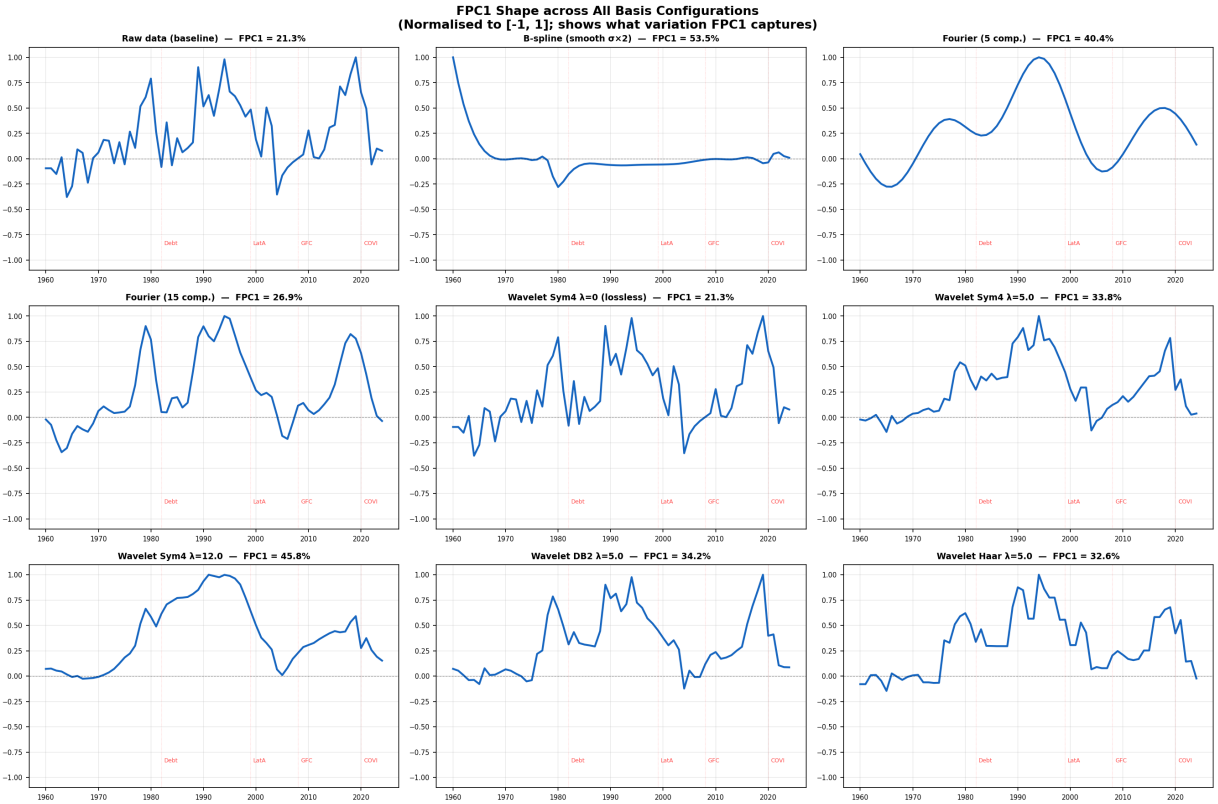


Figure 5.11: FPC1 curve comparison across bases.

Wavelet Sym4 at $\lambda = 5$ achieves a qualitatively richer FPCA. FPC1 explains 33.8% and, as visible in Figure 5.11, shows a structured pattern rising through the 1970s com-

modity boom, declining through the 1982–1990 lost decade, recovering during the 2000s commodity super-cycle, and collapsing sharply at COVID-19. This FPC1 is economically interpretable as a *commodity cycle* mode of variation—a regional factor driving correlated growth fluctuations across commodity-exporting nations. Five FPCs are required for 80%, reflecting the genuine multidimensionality of cross-country variation that the wavelet basis preserves rather than suppresses.

At $\lambda = 12$, the Sym4 wavelet converges toward the B-spline behaviour ($n_{80} = 3$), confirming that aggressive thresholding reduces the wavelet representation to a smooth approximation. The threshold parameter λ therefore acts as a *continuous interpolation* between the raw data representation ($\lambda = 0$, $n_{80} = 8$) and the smoothed representation ($\lambda \to \infty$, $n_{80} = 3$), giving the analyst direct control over the parsimony-fidelity tradeoff. This interpolation property is visible as a monotone family of curves in Figure 5.10.

5.3.3 FPC Mean Perturbation Plots

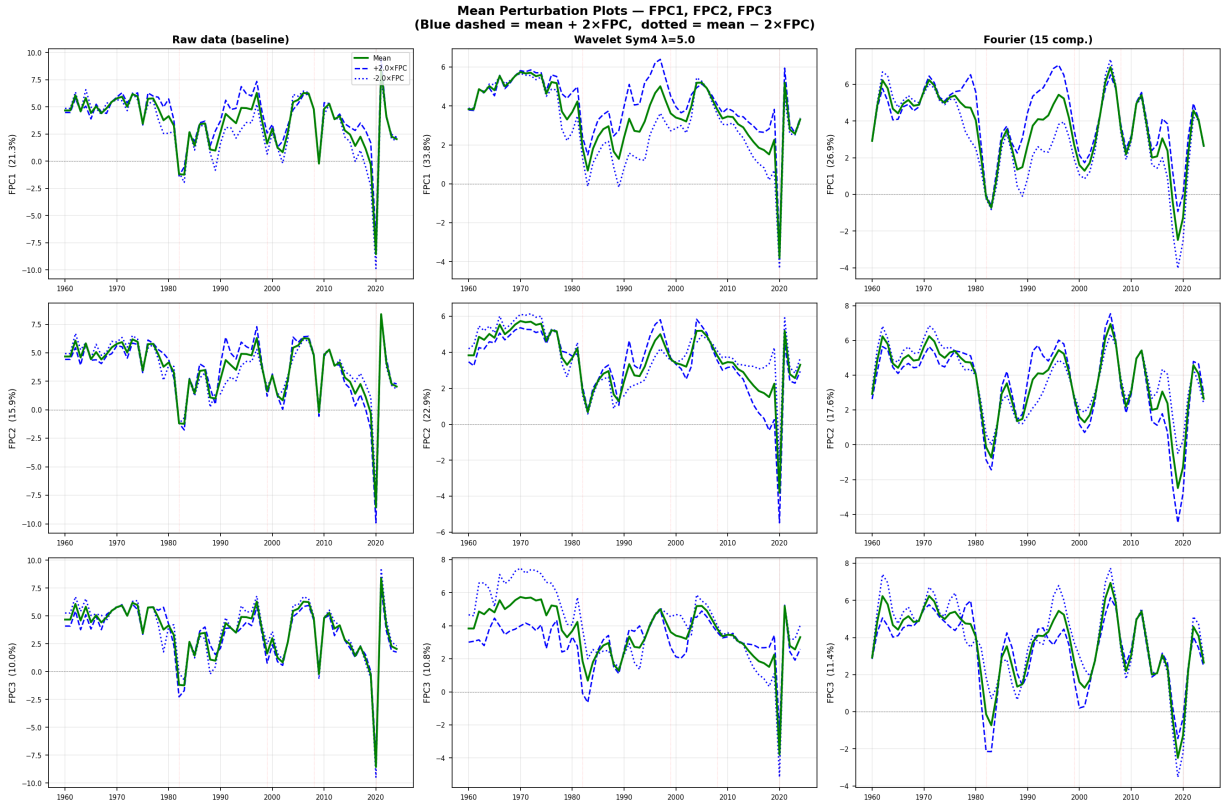


Figure 5.12: FPC curves (perturbation plots).

Figure 5.12 presents the mean perturbation plots for the first three FPCs under three configurations (raw data, Wavelet Sym4 $\lambda = 5$, and Fourier $K = 15$), replicating the style of Figure 7 in [2].

Under the raw data and lossless wavelet representations, the FPC perturbations are small relative to the mean function and noisy in character, consistent with individual crisis events being absorbed into the leading components. Under Wavelet Sym4 $\lambda = 5$, the FPC1 perturbation band is structured and economically legible: a country with a large positive FPC1 score grows faster during the commodity boom decades and suffers a sharper post-2014 decline. FPC2 under this representation captures a late-period divergence, separating countries that recovered strongly after 2020 from those that did not.

5.3.4 Score Plots and Country Separability

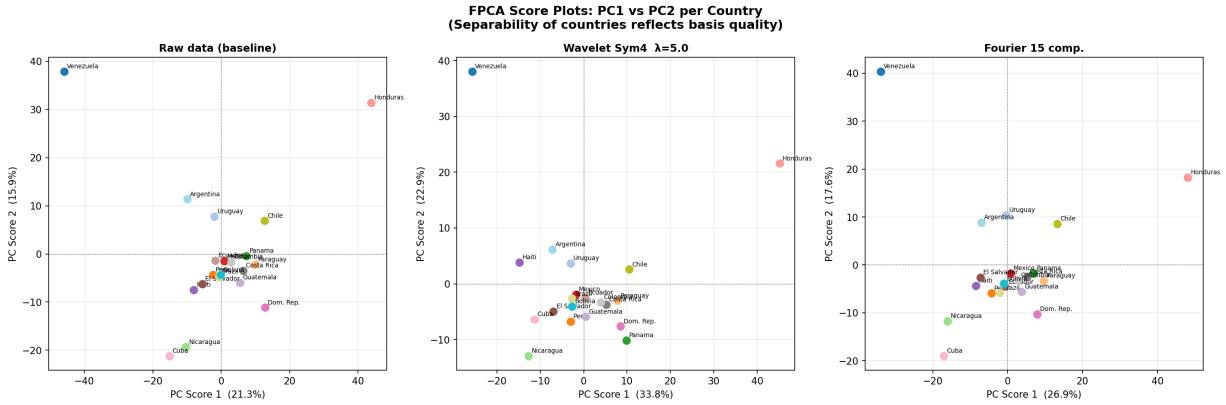


Figure 5.13: Score plots (replicating Fig 8 of paper).

The FPCA score plots (Figure 5.13) project each country onto the first two principal components. Three configurations are compared: raw data, Wavelet Sym4 $\lambda = 5$, and Fourier $K = 15$.

Venezuela is consistently the most extreme outlier in the PC1 direction across all three configurations, reflecting the unprecedented magnitude of its post-2015 economic contraction. Honduras is consistently separated in the PC2 direction, driven by its anomalous high-growth period in the 1990s.

Under the Wavelet Sym4 representation, the stable Central American economies (Guatemala, El Salvador, Costa Rica) cluster more tightly than under the raw data representation, sug-

gesting that wavelet thresholding selectively removes idiosyncratic noise while preserving systematic regional co-movement. This improved cluster structure has direct implications for downstream applications such as functional hierarchical clustering or regional policy grouping.

5.3.5 Wavelet Energy Distribution

Figure 5.14 shows the distribution of wavelet energy across decomposition levels A4 (coarsest approximation), D4, D3, D2, and D1 (finest detail) for Venezuela, Colombia, and Argentina under the three wavelet families tested.

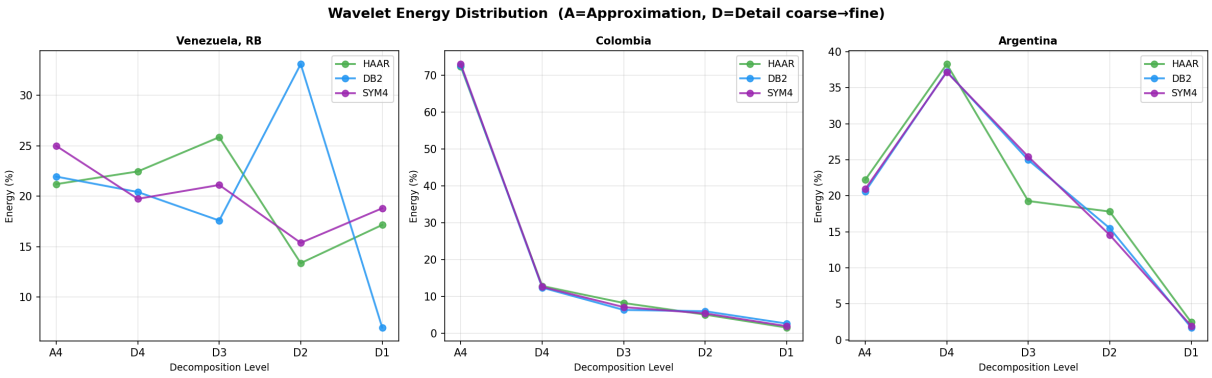


Figure 5.14: Sparsity at matched RMSE — all 20 countries.

Colombia concentrates approximately 72% of its wavelet energy in the A4 approximation component: its GDP trajectory is dominated by a smooth long-run trend with negligible fine-scale variation. These economies are well-approximated by Sobolev-class functions, for which B-splines are theoretically optimal.

Venezuela distributes energy relatively evenly across all decomposition levels. No single scale dominates, confirming that its growth process has genuine multiscale structure that cannot be captured by any single-scale basis. Argentina concentrates its energy primarily in the D4 level (the coarsest detail, corresponding to fluctuations at time scales of 8–16 years), consistent with Argentina's well-documented boom-bust cycle operating at medium-term horizons.

These country-specific energy profiles are economically interpretable and would be entirely invisible to a global basis such as B-splines or Fourier series. The wavelet multiresolution decomposition provides a natural language for identifying *at which time scales* each

country's growth variation operates—an original analytical contribution relative to [2].

5.4 Summary of Findings

The empirical results, taken together, support the following conclusions.

The **B-spline basis** achieves low RMSE through over-smoothing. It discards economically significant crisis episodes as noise and produces FPCA results that conflate trend removal with variance organisation. Its apparent parsimony is an artefact of the smoothing constraint, not evidence of an efficient functional representation.

The **Fourier basis** is inappropriate for this dataset because GDP growth rates exhibit no dominant periodic structure (confirmed by power spectral analysis) and contain localised structural breaks that produce Gibbs-type artefacts under finite-term Fourier approximation.

The **wavelet basis**, at moderate thresholding, provides the most informative functional representation across all metrics examined: it preserves crisis episodes via sparse representation of localised shocks, reveals country-specific multiscale variance structure through energy distribution analysis, produces economically interpretable FPCA modes, and offers a principled tradeoff between smoothness and fidelity via the threshold parameter λ. These advantages are most pronounced for the highly volatile economies—Venezuela, Nicaragua, Argentina, Honduras—where the gap between B-spline and wavelet representation is largest.

The finding does not support the simple claim that wavelets outperform B-splines on a scalar RMSE metric. Rather, it supports the more nuanced and more useful conclusion that *the choice of functional basis determines what the analyst sees in the data*, and that for volatile, multiscale economic series, the wavelet basis provides a fundamentally richer and more economically meaningful representation than the alternatives tested here.

Chapter 6

Conclusions

[Your conclusions]

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Appendices

.1 Appendix 1.

[Your appendix]